

Class-11th + 12th



EaJEE NOTES

for

ORGANIC CHEMISTRY

Basic से
Kattar Advanced की feel
बाबा के साथ



Rohit Agrawal
(RA Baba)



Shubh Karan Choudhary
(SKC Sir)

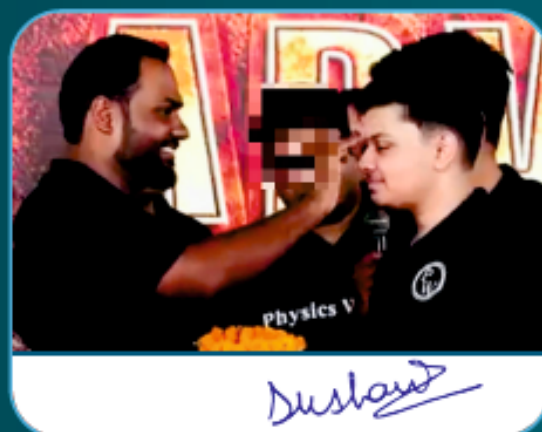
BACKLOG KILLER ❤️
Blue Book



Testimonials

- 1.** I had a great experience learning organic chemistry under Rohit Aggarwal Sir. He teaches the concepts deeply in a way that can be applied to all JEE problems and even beyond without much effort. I can attest to the effectiveness and completeness of his notes and in my preparation they were the most important resource for the subject which I followed diligently, and it had a very positive impact in my performance. Additionally, he provided a brilliant selection of problems. The problem sets cover all cases that are relevant for the exam and help in solidifying the concepts even better. The notes along with the problems provide a complete course for JEE organic chemistry. I would recommend it to any serious aspirant who wants to learn organic chemistry in an intuitive and comprehensive way.

**Sushant Padha,
AIR 52, JEE 2024**



Sushant

**Sparsh Patel,
AIR 283, JEE 2025.**



Patel

- 2.** It was a privilege to study organic chemistry under the guidance of Rohit Aggarwal Sir. He teaches organic chemistry at the best level. Thorough practice of his notes and practice questions proved instrumental during the exam. His materials progress systematically from basic principles to advanced topics. The structure and order of his notes and questions offers a wide range of applications, giving a deep understanding of the subject. Each theoretical topic is explained by detailed mechanisms, helping us understand each concept and not simply mug up. I wholeheartedly recommend his resources to any student seeking to excel in the subject and the examination.

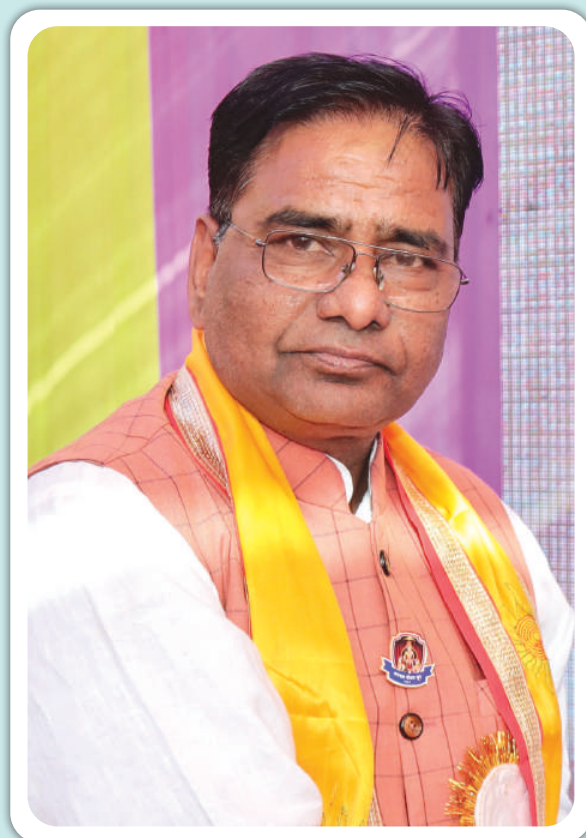
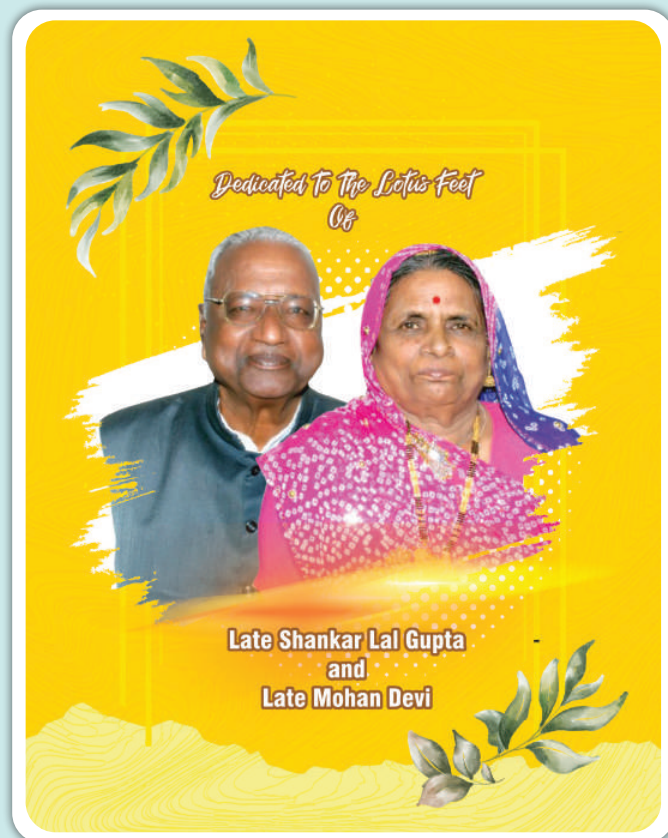
- 3.** Baba's class notes are more than sufficient for JEE Advanced.
Maine unke class notes ko chhodkar or kuch bhi extra nahi padha for my organic chemistry.
His class questions are top-level with everything and every concept you need for your exam.

**Karan Saw Adv. AIR 995,
Neet AIR 592, JAT AIR 55**



Saw

This Book is Dedicated To.....



“ये पुस्तक दादा-दादी के स्नेह और
पिता के मार्गदर्शन को समर्पित है-
जिन्होंने मेरे जीवन को अर्थ और दिशा दी।”

Pravachan



“सभी Lindanes को प्रवचन”

- ▶ Nomoskar Lindanes, कैसन बा, आप सभी को BB की बधाई।
- ▶ बाबा के इस प्रसाद में, pure basic, NCERT, Short notes और Kattar Advanced का पूरा तड़का हैं।
- ▶ इस Book को केवल Reading में Use ना करें और सबसे पहले white pages या काँपी बना लें।
- ▶ कोई भी Mechanism पहले खुद Try करे फिर उसके BTS points को अच्छे से पढ़े वरना सावधानी हटी दुर्घटना घटी।
- ▶ Reaction अच्छे से समझने के बाद सारे Question करे चाहे आप JEE के हो या NEET के।
- ▶ मैंने PYQ भी कुछ add किये हैं और सारे PYQ आपको QR Code में दे रहा हूँ ताकि COST कम हो जाए।
- ▶ Extra Objective & Subjective भी दिए हैं So समय मिले तो करना, otherwise अगर Book भी करली तो भी NORA का लेवल high रहेगा।
- ▶ Last में स्वस्थ रहे, मस्त रहे, व्यस्त रहे और चार लोगों की इज्जत हमेशा करें माता, पिता, गुरु और माता।

1,5 - epoxy
Pentane &
Aaalo0000
---∞



Acknowledgement

Rohit Sir sincerely thanks Alakh Pandey Sir and Prateek Sir, for creating a platform that makes quality education accessible to millions. He also expresses gratitude to Sanyam Badola Sir, for his unwavering support and guidance.

I (Rohit) gratefully acknowledge the encouragement and immense support of my mother Sunita Agrawal, father Bhupendra Agrawal, sister Richa Agrawal and daughter Yati Agrawal. They inspired me all the time during preparation of this book.

I (Shubhkaran) sincerely acknowledge the support of my mother Chandrapati Choudhary, father Baljeet Singh, sister Saroj, brother Vishwanath Choudhary, wife Anandi Choudhary and charming son Kanish Choudhary, Tavish Choudhary.

Our special thanks to guru 'Akshay Choudhary' for shaping this book.

Special thanks to Jyoti Rakheja, Content Faculty at PW Publication, for her dedicated efforts in designing and shaping this book.

I'm also grateful to Deepak Kumar and Rakhi Sharma (Book Compositor) for their continuous efforts.

1

Nomenclature of Organic Compounds

Definition

1. **Hydrocarbons:** Compound formed by carbon and hydrogen

ex: CH_4 , $\text{H}_3\text{C}-\text{CH}_3$,  etc.
Alternate π -bond

2. **Hydrocarbons derivatives:** Compound formed by carbon, Hydrogen and hetero atoms.

ex: CH_3Cl , CH_3-OH , $\text{CH}_3-\text{CH}_2-\text{OH}$ etc.

3. **Organic chemistry:** Branch of science which deals with hydrocarbon and its derivative

➤ Carbon

$1s^2 2s^2 2p^2$

G.S $\Rightarrow$

$\uparrow\downarrow$	$\uparrow$	1	1	
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E.S $\Rightarrow$

$\uparrow\downarrow$	1	1	1	1
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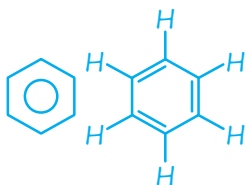
4 bonds (tetravalency)



Catenation

Practice Questions

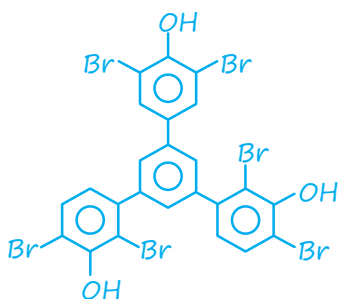
1. No. of CH_2 -group present in benzene is:



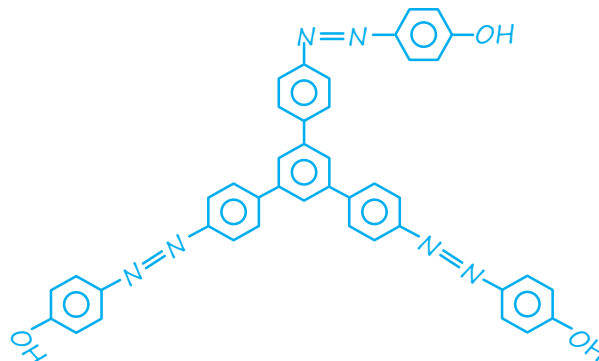
2. Number of $-\text{CH}_2$ group and hetero atom present in following compound is:



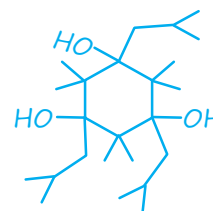
3. No. of hetero atom present in given compound.



4. No. of hetero atom + no. of C $\Rightarrow$??



5. Number of $-\text{CH}_3$ group present in following compound is:

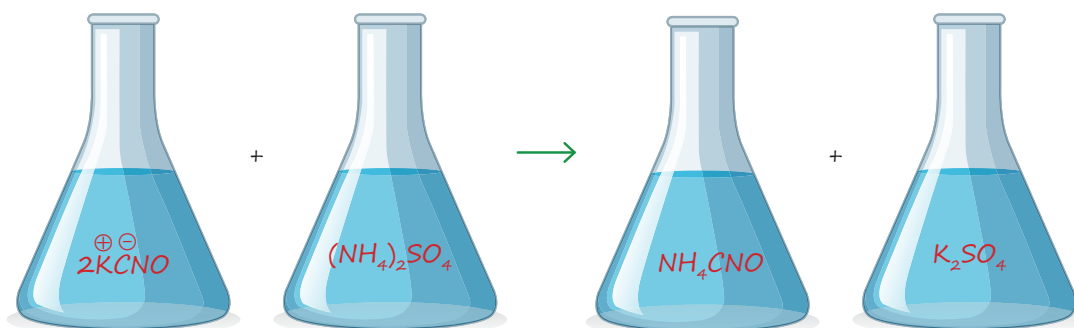


History

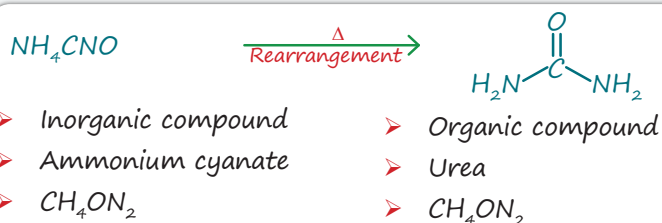
1. Berzelius was a Swedish scientist who in 1815 proposed that organic compounds could only be produced by some special force which must be existing in living organism and could not be prepared in a laboratory.

This force was called vital force and this theory came to be known as vital force theory

2. Experiment: F. wholer



Reaction



Observation

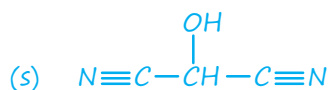
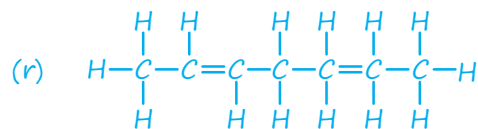
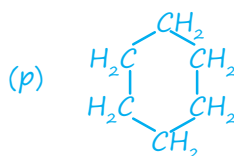
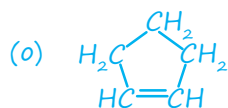
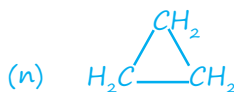
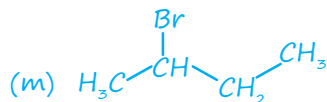
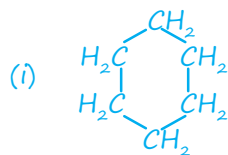
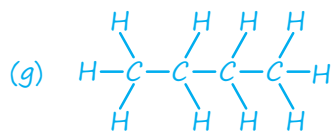
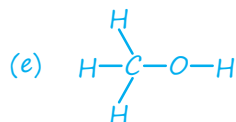
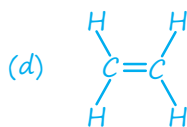
- Organic compounds can be obtained in lab
- Phenomenon of Isomerism

Representation of Organic Compound

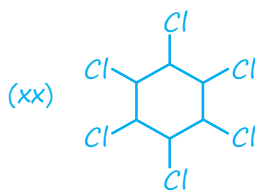
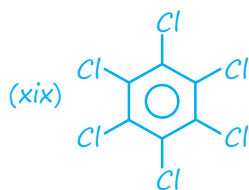
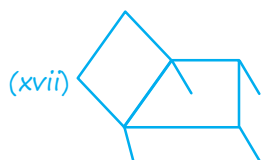
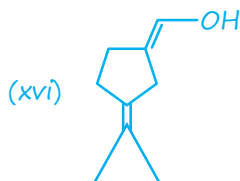
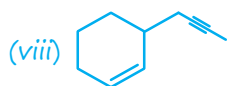
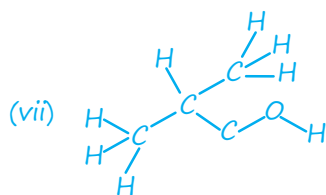
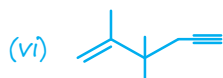
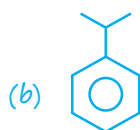
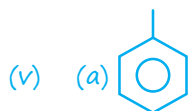
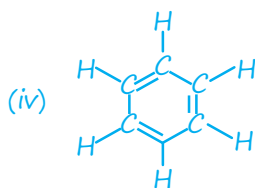
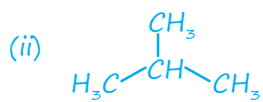
Molecular formula	Structural formula	Condensed formula	Bond line formula
➤ No. of C, no. of H, no. of hetero atoms ➤ $C_xH_yO_zN_w$	➤ All bonds are shown	➤ No bonds are shown ➤ Repeating units and branches as shown in (.) $H_3C-CH_2-\underset{\substack{ \\ CH_3}}{CH}-CH_2-$ $-CH_2-CH_2-CH_3$ $H_3CCH_2CCH(CH_3)$ $(CH_2)_3CH_3$	➤ C-H bond is never shown ➤ C ⇒ represent by.

Practice Questions

6. Find structural, condensed and bond line formula.



7. Draw the remaining representation of the following compounds?



Hybridization

Hybridisation $\Rightarrow$ number of σ + number of lone-pair

No. of σ	No. of lone-pair	Hybridisation	Shape
4	0	SP^3	Tetrahedral
3	1	SP^3	Tetrahedral
2	2	SP^3	Tetrahedral
3	0	SP^2	Planer
2	1	SP^2	Planer
2	0	SP	Linear
1	1	SP	Linear

Degree of atom/group

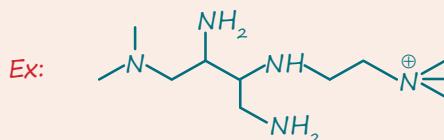
Carbon attached to how many atom.

4C $\longrightarrow$	Quaternary (4°)
3C $\longrightarrow$	Tertiary (3°)
2C $\longrightarrow$	Secondary (2°)
1C/OC $\longrightarrow$	Primary (1°)



- Hydrogen: degree of (H) = degree of attached(c)
- 4°H never exist
- degree of (H), (OH), (X), (NO_2) are same as attached degree of (C)

Amine: (N) $\Rightarrow$ same as (C)

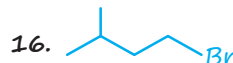
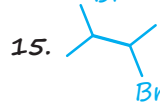
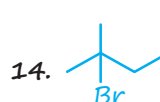
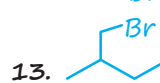


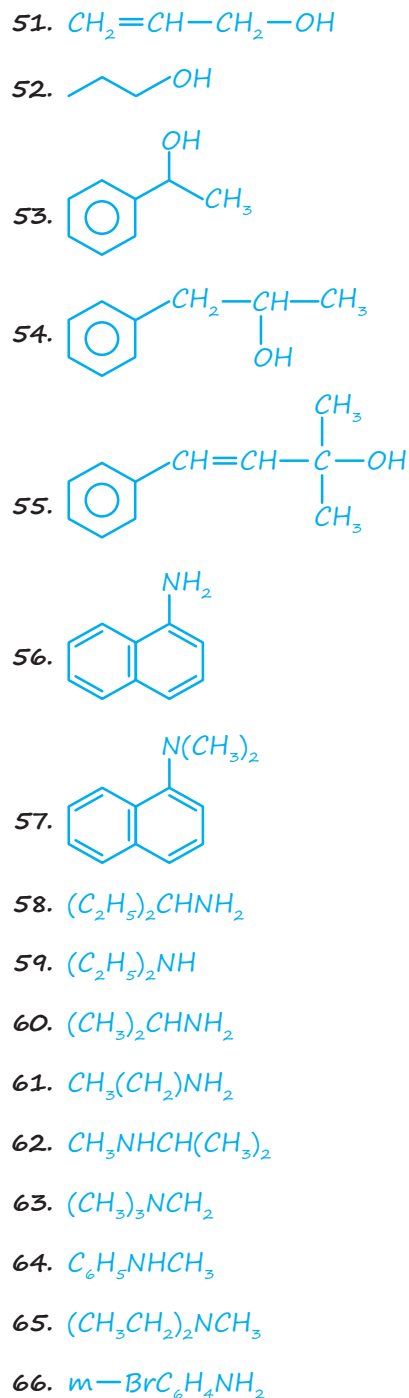
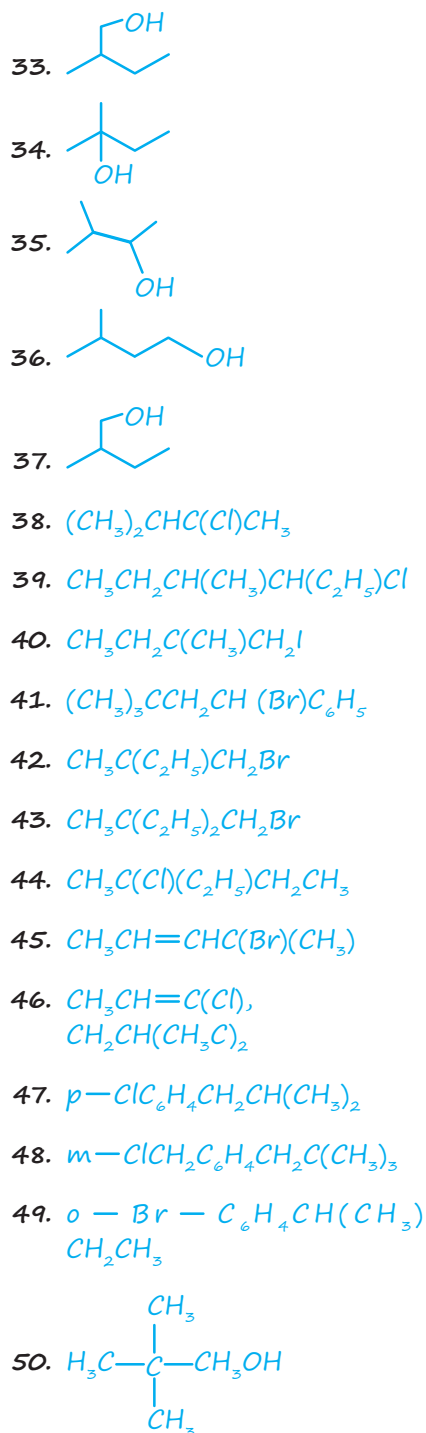
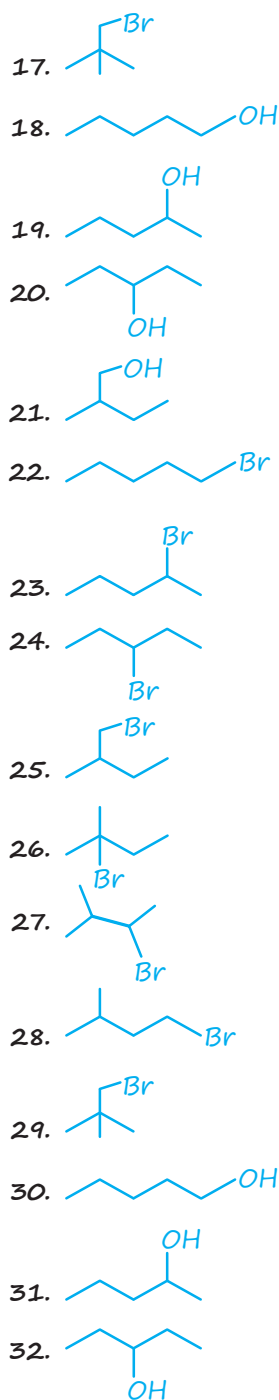
Practice Questions

8. Find σ , π and hybridisation of carbon?

- (a) CH_3Cl (b) $(\text{CH}_3)_2\text{CO}$
 (c) $\text{H}_3\text{C-CN}$ (d) HCONH_2
 (e) $\text{H}_3\text{CCH}=\text{CHCN}$ (f) CH_3F
 (g) HCNO (h) HCN
 (i) $\text{H}_2\text{C}=\text{CH}-\text{CH}_3$ (j) $\text{HC}\equiv\text{CCH}=\text{CHCH}_3$

9. Find σ , π and hybridisation of carbon in Qus. 8.
Find degree of following halide, alcohol, amine?





67. Draw the structure of alkane having least molecular mass that contains only:

- | | | |
|---|---|---|
| (a) 1° and 2° | (b) 1° and 3° | (c) 1° and 4° |
| (d) 1° , 2° and 3° | (e) 1° , 3° and 4° | (f) 1° , 2° , 3° , and 4° |

68. Draw the structure of cycloalkane having least molecular mass that contains only:

- | | | |
|---|---|---|
| (a) 2° | (b) 1° and 3° | (c) 1° and 4° |
| (d) 1° , 2° and 3° | (e) 1° , 3° and 4° | (f) 1° , 2° , 3° and 4° |

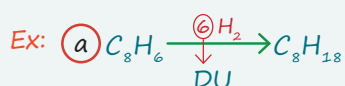
Degree of Unsaturation/Double Bond Equipment/Index of Hydrogen Deficiency

The presence of p bonds or rings within a molecule is indicated by a quantity called degree of unsaturation.

Definition: Deficiency of 2H atoms with respect to fully saturated acyclic hydrocarbon is equivalent to One DU. It is also known as Hydrogen Deficiency Index (HDI) or Double Bond Equivalent (DBE).

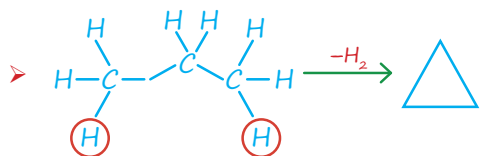
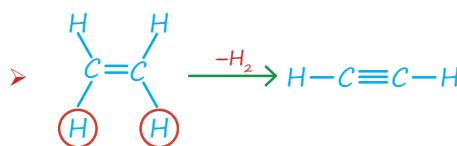
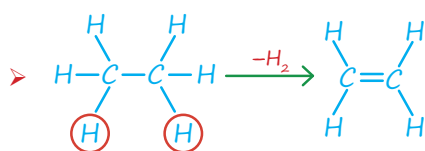
Hydrocarbon

➤ Alkane $\Rightarrow C_n H_{2n+2}$ ➤ Number of H_2 molecular required to produce open chain alkane is DU/DBE/HDI.

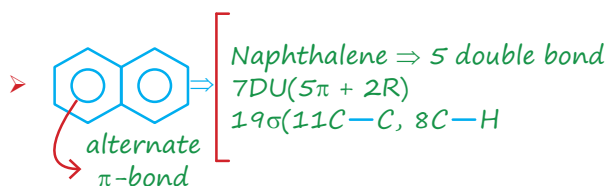
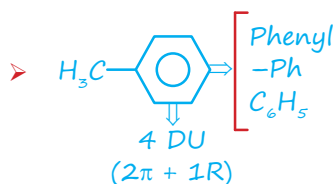
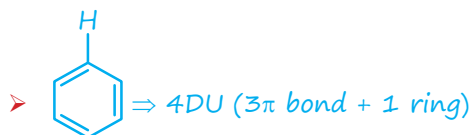


(b) $C_xH_y \Rightarrow \xrightarrow{2x+2-y} \boxed{x+1-y/2}$

Structures

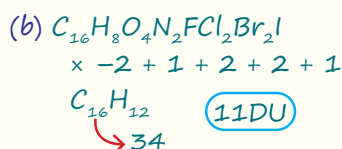
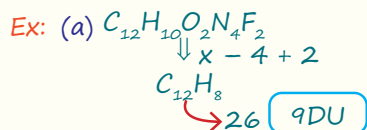


1 double bond = 1DU
1 Ring = 1DU
1-CN = 2DU ($-C \equiv N$)
1-Ph = 4DU

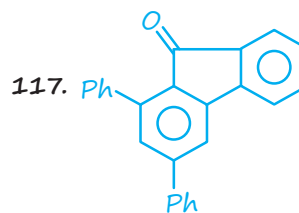
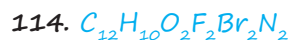
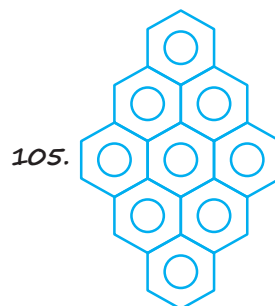
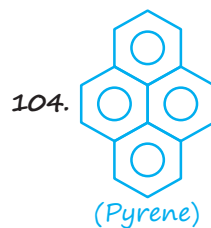
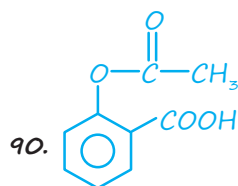
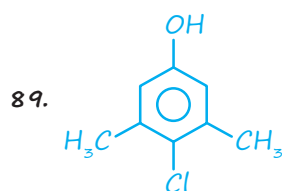
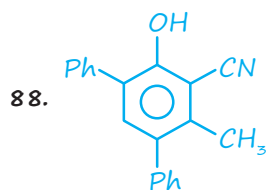
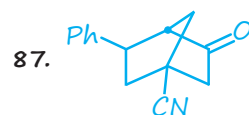
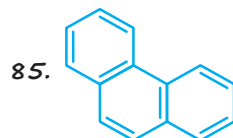
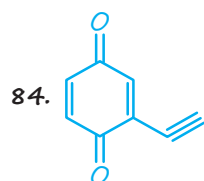
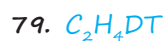


Hetero atoms

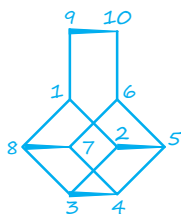
N	O	X
-H	Ignore	+H



Practice Questions



119. Basketane



120. Churhane



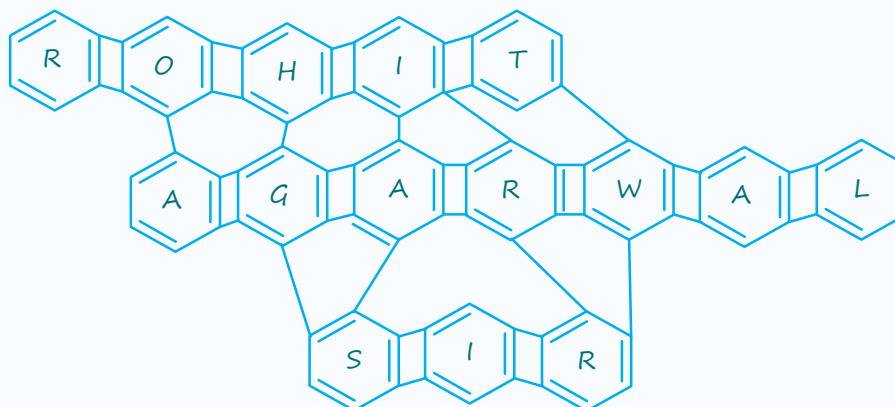
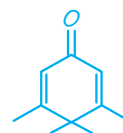
122. Housane



121. Windowane



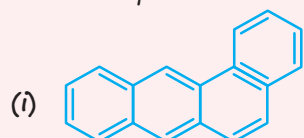
123. Penguinone



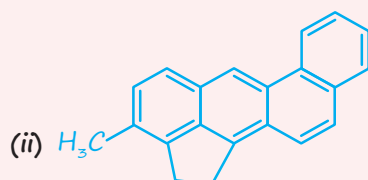
Baba Tea-Stall (BTS)

Carcinogenicity and toxicity

Benzene and poly-nuclear hydrocarbons containing more than two benzene rings fused together are toxic and said to possess cancer producing (carcinogenic) property.



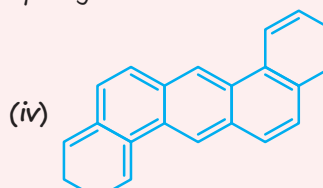
1, 2 Benzanthracene



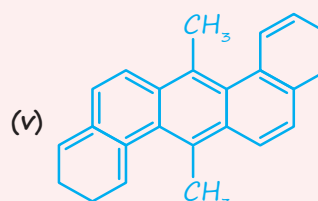
3-methylcholanthrene



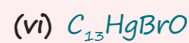
1, 2-Benzpyrene



1, 2, 5, 6-Dibenzanthracene



1, 2, 5, 6-Dibenzanthracene



Functional Group

S.N.	Functional Group	Structure
1.	Carboxylic acid	$\begin{array}{c} \text{O} \\ \parallel \\ -\text{C} \\ \backslash \\ \text{OH} \end{array}, -\text{CO}_2\text{H}$
2.	Sulphonic acid	$\begin{array}{c} \text{O} \\ \parallel \\ -\text{S}-\text{OH} \\ \parallel \\ \text{O} \end{array}, -\text{SO}_3\text{H}$
3.	Acid anhydride	$\begin{array}{c} \text{O} \qquad \text{O} \\ \parallel \quad \parallel \\ -\text{C}-\text{O}-\text{C}- \end{array}$
4.	Ester	$\begin{array}{c} \text{O} \\ \parallel \\ -\text{C} \\ \backslash \\ \text{O}- \end{array}$
5.	Acid halide	$\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{X}, -\text{COX} \end{array}$
6.	Acid amide	$\begin{array}{c} \text{O} \\ \parallel \\ -\text{C} \\ \backslash \\ \text{NH}- \end{array}$
7.	Cyanide	$-\text{C}\equiv\text{N}, -\text{CN}$
8.	Isocyanides	$(-\overset{\oplus}{\text{N}}\equiv\overset{\ominus}{\text{C}}, -\text{NC})$
9.	Aldehyde	$\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{H}, -\text{CHO} \end{array}$
10.	Ketone	$\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-, -\text{CO}- \end{array}$
11.	Alcohol	$-\text{OH}$
12.	Thiol	$-\text{SH}$
13.	Amines	$(-\overset{1^\circ}{\text{NH}}_2-\overset{2^\circ}{\text{NH}}-, -\overset{3^\circ}{\text{N}}\text{<})$
14.	Alkene	$-\text{C}=\text{C}-$
15.	Alkyne	$-\text{C}\equiv\text{C}-$
16.	Phenyl	

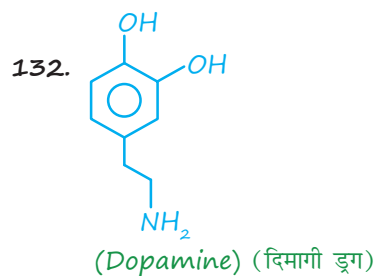
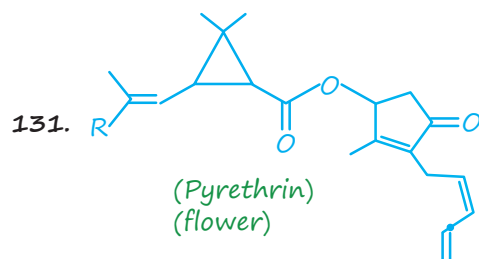
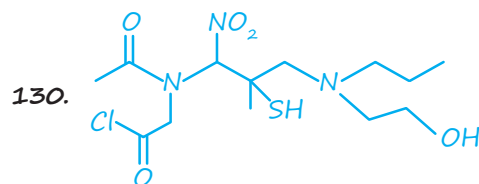
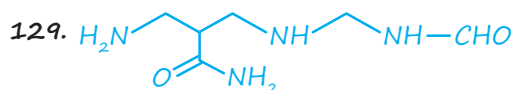
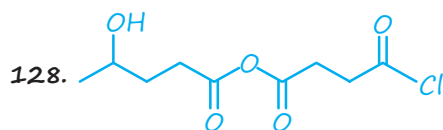
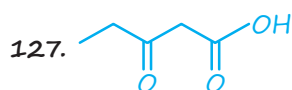
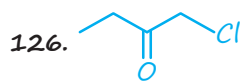
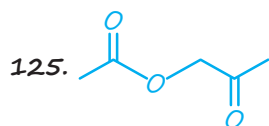
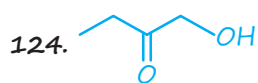
17.	Nitro	$-\text{NO}_2$
18.	Nitroso	$-\text{NO}$
19.	Thioether	$-\text{S}-$
20.	Ether	$-\text{O}-$
21.	Halide	$-\text{X}$
22.	azo	$-\text{N}=\text{N}$
23.	Imine	$-\text{CH}=\text{NH}$
24.	Imide	$\begin{array}{c} \text{O} \quad \text{NH} \quad \text{O} \\ \parallel \quad \mid \quad \parallel \\ -\text{C} - \text{NH} - \text{C}- \\ \parallel \quad \mid \quad \parallel \\ \text{O} \quad \text{O} \end{array}$
25.	enol	$-\text{CH}=\text{CH}-\text{OH}$
26.	enamine	$-\text{CH}=\text{CH}-\text{NH}_2$
27.	Aldol	$\begin{array}{c} \text{OH} \\ \\ \text{R}-\text{CH}-\text{CH}_2-\text{CH}_2-\text{C}-\text{H} \\ \\ \text{O} \end{array}$

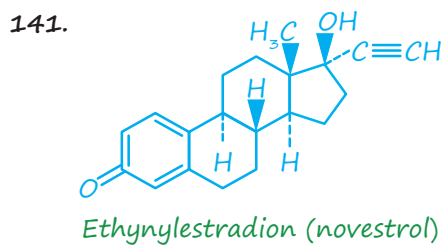
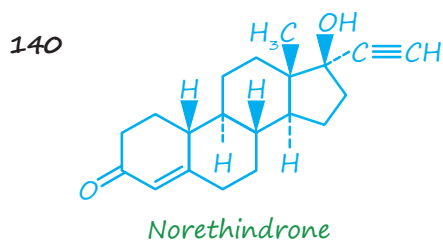
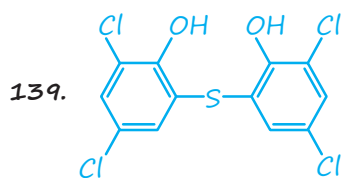
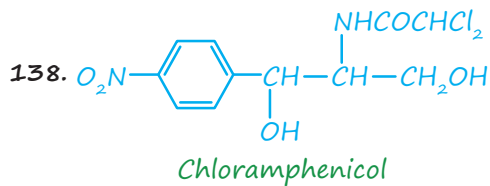
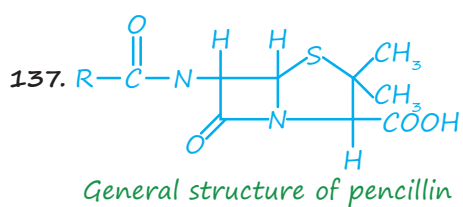
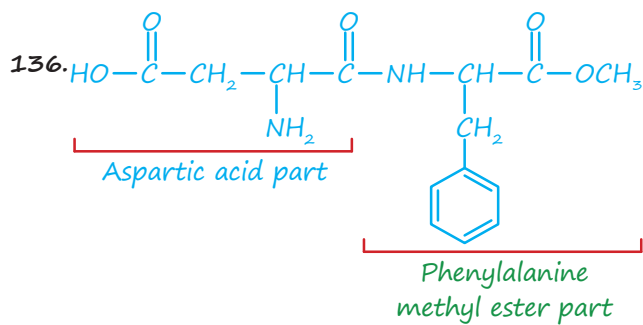
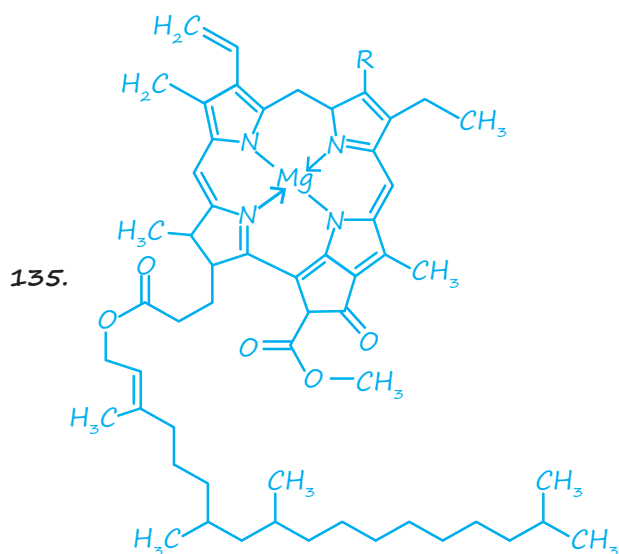
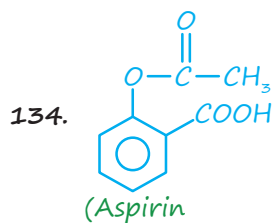
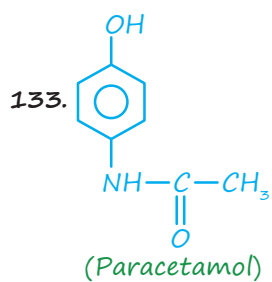
R—OH Points

- 1° , 2° , 3° amine are different functional group. ➤ 1° , 2° , 3° alcohol are same functional group.
- ⇒ Responsible for any chemical reaction. ➤ 1° , 2° , 3° amide are different functional group.
- Alcohol and phenol are different functional group.

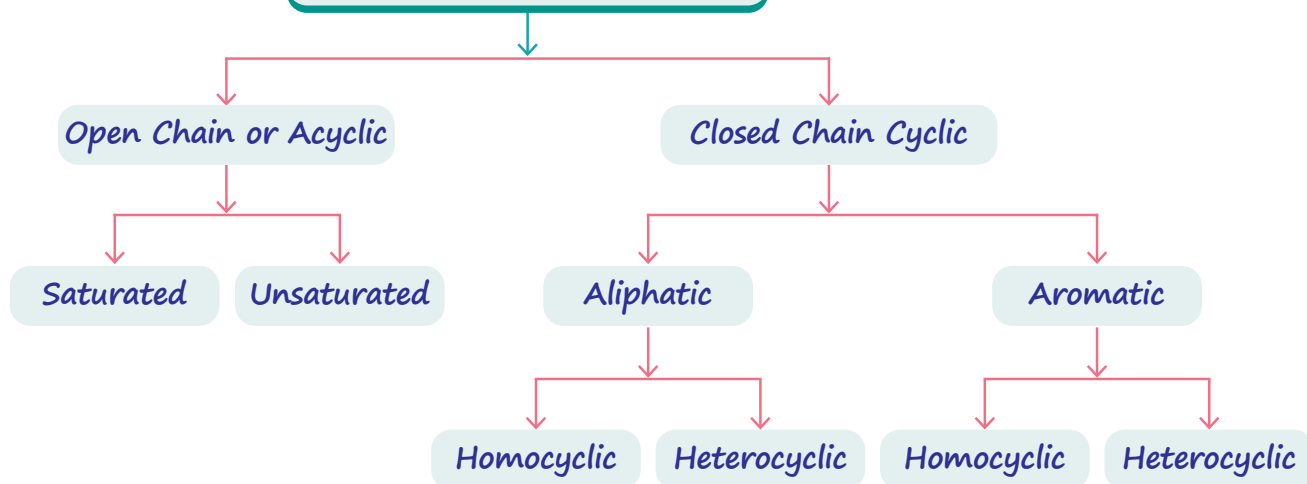
Practice Questions

Find different functional group?



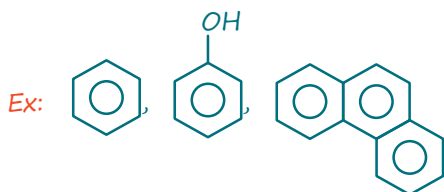


Classification of Organic Compounds



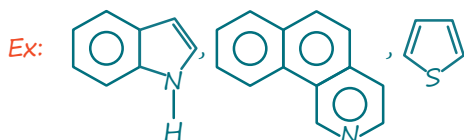
➤ Homocyclic Compound

- When all atoms of all rings are same then compound is known as homocyclic compound.



➤ Heterocyclic compound.

- When at least one atom of all ring is different then compound is known as heterocyclic compound.



Practice Questions

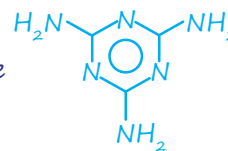
142. Which of the following are heterocyclic compounds?

(a) Benzene

(b) Phenol

(c) Toluene

(d) Melamine



(e) Aspirin

(f) Naphthalene

(g) Cumene

(h) Pyridine



(i) Paracetamol

(j) Pyrrole



(k) Furan



(l) Anthracene

143. Which of the following functional groups are possible for given general formula?

(a) $C_nH_{2n+2}O$

(b) $C_nH_{2n}O$

(c) $C_nH_{2n}O_2$

(d) $C_nH_{2n-2}O_2$

144. For the given DU, which structure are possible?

(a) 1

(b) 2

(c) 3

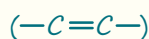
(d) 4

General Formula

1. Alkane



2. Alkene



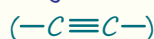
$$DU = 1$$

$$H = 2n + 2 - 2$$

$$= 2$$



3. Alkyne

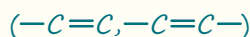


$$DU = 2$$

$$H = 2n + 2 - 4 = 2n - 2$$



4. Alkadiene

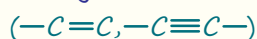


$$DU = 2$$

$$H = 2n + 2 - 4 = 2n - 2$$



5. Alkenyne



$$DU = 3$$

$$H = 2n + 2 - 6 = 2n - 4$$

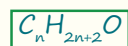


6. Alcohol

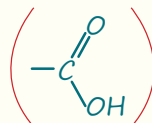


$$DU = 0$$

$$H = 2n + 2$$

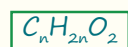


7. Carboxylic acid

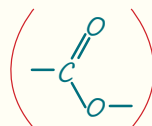


$$DU = 1$$

$$H = 2n + 2 - 2 = 2n$$

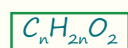


8. Ester



$$DU = 1$$

$$H = 2n + 2 - 2 = 2n$$

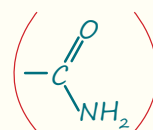


9. Aldehyde

10. acid anhydride

11. ketone

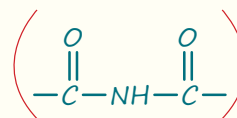
12. Acid amide



13. Amine



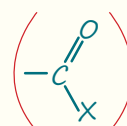
14. Imide



15. Azo



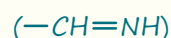
16. Acid halide



17. Halide



18. Halide



19. Cyanide



Homologous Series

Homologous series may be defined as a series of similarly constituted compounds in which the members possess the same functional group, have similar chemical characteristics and have a regular gradation in their physical properties. The two consecutive members differ in their molecular formula by CH_2 .

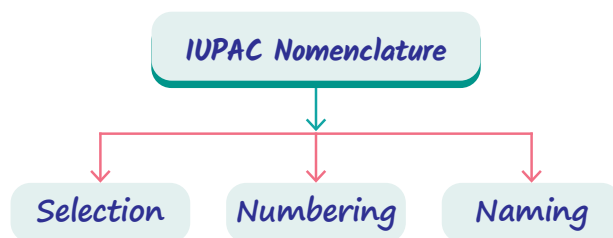
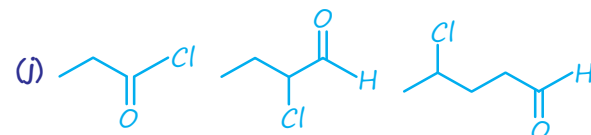
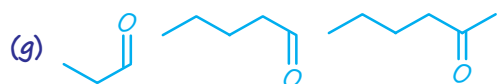
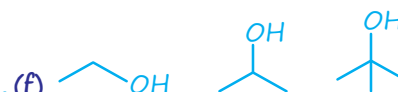
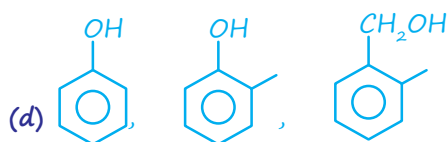
- Same general formula
- Same functional group
- Same degree of unsaturation
- Same chemical properties
- Different physical properties
- Difference of $-CH_2$ (14gm) in consecutive member.

Practice Questions

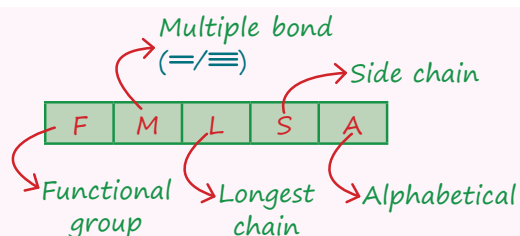
145. Find first 3 member's?

- | | |
|---------------------|-------------------|
| (a) Carboxylic acid | (b) Ester |
| (c) Aldehyde | (d) Ketone |
| (e) Acid anhydride | (f) 1° acid amide |
| (g) Acid halide | (h) Alcohol |

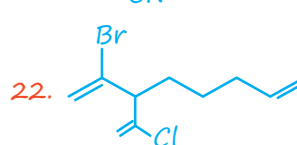
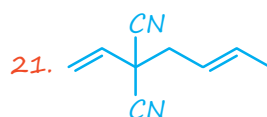
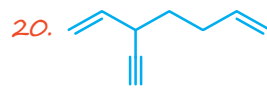
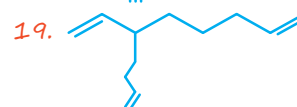
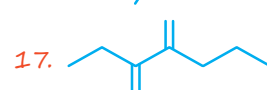
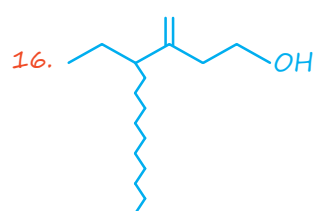
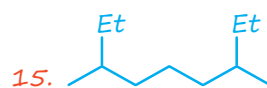
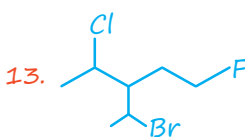
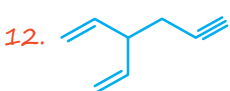
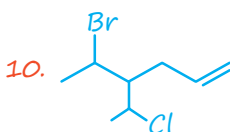
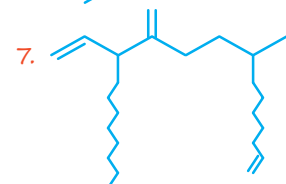
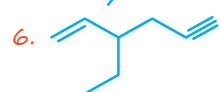
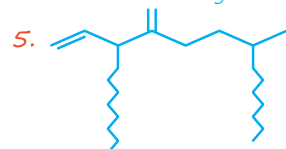
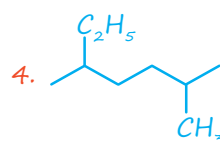
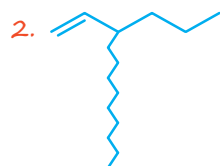
146. Which of their following is (are) pairs of same homologous series?



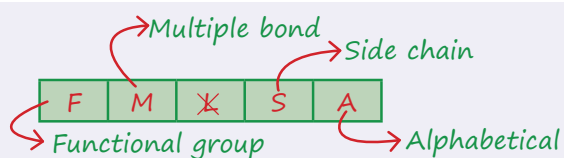
Selection of Parent Chain



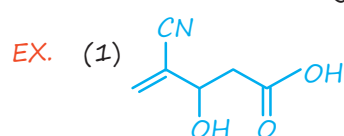
Selection



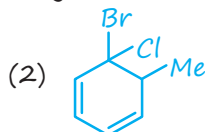
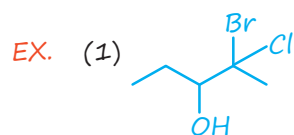
Numbering of Parent Chain



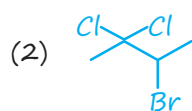
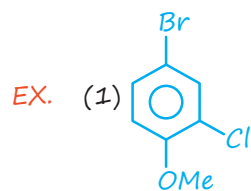
➤ **Boss rule:** Least number is given to carbon of functional.



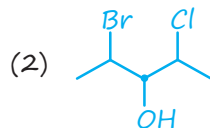
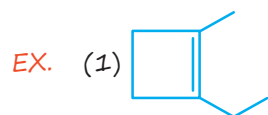
➤ **Multiple bond rule:** Least number is given to carbon of multiple bond.



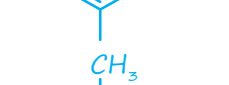
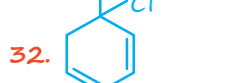
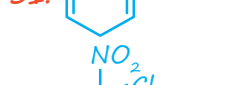
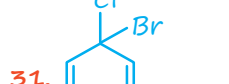
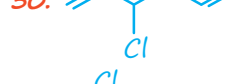
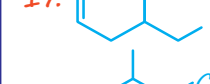
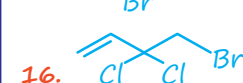
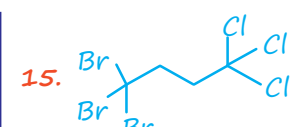
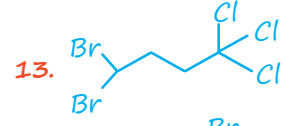
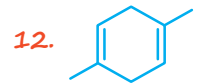
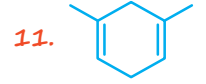
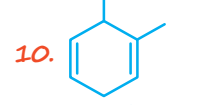
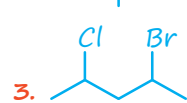
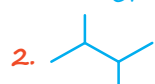
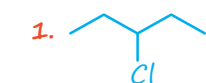
➤ **Lowest locant or substitutes rule:** If F & M are same then given least numbers to substituents.



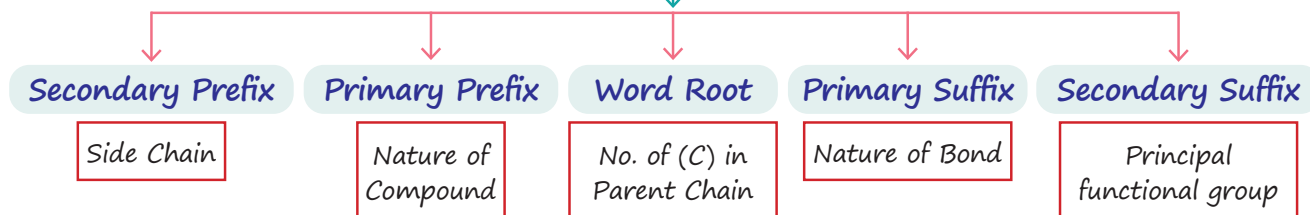
➤ **Alphabetical rule:** If all factors are same then alphabetical order will be considered.



IUPAC



IUPAC Nomenclature



Word Root: Total carbon in parent chain

- | | | |
|-------------|-------------|------------------|
| ➤ 1C ⇒ meth | ➤ 5C ⇒ pent | ➤ 9C ⇒ Non |
| ➤ 2C ⇒ Eth | ➤ 6C ⇒ Hex | ➤ 10C ⇒ Dec |
| ➤ 3C ⇒ Prop | ➤ 7C ⇒ Hept | ➤ 11C ⇒ Undec |
| ➤ 4C ⇒ But | ➤ 8C ⇒ Oct | ➤ 20C ⇒ Eicosane |

Primary suffix: Nature of bonds

- | | |
|--------------------------|--|
| ➤ Only single bond ⇒ ane | ➤ Any double bond ⇒ ene |
| ➤ any triple bond ⇒ yne | ➤ Double and triple bond ⇒ enyne, (P-en-q-yne) |

Note

1. Always written with respective position and in alphabetical order.
2. In case of vowel and y, e of suffix will be removed.
3. If di, tri, tetra-is used with MB the 'a' is added to word root.

Try Yourself

- 1.
- 2.
3. $\text{CH}_3-\text{C}=\text{CH}$
- 4.
5. $\text{CH}_2=\text{CH}-\text{CH}_2-\text{C}\equiv\text{CH}$
6. $\text{HC}\equiv\text{C}-\text{CH}_2-\text{CH}_2-\text{C}\equiv\text{CH}$

Primary prefix: Nature of compound.

- | | | | |
|------------|--------------|-----------|-------------|
| ➤ Cyclic ⇒ | ➤ Bicyclic ⇒ | ➤ Spiro ⇒ | ➤ Benzene ⇒ |
|------------|--------------|-----------|-------------|

Try Yourself

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.

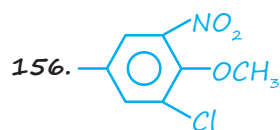
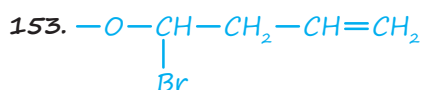
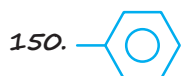
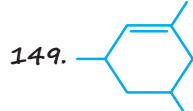
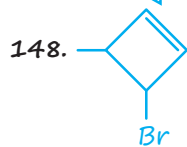
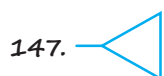
Secondary prefix: Side chain in compound

- | | | | |
|--|--|--|--|
| ➤ $-\text{F} \Rightarrow$ | ➤ $-\text{Cl} \Rightarrow$ | ➤ $-\text{Br} \Rightarrow$ | ➤ $-\text{I} \Rightarrow$ |
| ➤ $-\text{NO}_2 \Rightarrow$ | ➤ $-\text{NO} \Rightarrow$ | ➤ $-\text{CH}_3 \Rightarrow$ | ➤ $-\text{CH}_2-\text{CH}_3 \Rightarrow$ |
| ➤ $-\text{CH}=\text{CH}_2 \Rightarrow$ | ➤ $-\text{C}\equiv\text{CH} \Rightarrow$ | ➤ $-\text{CH}_2-\text{CH}=\text{CH}_2 \Rightarrow$ | |
| ➤ $-\text{CH}(\text{CH}_3)-\text{CH}_2=\text{CH}_2\text{Br}$ | | | |

'yl' is added for carbon chain

Always written with respective position in alphabetical order

Practice Questions

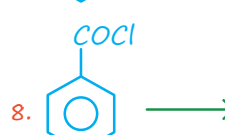
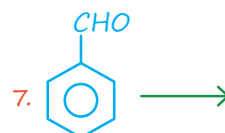
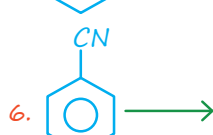
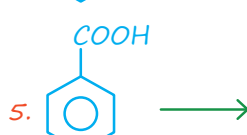
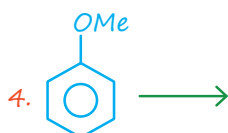
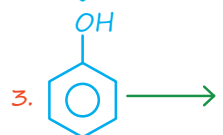
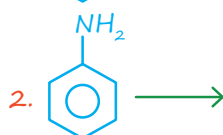
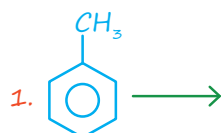


Secondary suffix

Functional group	Structure	Prefix	Suffix	Suffix*
Carboxylic	$-\text{COOH}$	Carboxy	—oic acid	Carboxylic acid
Sulphonic acids	$-\text{SO}_3\text{H}$	Sulpho	—Sulphonic acid	—
Carboxylic anhydrides	$\begin{array}{c} \text{O} \quad \text{O} \\ \parallel \quad \parallel \\ -\text{C}-\text{O}-\text{C}- \end{array}$	—	—oic anhydride	—carboxylic anhydride
Carboxylic esters	$-\text{COOR}$	Alkoxycarbonyl or Alkanoyl oxy	—oate	—carboxylate
Acid halides	$-\text{COCl}$	Halocarbonyl	—oyl halide	—carbonyl halide
Amides	$-\text{CONH}_2$	Cyano	—nitrile	—carbonitrile
Aldehydes	$-\text{CHO}$	Formyl or oxo	—al	—carbaldehyde
Ketones	$=\text{O}$	Oxo or keto	—one	—
Alcohols	$-\text{OH}$	hydroxy	—ol	—
Thiols	$-\text{SH}$	mercapto	—thiol	—
Amines	$-\text{NH}_2$	amine	—amino	—

Note:

Mono substituted benzene has same common and IUPAC name.

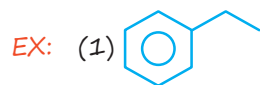


➤ di, tri, tetra ... are not used in alphabetical order and only CIN (Cyclo, iso, Neo) is used in alphabetical order.

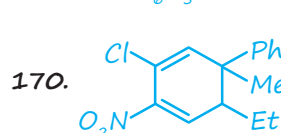
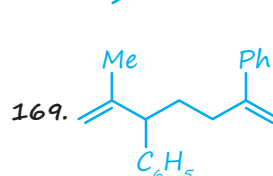
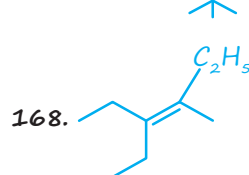
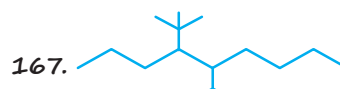
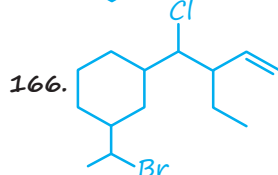
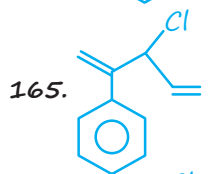
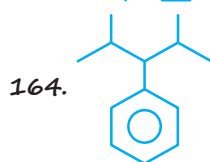
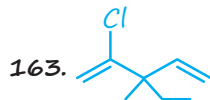
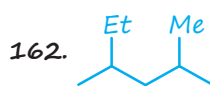
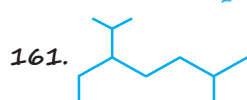
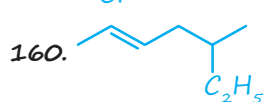
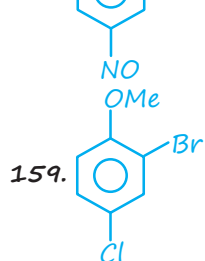
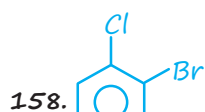
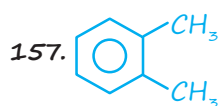
➤ If everything is same in open chain and closed chain then closed chain is preferred.



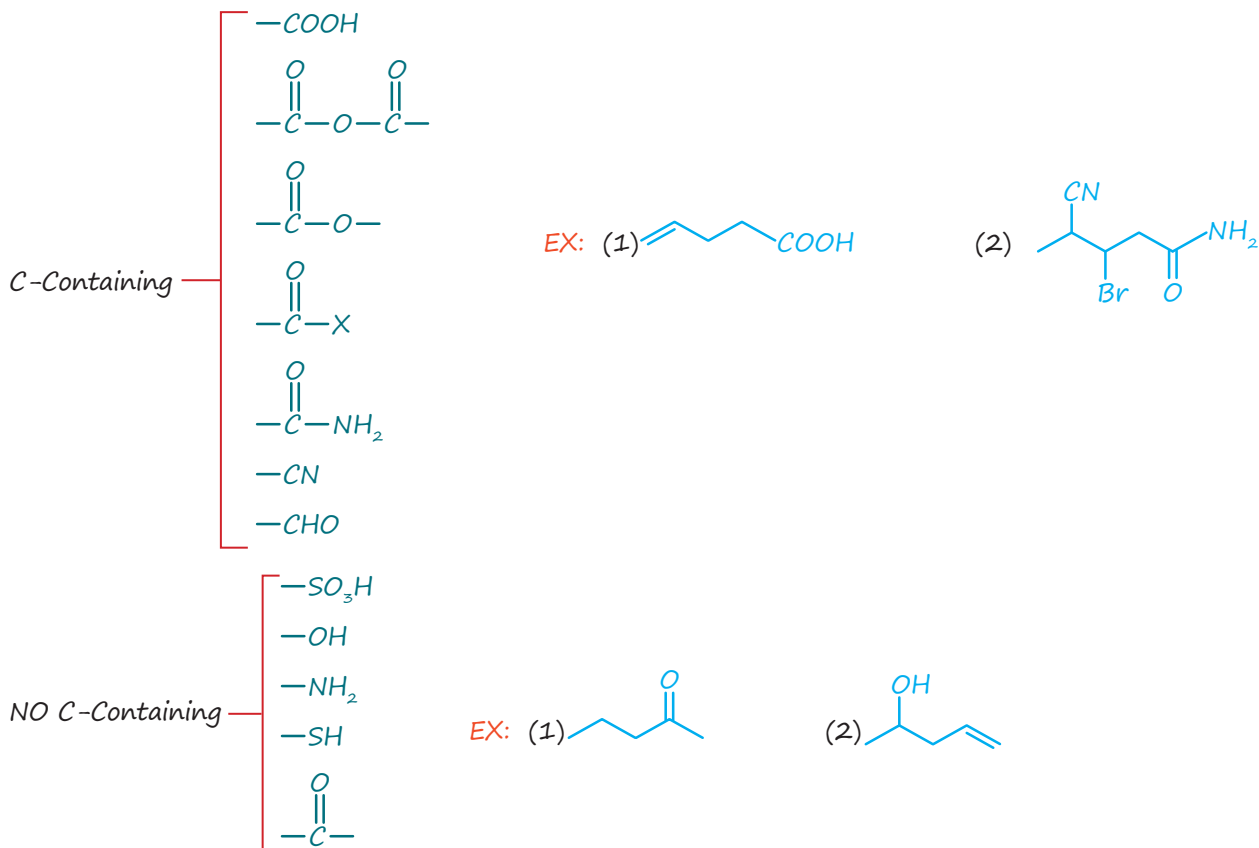
➤ Generally methyl and ethyl one considered as S.C. in benzene and in other cases benzene will be side chain.



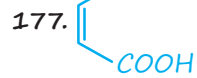
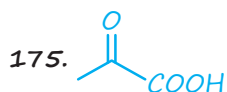
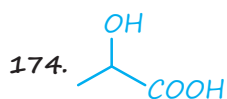
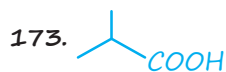
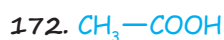
Practice Questions



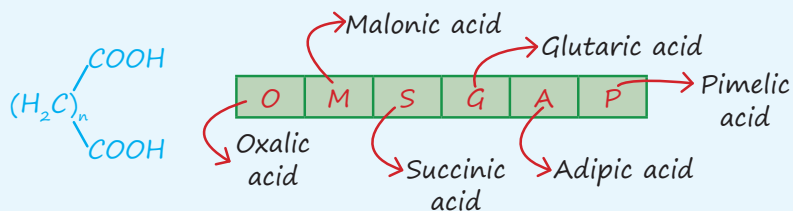
IUPAC of Functional Group



Practice Questions



Dicarboxylic acid



Practice Questions

179. O

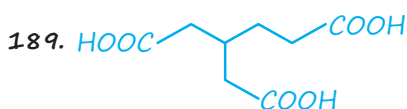
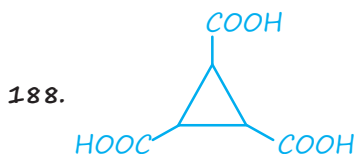
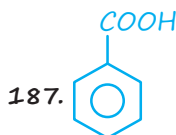
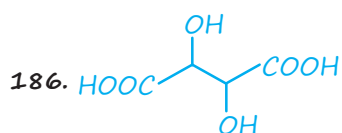
180. M

181. O

182. G

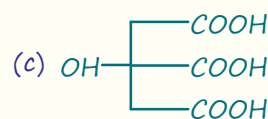
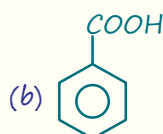
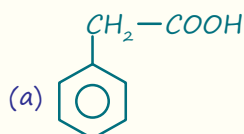
183. A

184. P



Note:

- Carbon of F.G. present in P.C. $\Rightarrow$
- Carbon of F.G. not present in P.C. and directly attached to P.C. $\Rightarrow$
- Carbon of F.G. present in side chain $\Rightarrow$
- If more than two same F.G. are present then rule (2) is applied but if rule (2) is not applicable then select max. F.G. in P.C. and other are treated as S.C.



1. Carboxylic acid

Prefix $\Rightarrow$ Carboxy

Suffix $\Rightarrow$ oic acid

Suffix* $\Rightarrow$ Carboxylic acid

2. Acid halide

Prefix $\Rightarrow$ halocarbonyl

Suffix $\Rightarrow$ oyl halide

Suffix* $\Rightarrow$ Carbonyl halide

3. Cyanide

Prefix $\Rightarrow$ Cyano

Suffix $\Rightarrow$ nitrile

Suffix* $\Rightarrow$ Carbonitrile

4. Aldehyde

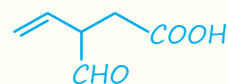
Prefix $\Rightarrow$ Formyl, OXO

Suffix $\Rightarrow$ al

Suffix* $\Rightarrow$ Carbaldehyde

If (C) of $-CHO$ is present in P.C. $\Rightarrow$ OXO

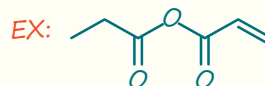
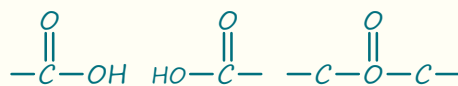
If (C) of $-CHO$ is not present in P.C. $\Rightarrow$ formyl



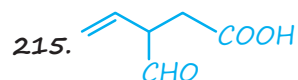
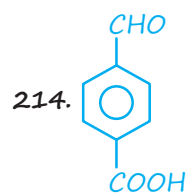
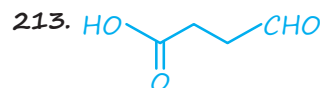
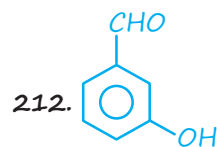
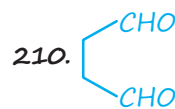
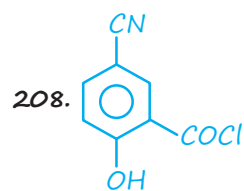
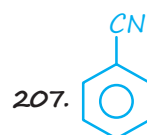
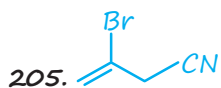
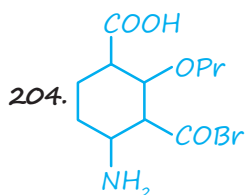
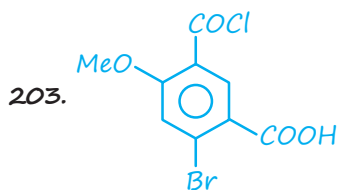
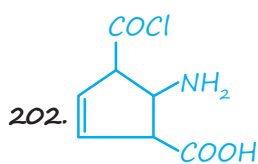
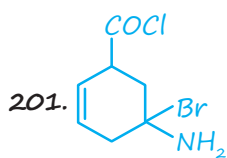
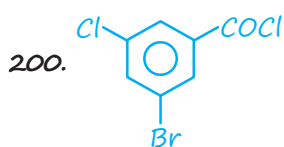
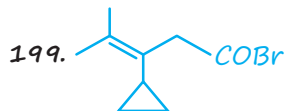
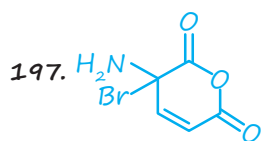
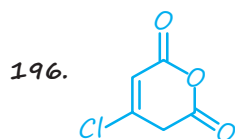
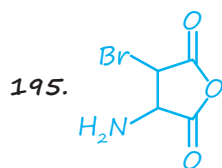
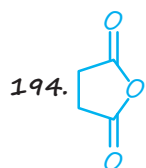
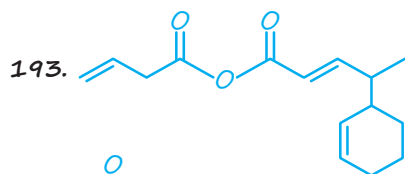
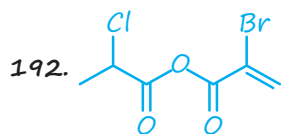
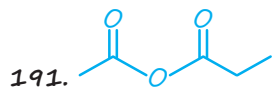
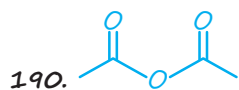
5. Acid Anhydride

Prefix $\Rightarrow$

Suffix $\Rightarrow$ oic anhydride

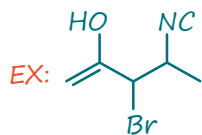


Practice Questions

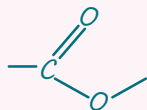


Ketone	Prefix	Keto/oxo	Suffix	one
Alcohol	Prefix	hydroxy	Suffix	ol
Thiol	Prefix	mercapto	Suffix	thiol
Sulphonic acid	Prefix	Sulpho	Suffix	Sulphonic acid
Amine	Prefix	amino	Suffix	amine
NC (Isonitrile)				

Least number is given to (C) of functional group.

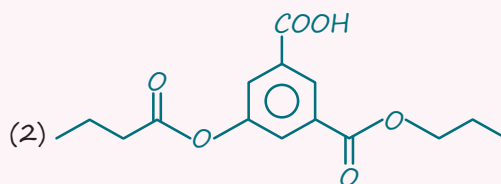
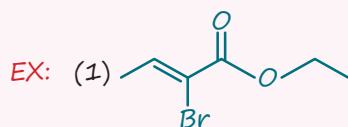
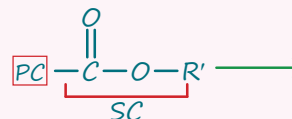
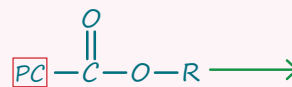


Ester

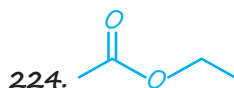
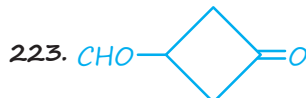
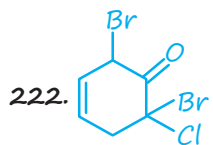
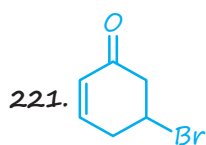
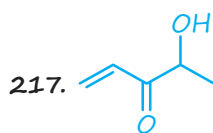
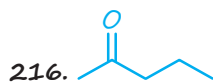


Suffix $\Rightarrow$ oate, Suffix* $\Rightarrow$ Carboxylate, Prefix $\Rightarrow$

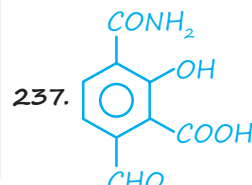
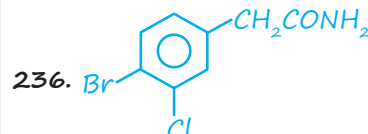
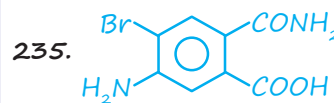
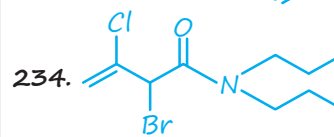
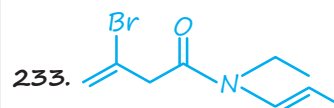
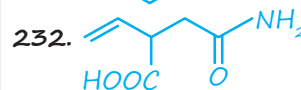
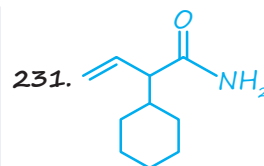
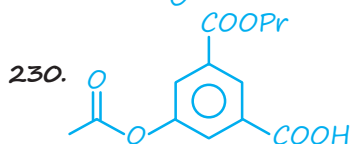
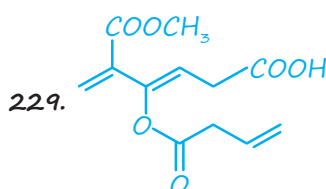
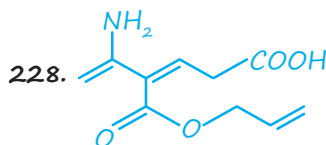
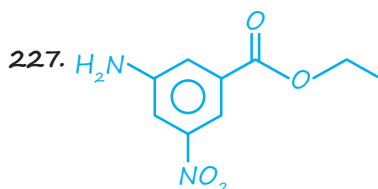
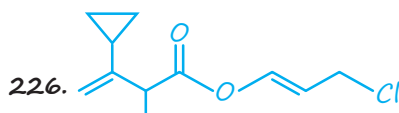
➤ Group/atom attached to oxygen of ester is always written first.

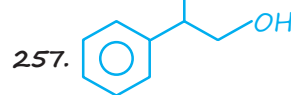
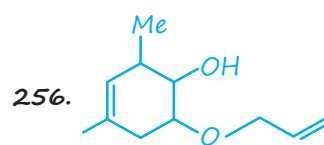
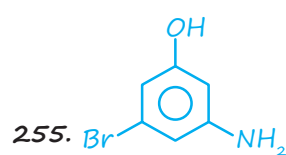
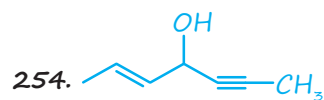
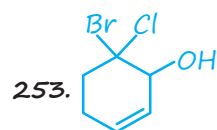
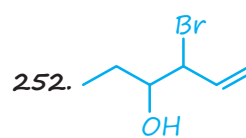
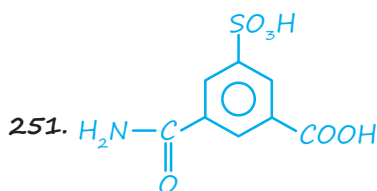
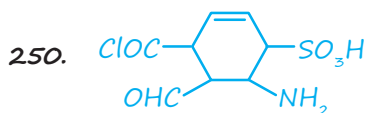
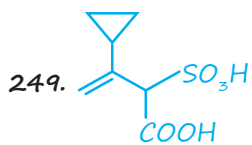
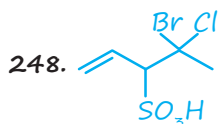
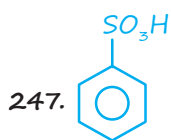
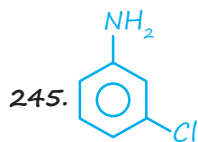
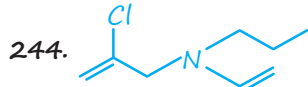
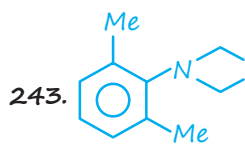
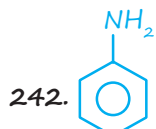
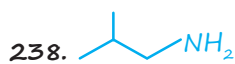


Practice Questions



225. 1st member





Cyclic Ether

➤ Cyclic ether ⇒

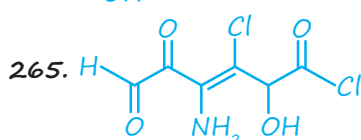
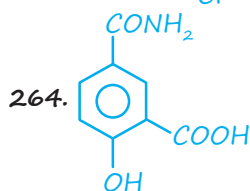
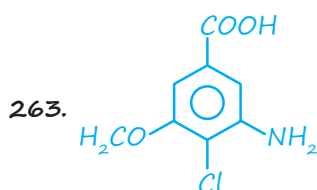
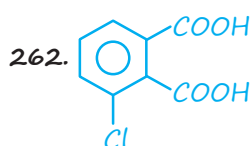
➤ Cyclic amide ⇒

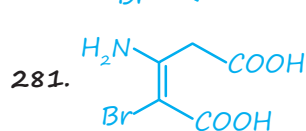
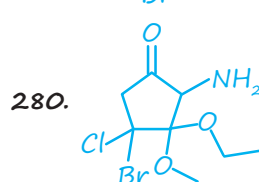
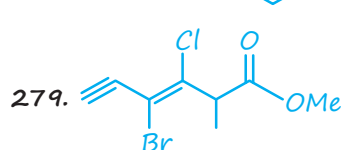
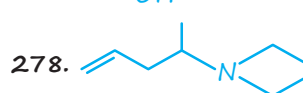
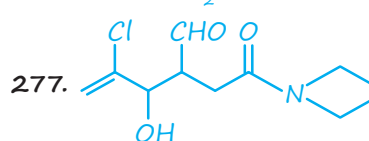
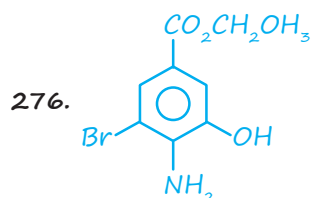
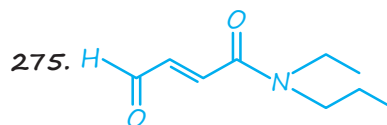
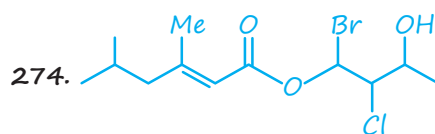
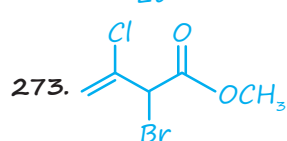
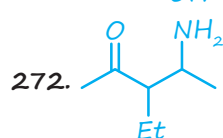
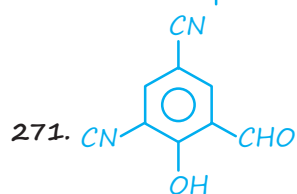
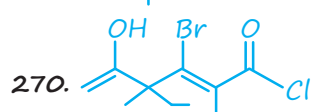
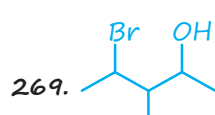
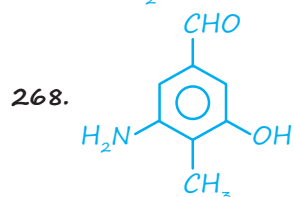
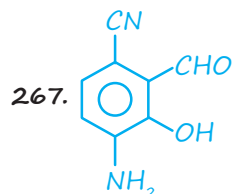
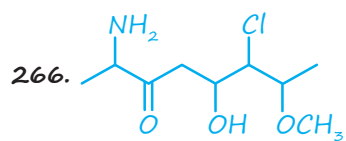
➤ Cyclic ester ⇒

EX:



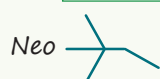
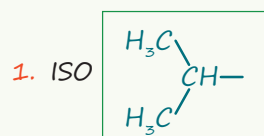
Practice Questions





Common Naming

ISO Octane



1. Isohexyl chloride
2. Neohexyl alcohol
3. Isobutene



5. Neohexyl and Isopentyl join

2. Normal (n)

Secondary (sec)

Tertiary (tert)

sec. butyl (tert butyl)

1. n-hexane

2. n-propyl bromide

3. sec. butylchloride

4. Tert-butanol

5. sec. butyl tert butyl ketone

6. ditert butyl ether

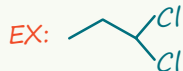
7. sec. pentyl-n-propyl ketone

3. Vinyl

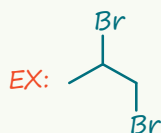
Allyl

1. Vinyl chloride
2. Allyl alcohol
3. Allyl and vinyl join
4. Allyl and Iso pentyl
5. No. of vinylic (C)
6. Vinyl and tert butyl join

4. Geminal (Germ)



Vicinal (Vic)



5. (a) Alkylidene

1. Ethylidene chloride
2. Ethylene dichloride
3. Ethylene oxide

(b) Alkylene

- 1.
- 2.
3. CH_2Cl_2

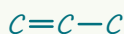
6. α , β , γ , δ

1. β -hydroxy acetic acid
2. γ -methyl cyclohexane carbaldehyde
3. methyl cyclohexane carbaldehyde

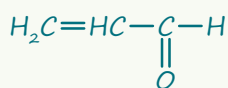
7. Benzyl, Benzal, Benzo, Phenyl

1. Benzyl alcohol
2. Benzyl chloride
3. Benzal chloride
4. Benzoic acid
5. o-substituted benzal chloride

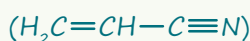
8. (a) Acryl



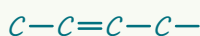
1. Acraldehyde



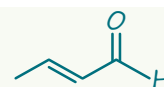
2. Acrylonitrile/Acrotein



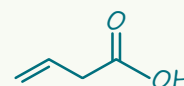
(b) Croton



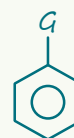
1. Crotonaldehyde



2. β -crotonic acid



9. o, m, p, IPSO



10. No. of carbon



1. Formaldehyde

2. Iodobutyric acid

3. β -Hydroxy valeric acid

4. α -amino butyraldehyde

5. Propionone

6. p-Chlorophenyl acetic acid

Practice Questions

Write the IUPAC name by joining given radicals:

282. Isopropyl and Isobutyl
283. Isopentyl and Neopentyl
284. Isopropyl and Neopentyl
285. Tertiary butyl and Neohexyl
286. Tert-butyl and Isobutyl
287. Allyl and Allyl
288. Vinyl and Allyl
289. Allyl and Isohexyl
290. Allyl and sec. butyl
291. Disec-butyl

Draw structure and write IUPAC names?

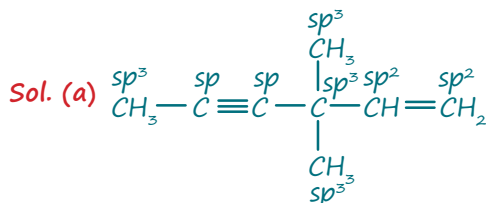
292. Acetone
293. Phenyl acetate
294. β -Hydroxyvaleric acid
295. Isobutyric acid
296. Ethylidene cyclopentane
297. Di tert butyl ketone
298. Sec. butyl aldehyde
299. Acetyl aldehyde
300. Valeric acid

Previous Year Questions

1. In 3,3-dimethylhex-1-ene-4-yne, there are _____ sp^3 , _____ sp^2 and _____ sp hybridised carbon atoms respectively:

[02 April, 2025 (Shift-II)]

- (a) 4, 2, 2 (b) 3, 3, 2
(c) 2, 4, 2 (d) 2, 2, 4



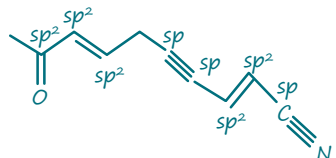
2. In the given structure, number of sp and sp^2 hybridized carbon atoms present respectively are:

[24 Jan, 2025 (Shift-II)]



- (a) 4 and 5 (b) 4 and 6
(c) 3 and 5 (d) 3 and 6

- Sol. (c)** sp and sp^2 hybridized carbon atoms in the structure can be shown as:



No. of sp hybridized carbon-3

No. of sp^2 hybridized carbon-5

3. The sum of sigma (σ) and pi (π) bonds in Hex-1,3-dien-5-yne is _____.

[29 Jan, 2025 (Shift-I)]



Total sigma bonds = 11

Total pi bonds = 4

Total sigma and pi bonds: $11 + 4 = 15$

Therefore, the sum of sigma and pi bonds in Hex-1,3-dien-5-yne is 15.

4. Total number of sigma (σ) _____ and pi (π) _____ bonds respectively present in hex-1-en-4-yne are:

[29 Jan, 2025 (Shift-II)]

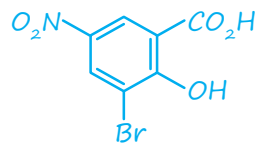
- Sol. (d)** Structure of hex-1-en-4-yne:



Total Bonds: Sigma (σ) bonds : 13

Pi (π) bonds: 3

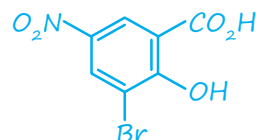
5. What is the correct IUPAC name of



[03 April, 2025 (Shift-II)]

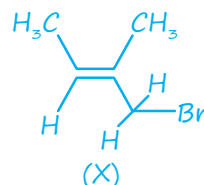
- (a) 3-Bromo-2-hydroxy-5-nitrobenzoic acid
(b) 3-Bromo-4-hydroxy-1-nitrobenzoic acid
(c) 2-Hydroxy-3-bromo-5-nitrobenzoic acid
(d) 5-Nitro-3-bromo-2-hydroxybenzoic acid

- Sol. (a)**



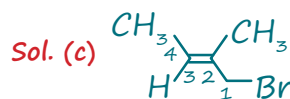
The correct IUPAC name of the given compound is 3-Bromo-2-hydroxy-5-nitro-Benzoic acid.

6. Which of the following is the correct IUPAC name of given organic compound (X) ?



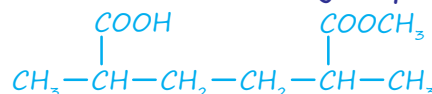
[07 April, 2025 (Shift-I)]

- (a) 2-Bromo-2-methylbut-2-ene
(b) 3-Bromo-3-methylprop-2-ene
(c) 1-Bromo-2-methylbut-2-ene
(d) 4-Bromo-3-methylbut-2-ene



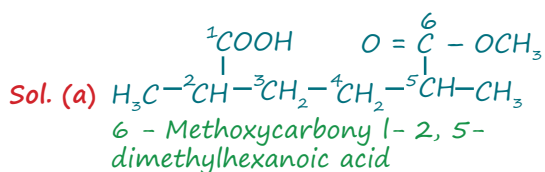
The IUPAC name of the compound is : 1-Bromo-2-methyl but-2-ene

7. The IUPAC name of the following compound is:



[22 Jan, 2025 (Shift-I)]

- (a) 6-Methoxycarbonyl-2,5-dimethylhexanoic acid.
(b) 2-Carboxy-5-methoxycarbonylhexane.
(c) Methyl-6-carboxy-2,5-dimethylhexanoate.
(d) Methyl-5-carboxy-2-methylhexanoate.

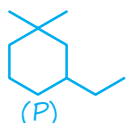


8. Cyclohexene  is _____ type of an organic compound. [27 Jan, 2024 (Shift-I)]

- (a) Benzenoid aromatic
- (b) Benzenoid non-aromatic
- (c) Acyclic
- (d) Alicyclic

Sol. (d)  is Alicyclic compound

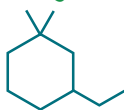
9. IUPAC name of following compound (P) is:



[27 Jan, 2024 (Shift-I)]

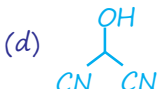
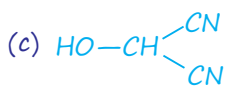
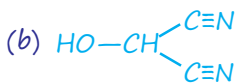
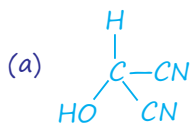
- (a) 1-Ethyl-5, 5-dimethylcyclohexane
- (b) 3-Ethyl-1,1-dimethylcyclohexane
- (c) 1-Ethyl-3, 3-dimethylcyclohexane
- (d) 1, 1-Dimethyl-3-ethylcyclohexane

Sol. (b) IUPAC name of given compound is

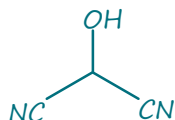


3-ethyl 1, 1-dimethylcyclohexane

10. Bond line formula of $\text{HOCH}(\text{CN})_2$ is: [27 Jan, 2024 (Shift-II)]



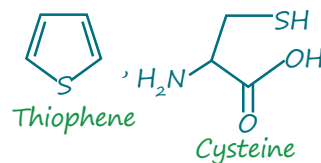
Sol. (d) Bond line formula of compound $\text{CH}(\text{OH})(\text{CN})_2$ is



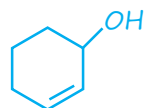
11. Number of compounds among the following which contain sulphur as heteroatom is ____.

Furan, Thiophene, Pyridine, Pyrrole, Cysteine, Tyrosine. [29 Jan, 2024 (Shift-I)]

Sol. [2] Thiophene and Cysteine contain sulphur as shown in structures given below



12. According to IUPAC system, the compound



is named as

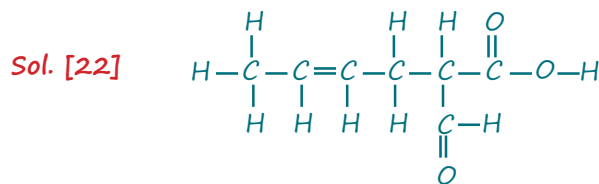
[29 Jan, 2024 (Shift-II)]

- (a) Cyclohex-1-en-2-ol
- (b) 1-Hydroxyhex-2-ene
- (c) Cyclohex-1-en-3-ol
- (d) Cyclohex-2-en-1-ol

Sol. (d) IUPAC name: Cyclohex-2-en-1-ol

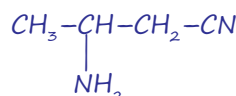
13. The total number of 'Sigma' and Pi bonds in 2-formylhex-4-enoic acid is ____.

[29 Jan, 2024 (Shift-II)]



Total number of bonds are 22

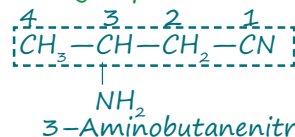
14. IUPAC name of following compound is



[30 Jan, 2024 (Shift-II)]

- (a) 2-Aminopentanenitrile
- (b) 2-Aminobutanenitrile
- (c) 3-Aminobutanenitrile
- (d) 3-Aminopropanenitrile

Sol. (c) Priority is given to cyano group as compare to amino group.



15. Given below are two statements:

Statement-I: IUPAC name of $\text{HO}-\text{CH}_2-(\text{CH}_2)_3-\text{CH}_2-\text{COCH}_3$ is 7-hydroxyheptan-2-one.

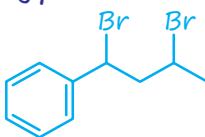
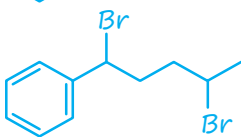
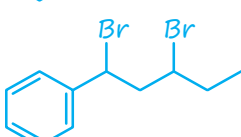
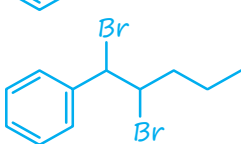
Statement-II: 2-oxoheptan-7-ol is the correct IUPAC name for above compound.

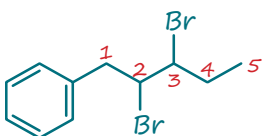
In the light of the above statements. Choose the most appropriate answer from the options given below. [31 Jan, 2024 (Shift-I)]

- (a) Statement-I is correct but Statement-II is incorrect.
 (b) Both Statement-I and Statement-II are incorrect.
 (c) Both Statement-I and Statement-II are correct.
 (d) Statement-I is incorrect but Statement-II is correct.

Sol. (a) 7-Hydroxyheptan-2-one is correct IUPAC name of given compound.

16. Identify structure of 2, 3-dibromo-1-phenylpentane. [31 Jan, 2024 (Shift-II)]

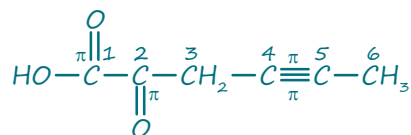
- (a) 
 (b) 
 (c) 
 (d) 

Sol. (c) 
 2, 3-dibromo -1-phenylpentane

17. The total number of 'sigma' and 'Pi' bonds in 2-oxohex-4-ynoic acid is ____.

[04 April, 2024 (Shift-II)]

Sol. [18] The structure is:



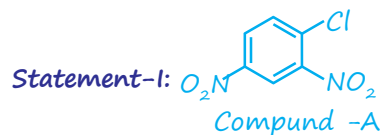
2-Oxohex-4-ynoic acid

Number of σ -bonds = 14

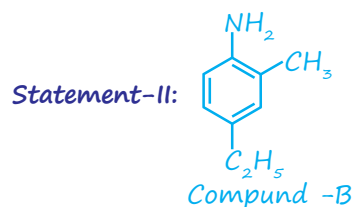
Number of π -bonds = 4

Number of σ & π -bonds = 18

18. Given below are two statements:



IUPAC name of Compound A is 4-chloro-1,3-dinitrobenzene:



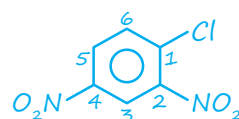
IUPAC name of Compound B is

4-ethyl-2-methylaniline.

In the light of the above statements, choose the most appropriate answer from the options given below: [08 April, 2024 (Shift-I)]

- (a) Both Statement-I and Statement-II are correct
 (b) Statement-I is incorrect but Statement-II is correct
 (c) Statement-I is correct but Statement-II is incorrect
 (d) Both Statement-I and Statement-II are incorrect

Sol. (b) Statement I:

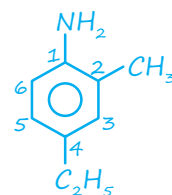


IUPAC name

$\Rightarrow$ 1-Chloro-2, 4-dinitrobenzene

Hence, statement-I is incorrect

Statement-II:

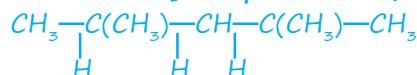


IUPAC name

$\Rightarrow$ 4-Ethyl-2-methylaniline

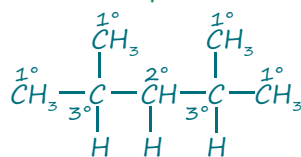
Hence, statement-II is correct

19. In the given compound, the number of 2° carbon atom/s is _____. [08 April, 2024 (Shift-I)]



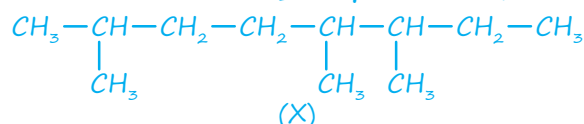
- (a) Three (b) One
 (c) Two (d) Four

Sol. (b) There is only one 2° carbon atom present in this compound.



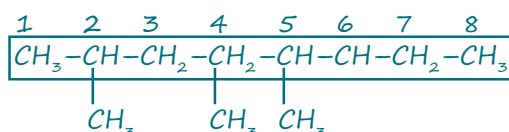
20. IUPAC name of following hydrocarbon (X) is:

[08 April, 2024 (Shift-II)]



- (a) 2-Ethyl-3,6-dimethylheptane
- (b) 2-Ethyl-2,6-diethylheptane
- (c) 2,5,6-Trimethyloctane
- (d) 3,4,7-Trimethyloctane

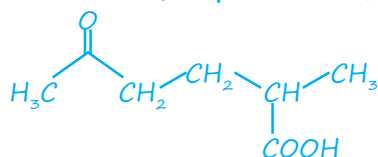
Sol. (c)



IUPAC name is 2,5,6-Trimethyloctane

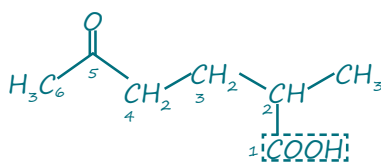
21. The correct IUPAC nomenclature for the following compound is

[8 April, 2023 (Shift-II)]



- (a) 5-Formyl-2-methylhexanoic acid
- (b) 2-Methyl-5-oxohexanoic acid
- (c) 2-Formyl-5-methylhexan-6-oic acid
- (d) 5-Methyl-2-oxohexan-6-oic acid

Sol. (b) In the given compound, the numbering will start from $-\text{COOH}$ group. Double bond O present at 5th position will named as 'oxo' in prefix.



IUPAC name of the above compound is:
2-Methyl-5-oxohexanoic acid

22. Match List-I with List-II.

[28 July, 2022 (Shift-I)]

List-I (Pair of Compounds)		List-II (Isomerism)	
(A)		(I)	Spiro compound
(B)		(II)	Aromatic compound
(C)		(III)	Non-planar Heterocyclic compound
(D)		(IV)	Bicyclo compound

Choose the correct answer from the options given below

- (a) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
- (b) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)
- (c) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
- (d) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

Sol. (c) $\rightarrow$ Non-planar heterocyclic compound

$\rightarrow$ Bicyclo compound

$\rightarrow$ Spiro compound

$\rightarrow$ Aromatic compound

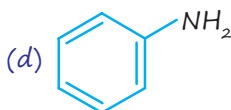
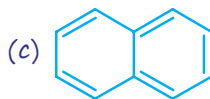
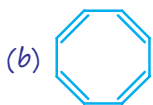
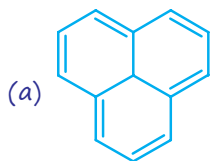
23. The correct decreasing order of priority of functional groups in naming an organic compound as per IUPAC system of nomenclature is
[26 July, 2022 (Shift-II)]

- (a) $-\text{COOH} > -\text{CONH}_2 > -\text{COCl} > -\text{CHO}$
- (b) $-\text{SO}_3\text{H} > -\text{COCl} > -\text{CONH}_2 > -\text{CN}$
- (c) $-\text{COOR} > -\text{COCl} > -\text{NH}_2 > \text{C}=\text{O}$
- (d) $-\text{COOH} > -\text{COOR} > -\text{CONH}_2 > -\text{COCl}$

Sol. (b) The decreasing order of priority of functional group is, $-\text{SO}_3\text{H} > -\text{COCl} > -\text{CONH}_2 > -\text{CN}$

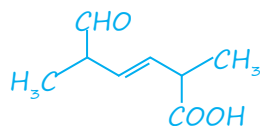
24. Which of the following is not an example of benzenoid compound?

[26 July, 2022 (Shift-II)]



Sol. (a,b) The structure consisting benzene ring is called benzenoid compound. Options (a) and (b) do not contain benzene ring, so they are non benzenoid compounds.

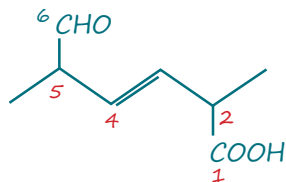
25. The IUPAC name for the following compound is:



[2 Sept, 2020 (Shift-I)]

- (a) 2, 5-Dimethyl-6-carboxy-hex-3-enal
 (b) 6-Formyl-2-methyl-hex-3-enoic acid
 (c) 2, 5-Dimethyl-6-oxo-hex-3-enoic acid
 (d) 2, 5-Dimethyl-5-carboxy-hex-3-enal

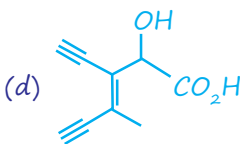
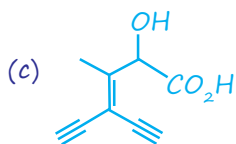
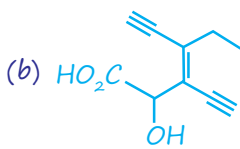
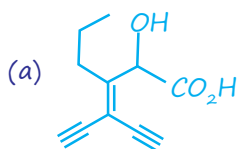
Sol. (c) In the given compound highest priority group is -COOH . Therefore, IUPAC name is given as,



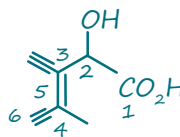
(2, 5-dimethyl-6-oxo-hex-3-enoic acid)

26. Which one of the following structures has the IUPAC name 3-ethynyl-2-hydroxy-4-methylhex-3-en-5-ynoic acid?

C-62.22W-24.61 UA-13.17 (JEE Adv. 2020)

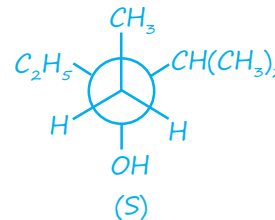
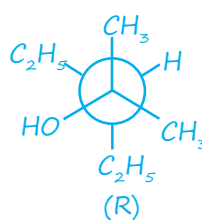
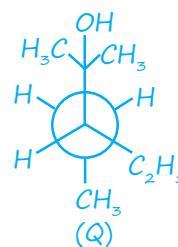
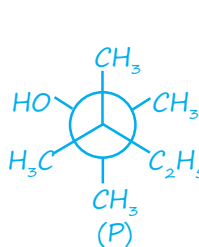


Sol. (d) For IUPAC naming, following numbering will be followed



3-ethynyl-2-hydroxy-4-methylhex-3-en-5-ynoic acid

27. Newman projections P, Q, R and S are shown below:

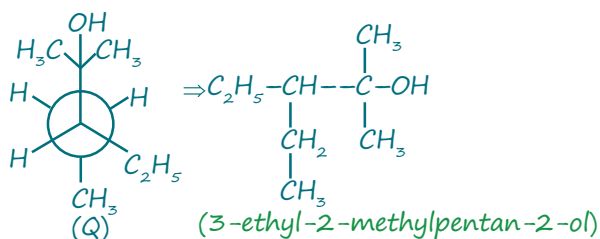
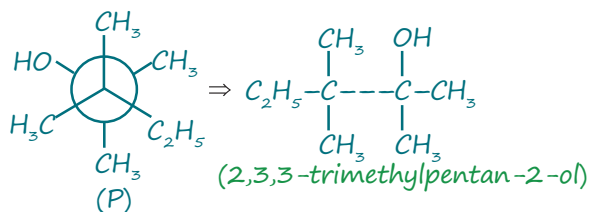


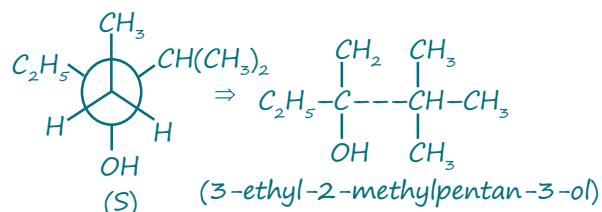
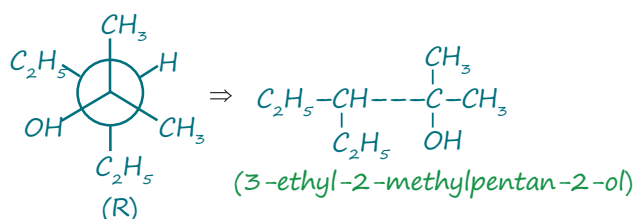
Which one of the following options represents identical molecules?

C-32.04 W-38.77 UA-29.19
(JEE Adv. 2020)

- (a) P and Q
 (b) Q and S
 (c) Q and R
 (d) R and S

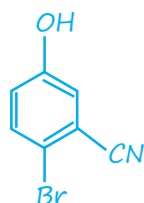
Sol. (c) IUPAC name of P, Q, R and S are:





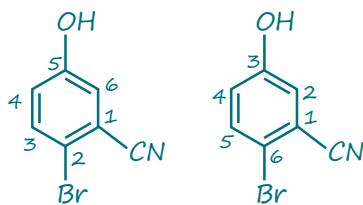
IUPAC name of compounds Q and R are same, thus the correct option (c).

28. The IUPAC name of the following compound is (IIT JEE 2009)



- (a) 4-bromo-3-cyanophenol
- (b) 2-bromo-5-hydroxybenzonitrile
- (c) 2-cyano-4-hydroxybromobenzene
- (d) 6-bromo-3hydroxybenzonitrile

Sol. (b) -CN has highest priority than -OH.



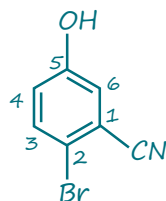
Numbering 1 Numbering 2

From numbering 1 we get sum of locants = 1+2+5=8

And from numbering 2 sum of locant = 1+3+6 = 10

So from the lowest sum rule numbering 1 is correct.

Hence the IUPAC name is 2-bromo-5-hydroxybenzonitrile



29. The IUPAC name of C_6H_5COCl is

(IIT JEE 2006)

- (a) benzoyl chloride
- (b) benzene chloro ketone
- (c) benzene carbonyl chloride
- (d) chloro phenyl ketone

Sol. (a) C_6H_5COCl is known as benzoyl chloride by IUPA(c)

30. The IUPAC name of the compound $CH_2 = CH - CH(CH_3)_2$ is (IIT JEE 1987)

- (a) 1, 1-dimethyl-2-butene
- (b) 3-methyl-1-butene
- (c) 2-vinyl propane
- (d) None of these

Sol. (b) $CH_2 = CH - CH(CH_3)_2$

In this structure functional group is alkene. And the longest chain is of 4 carbon. So the IUPAC name is 3-methyl-1-butene.

31. The bond between carbon atom (1) and carbon atom (2) in compound

$N \equiv C - CH = CH_2$ involves the hybridisation as (IIT JEE 1987)

- (a) sp^2 and sp^2
- (b) sp^3 and sp
- (c) sp and sp^2
- (d) sp and sp

Sol. (c) The bond between carbon atom (1) is sp as it is attached to triple bond and carbon atom (2) is sp^2 , Since it is attached with a double bond.

32. The IUPAC name of the compound having the formula is (IIT JEE 1984)



- (a) 3,3,3-trimethyl-1-propene
- (b) 1,1,1-trimethyl-2-propene
- (c) 3,3-dimethyl-1-butene
- (d) 2,2-dimethyl-3-butene

Sol. (c)

33. The IUPAC name(s) of the following compound is (are)

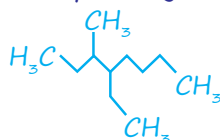
C-18.6 W-50.63 UA-6.71 PC-24.06
(JEE Adv. 2017)



- (a) 4-methylchlorobenzene
- (b) 4-chlorotoluene
- (c) 1-chloro-4-methylbenzene
- (d) 1-methyl-4-chlorobenzene

Sol. (b, c) 4-chlorotoluene and 1-chloro-4-methylbenzene both are correct answer.

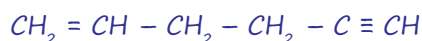
34. Name of the compound given below is: (2003)



- (a) 4-ethyl-3-methyloctane
- (b) 3-methyl-4-ethyloctane
- (c) 2,3-diethylheptane
- (d) 5-ethyl-6-methyloctane

Sol. (a)
4-ethyl-3-methyloctane

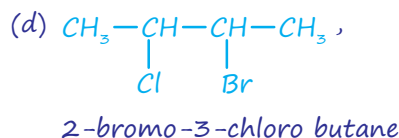
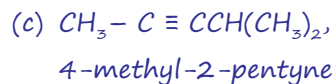
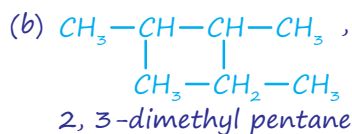
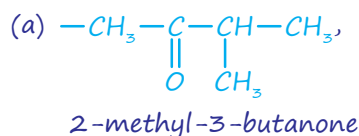
35. IUPAC name of the following is: (2002)



- (a) 1,5-hexenyne
- (b) 1-hexene-5-yne
- (c) 1-hexyne-5-ene
- (d) 1,5-hexynene

Sol. (b) $\overset{1}{\text{CH}_2} = \overset{2}{\text{CH}} - \overset{3}{\text{CH}_2} - \overset{4}{\text{CH}_2} - \overset{5}{\text{C}} \equiv \overset{6}{\text{CH}}$
1-hexene-5-yne.

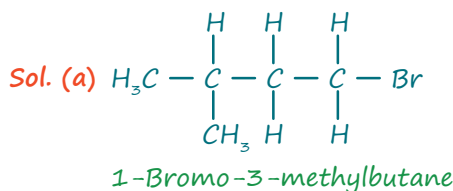
36. The incorrect IUPAC name is: (2001)



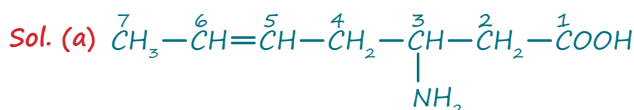
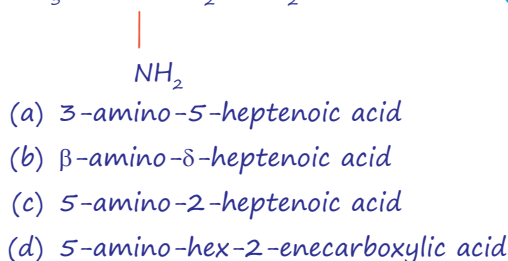
Sol. (a)

37. The IUPAC name of $(\text{CH}_3)_2\text{CH} - \text{CH}_2 - \text{CH}_2\text{Br}$ is: (1996)

- (a) 1-bromo-3-methylbutane
- (b) 2-methyl-3-bromopropane
- (c) 1-bromopentane
- (d) 2-methyl-4-bromobutane



38. The IUPAC name for $\text{CH}_3\text{CH} = \text{CHCH}_2\text{CHCH}_2\text{COOH}$ is: (1995)



As $-\text{COOH}$ group is highest priority group, it is numbered one. So, the IUPAC name is 3-amino-5-heptenoic acid.

Short Notes

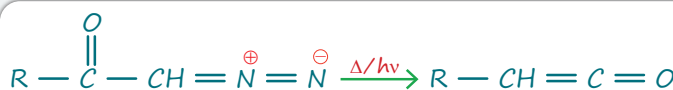
Chemistry of Nitrenes

Nitrenes are highly reactive species with a neutral nitrogen atom having only six valence electrons (like a nitrogen radical). They are electron-deficient and behave like reactive intermediates in many organic reactions. Nitrenes can be generated by the loss of nitrogen gas (N_2) from azides or by photolysis of certain compounds.

Wolff Rearrangement

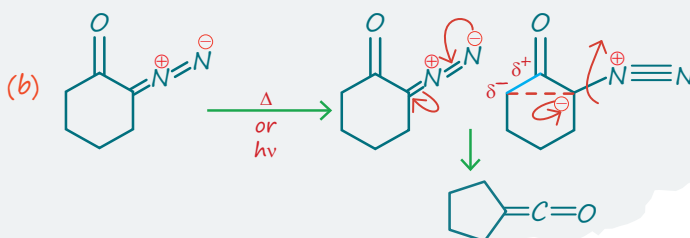
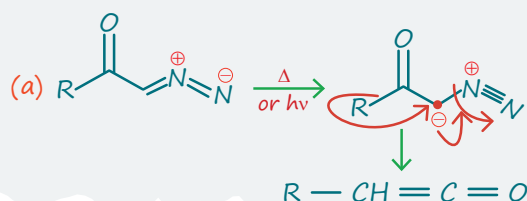
Wolff rearrangement involves the transformation of α -diazoketone into a ketene, typically initiated by heat or light (photolysis).

Reaction



Wolff flashes,
Eistert extends

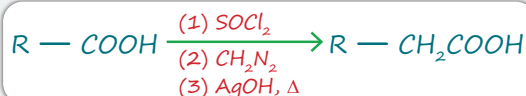
Mechanism



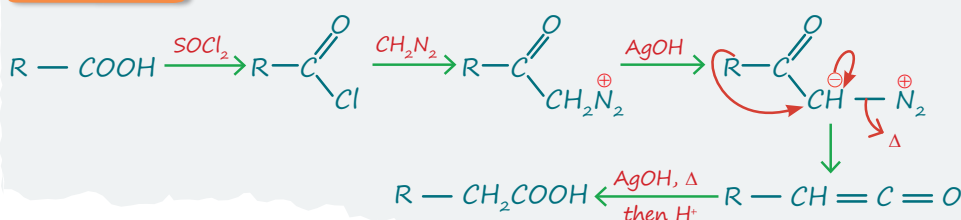
Arndt Eistert rearrangement

Conversion of carboxylic acids to higher homologous acids via diazoketone formation and Wolff rearrangement.

Reaction



Mechanism



Arndt-Eistert
rearrangement opens
doors for creating bigger
acids.



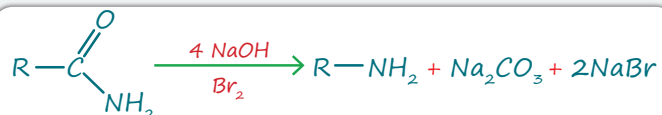
Preparation of Amines

Hoffmann Bromamide Rearrangement

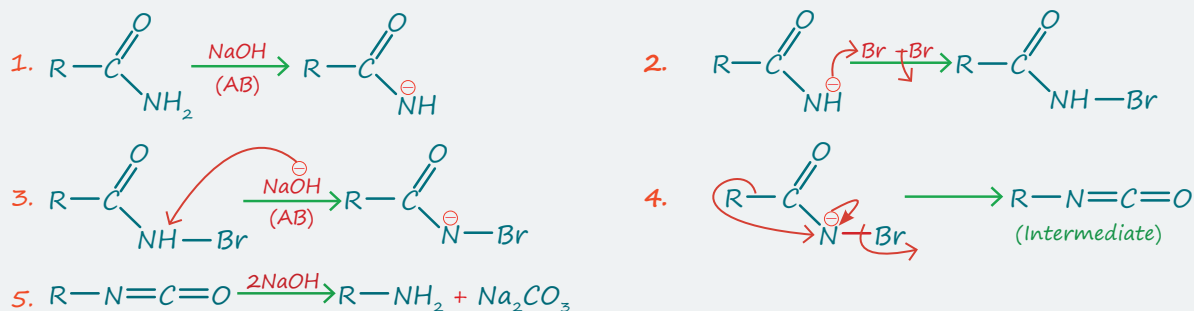
Primary amides react with Br_2 and NaOH to give primary amines with one carbon less via formation of an isocyanate intermediate.



Reaction



Mechanism

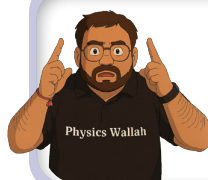
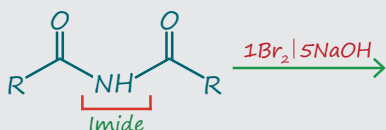


Baba Tea-Stall (BTS)

- (i) 1 mole of Br_2 and 4 mol of NaOH is taken.
- (ii) Migration at e^- deficient 'N' (Nitrene like) takes place. by alkyl or aryl group.
- (iii) Intermediate Obtained is $\text{R}-\text{N}=\text{C}=\text{O}$
- (iv) Rearrangement is completely intramolecular process.
- (v) During rearrangement, configuration never changes.
- (vi) In the presence of NaOH , Br_2 reacts, to form (NaOBr).

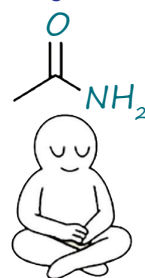
? Try Yourself

➤ For Imide (For JEE Advanced)



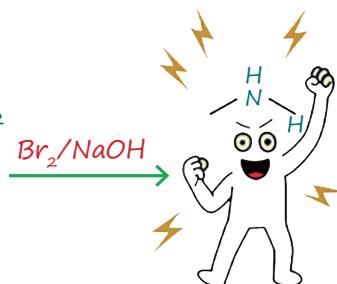
Hoffmann Shortcut:
"Hoffmann doesn't do extra — he deletes carbon like unnecessary drama."

Jugaad



Amide before
 Br_2/NaOH :

"I'm stable, I'm chill."



After reaction:

"I lost a carbon and gained a new identity — AMINE MODE ACTIVATED!"

Recap: Reaction Comparison

Feature	Curtius	Schmidt	Lossen	Hoffmann
Starts From	Acyl Azide	Carboxylic Acid	Hydroxamic Acid	Amide
Intermediate	Isocyanate	Isocyanate	Isocyanate	Isocyanate
Final Product	1° Amine	1° Amine	1° Amine	1° Amine
Carbon Loss?	Yes	Yes	Yes	Yes



One Intermediate to Rule Them All

Acyl Azide $\rightarrow$ Curtius $\rightarrow$ Isocyanate $\rightarrow$ Amine

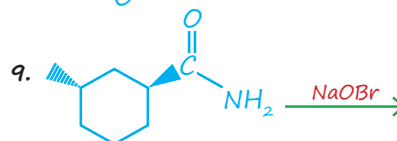
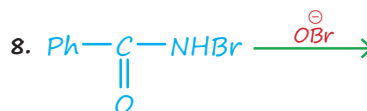
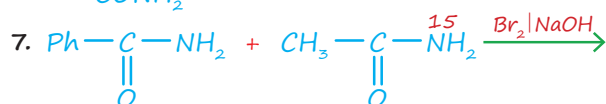
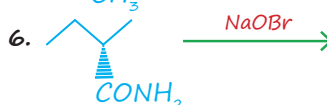
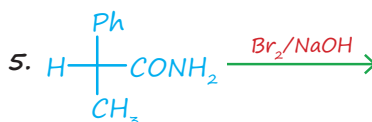
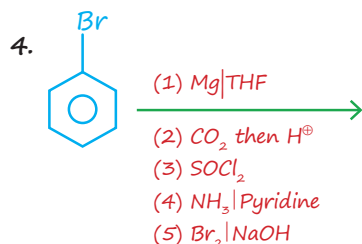
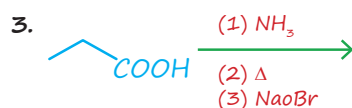
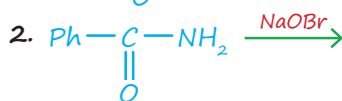
Carboxylic Acid $\rightarrow$ Schmidt $\rightarrow$ Isocyanate $\rightarrow$ Amine

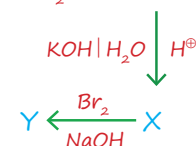
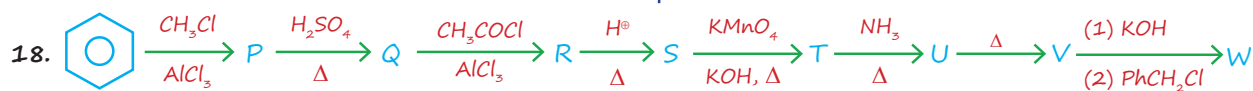
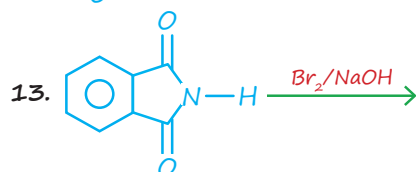
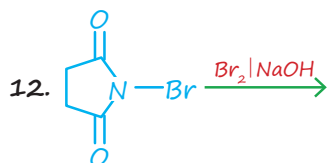
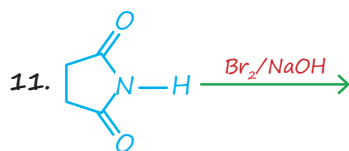
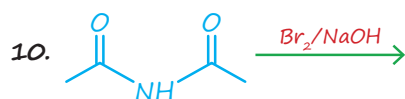
Hydroxamic Acid $\rightarrow$ Lossen $\rightarrow$ Isocyanate $\rightarrow$ Amine

Amide $\rightarrow$ Hoffmann $\rightarrow$ Isocyanate $\rightarrow$ Amine

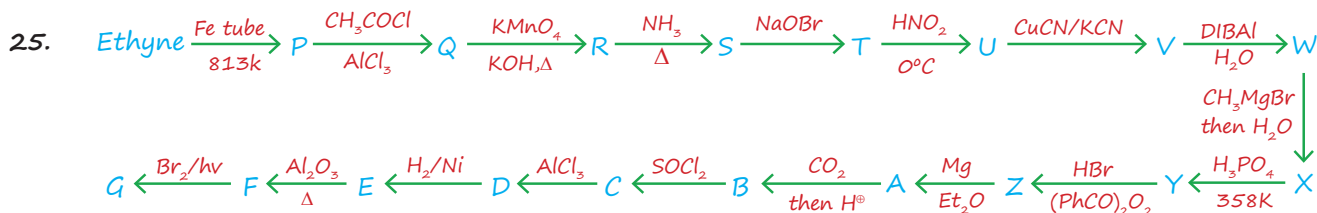
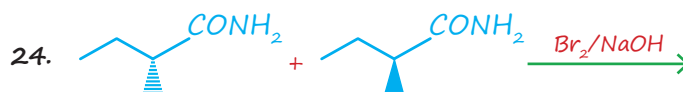
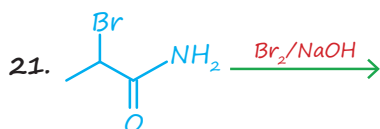


Practice Questions



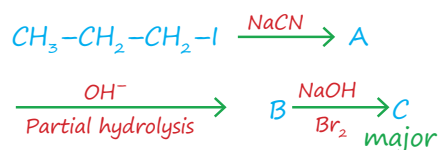


20. Find mass of 'U'



Previous Year Questions

1. Identify the major product C formed in the following reaction sequence: (2024)



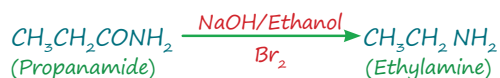
- (a) butanamide
(b) α -bromobutanoic acid
(c) propylamine
(d) butylamine

Sol.(c) Propylamine

4. Reaction of propanamide with ethanolic sodium hydroxide and bromine will give (2020-Covid)

- (a) Methylamine (b) Propylamine
(c) Aniline (d) Ethylamine

Sol. (d) The reaction is Hoffmann bromide reaction



6. Acetamide is treated with the following reagents separately. Which one of these would yield methyl amine? (2010 Pre)

- (a) PCl_5 (b) $\text{NaOH}-\text{Br}_2$
(c) Soda lime (d) Hot conc. H_2SO_4

Sol. (b) $\text{NaOH}-\text{Br}_2$

8. Intermediates formed during reaction of $\text{R}-\text{C}(=\text{O})-\text{NH}_2$ with Br_2 and KOH are: (2001)

- (a) RCONHBr and RNCO
(b) RNHCOBr and RNCO
(c) $\text{RNH}-\text{Br}$ and RCONHBr
(d) RCONBr_2

Sol. (a) RCONHBr and RNCO

9. Amides can be converted into amines by reaction named after: (1999)

- (a) Perkin (b) Claisen
(c) Hoffmann (d) Kekule

Sol. (c) Amide can be converted into amine by Hoffmann - Bromamide reaction.

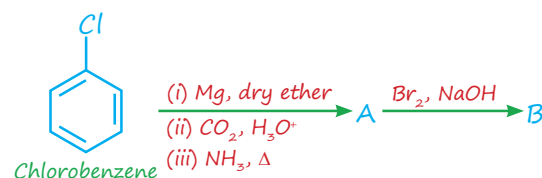
10. Indicate which nitrogen compound amongst the following would undergo Hoffmann reaction

(i.e., reaction with Br_2 and strong KOH) to furnish the primary amine ($\text{R}-\text{NH}_2$)? (1989)

- (a) RCONHCH_3 (b) RCOONH_4
(c) RCONH_2 (d) $\text{R}-\text{CO}-\text{NHOH}$

Sol. (c) The amide ($-\text{CONH}_2$) group is converted into primary amino group ($-\text{NH}_2$) by Hoffmann's bromamide reaction.

5. Consider the following sequence of reactions:

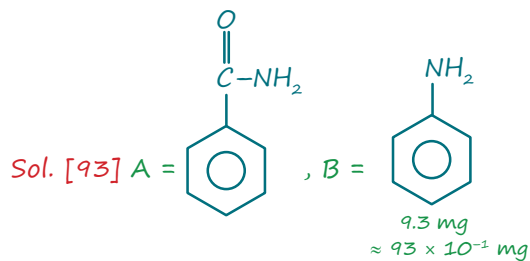


11.25 mg of chlorobenzene will produce $\times 10^{-1}$ mg of product B.

(Consider the reactions result in complete conversion.)

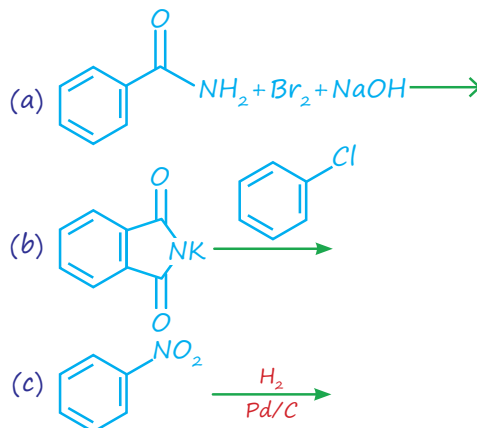
[Given molar mass of $\text{C}, \text{H}, \text{O}, \text{N}$ and Cl as 12, 1, 16, 14 and 35.5 g mol^{-1} respectively]

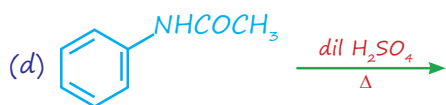
[28 Jan, 2025 (Shift-I)]



8. How many of the transformation given below would result in aromatic amines?

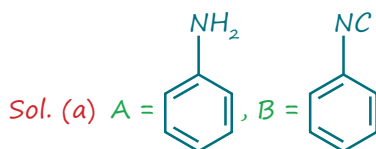
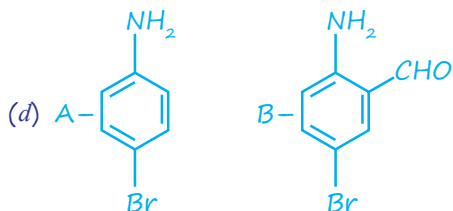
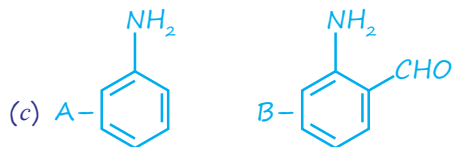
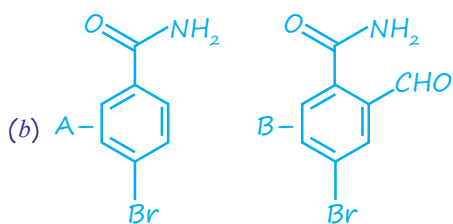
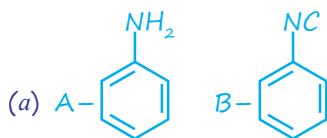
[31 Jan, 2023 (Shift-I)]





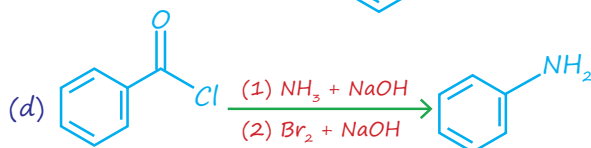
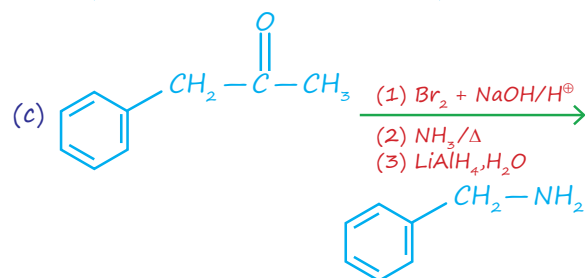
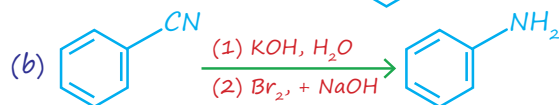
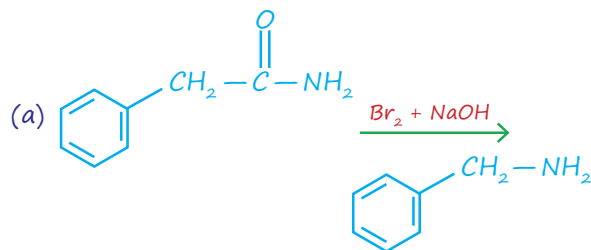
Sol. [3] Aromatic amines will be produced in the reactions (a), (c) and (d).

15. Hoffmann bromamide degradation of benzamide gives product A, which upon heating with CHCl_3 and NaOH gives product B. The structures of A and B are: [17 March, 2021 (Shift-I)]

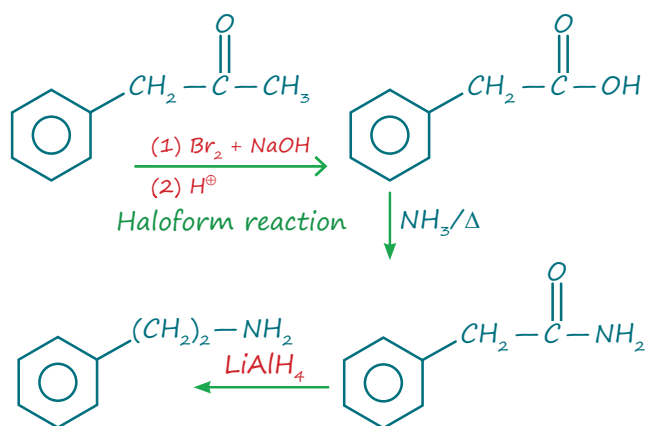


16. Which of the following reactions DOES NOT involve Hoffmann bromamide degradation?

[16 March, 2021 (Shift-I)]



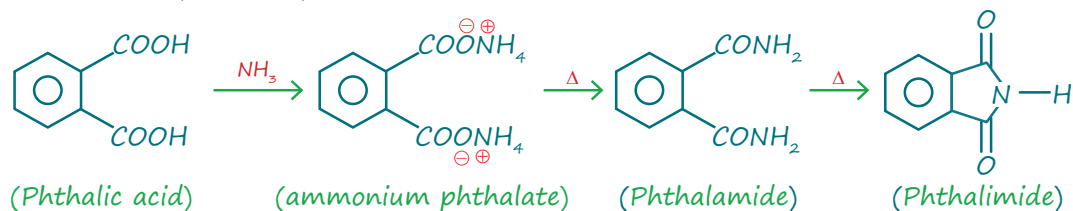
Sol. (c)



Gabriel Phthalimide Synthesis

➤ Formation of phthalimide from phthalic acid.

Phthalic acid reacts with ammonia to form ammonium phthalate, which on heating gives phthalamide, and further heating produces phthalimide.



➤ Gabriel Phthalimide Amine Synthesis:

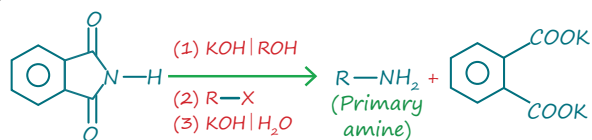
Gabriel Phthalimide Amine Synthesis is used to prepare primary amines ($R-NH_2$).

Phthalimide reacts with KOH and an alkyl halide ($R-X$) through nucleophilic substitution, followed by hydrolysis, to give the primary amine and potassium phthalate.

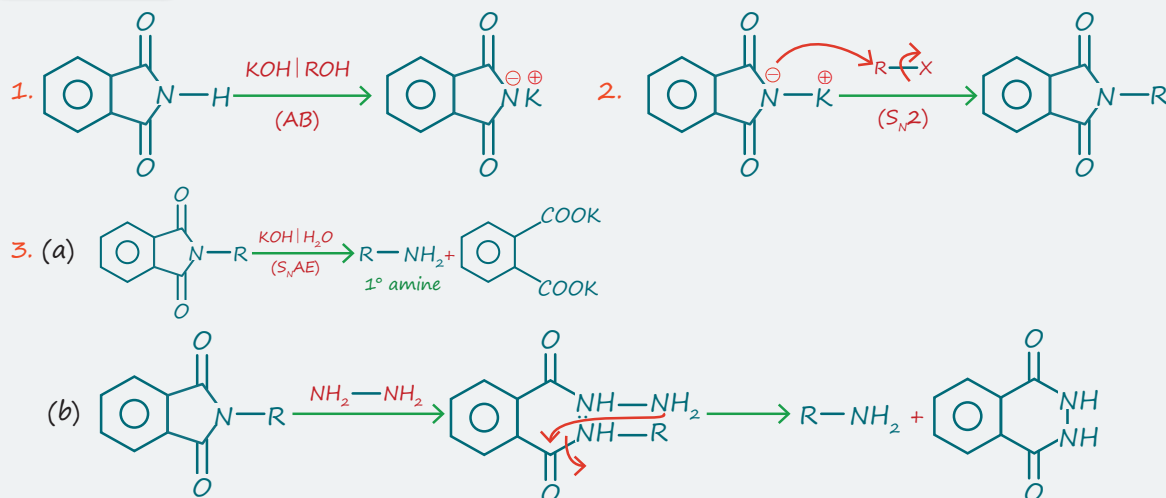
? Try Yourself



Reaction



Mechanism



Gabriel only admits alkyl kids. Aryl halides? Rejected at the gate.

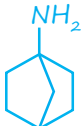
Baba Tea-Stall (BTS)

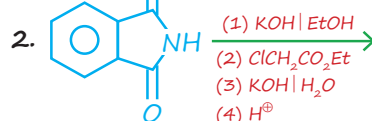
- | | |
|--|--|
| (i) Best method to synthesize primary amine. | (v) Inversion takes place. |
| (ii) For acid base, alcoholic KOH is taken. | (vi) Only H_2O must not be taken, either acidic or basic medium is taken for hydrolysis. |
| (iii) $R-X$ must be 1° or 2° halide. | (vii) Reaction is used prepare amino acid. |
| (iv) S_N2 mechanism. | |



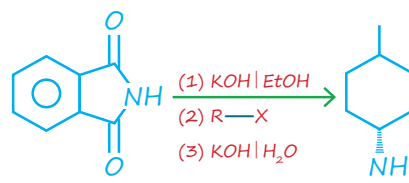
Practice Questions

1. Which of the following amines can be obtained good yield by Gabriel Phthalimide Synthesis?

- (a) PhNH_2 (b) CH_3NH_2
 (c)  (d) $(\text{CH}_3)_2\text{NH}$
 (e) CH_3CONH_2 (f) 
 (g) PhNHMe (h) PhCH_2NH_2
 (i) 



3.



6. The source of nitrogen in Gabriel synthesis of amines is _____. [NCERT Exemplar]

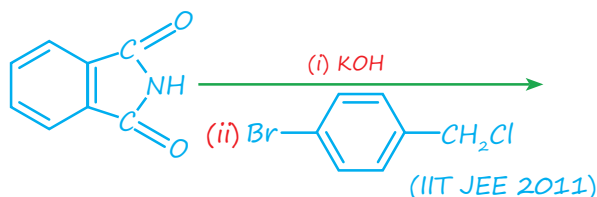
- (a) Sodium azide, NaN_3
 (b) Sodium nitrite, NaNO_2
 (c) Potassium cyanide, KCN
 (d) Potassium phthalimide, $\text{C}_6\text{H}_4(\text{CO})_2\text{NK}^+$

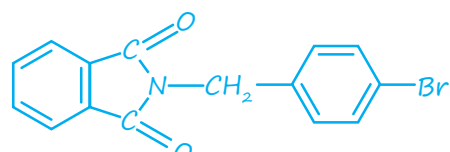
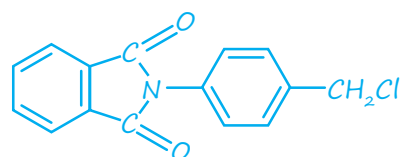
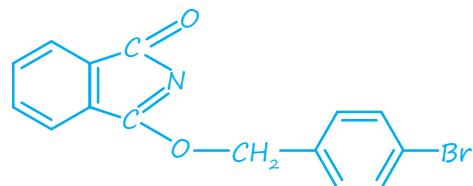
7. Which of the following amines can be prepared by Gabriel synthesis. [NCERT Exemplar]

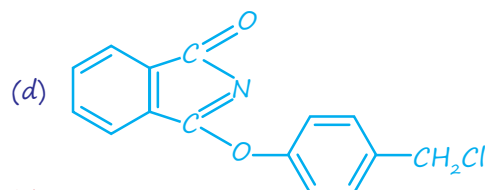
- (a) Isobutyl amine
 (b) 2-Phenylethylamine
 (c) N-methylbenzylamine
 (d) Aniline

Previous Year Questions

22. The major product of the following reaction is



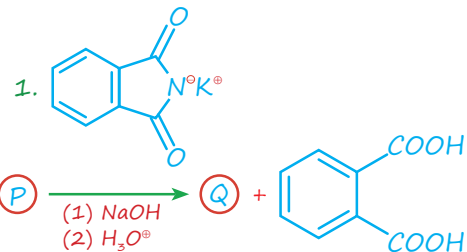
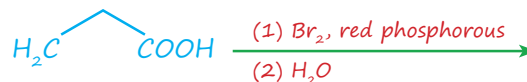
- (a) 
 (b) 
 (c) 



Sol. (a)

50. Considering the reaction sequence given below, the correct statement(s) is/are:

C2.05 W20.77 UA47.80 (JEE Adv. 2022)

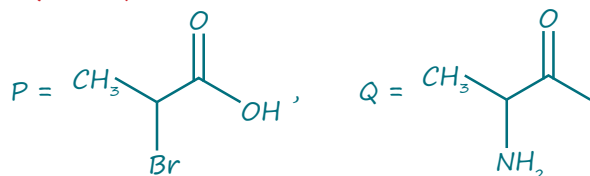


- (a) (P) can be reduced to a primary alcohol using NaBH_4
 (b) Treating (P) with conc. NH_4OH solution followed by acidification gives (Q).

(c) Treating (Q) with a solution of NaNO_2 in aq. HCl liberates N_2

(d) (P) is more acidic than $\text{CH}_3\text{CH}_2\text{COOH}$.

Sol. (b, c, d)



5. Method by which Aniline cannot be prepared is: (2015 Re)

- Potassium salt of phthalimide treated with chlorobenzene followed by hydrolysis with aqueous NaOH solution
- Hydrolysis of phenylisocyanide with acidic solution
- Degradation of benzamide with bromine in alkaline solution
- Reduction of nitrobenzene with H_2/Pd in ethanol

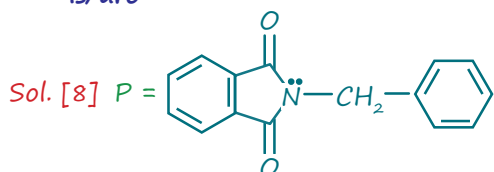
Sol. (a) Aryl halides are less reactive towards Gabriel phthalimide reaction.

7. Phthalimide is made to undergo following sequence of reactions.

[04 April, 2024 (Shift-II)]



Total number of π -bonds present in product 'P' is/are



9. Number of isomeric aromatic amines with molecular formula $\text{C}_8\text{H}_{11}\text{N}$, which can be synthesized by Gabriel Phthalimide synthesis is

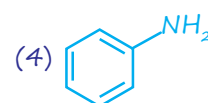
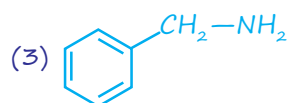
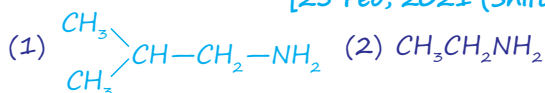
[6 April, 2023 (Shift-II)]

Sol. [*] Gabriel synthesis is used for the preparation of primary amine. Primary Aromatic amine can't be synthesised by this method because aryl halide do not undergo nucleophilic substitution reaction.

NTA Answer Key [5], PW Answer Key [6]

11. The total number of amines among the following which can be synthesized by Gabriel synthesis is:

[25 Feb, 2021 (Shift-II)]



Sol. [3] Primary amines can be synthesized by Gabriel synthesis. 3 Amines can be synthesized.

14. Given below are two statements, one is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A): Gabriel phthalimide synthesis cannot be used to prepare aromatic primary amines.

Reason (R): Aryl halides do not undergo nucleophilic substitution reaction.

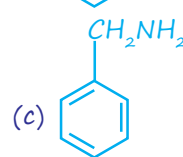
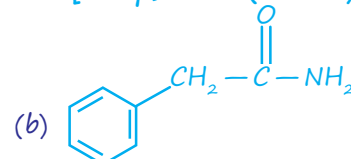
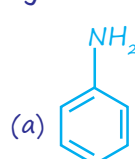
In the light of the above statements, choose the correct answer from the options given below:

[25 July, 2021 (Shift-I)]

- Both (A) and (R) true but (R) is not the correct explanation of (A).
- (A) is false but (R) is true.
- Both (A) and (R) true and (R) is correct explanation of (A).
- (A) is true but (R) is false.

Sol. (c) Both (A) and (R) true and (R) is correct explanation of (A).

22. Which of the following compounds can be prepared in good yield by Gabriel phthalimide synthesis? [6 Sep, 2020 (Shift-II)]



Sol. (c) Gabriel phthalimide synthesis gives 1° amine in good yield.

24. Which of the following amines can be prepared by Gabriel phthalimide reaction?

[8 Apr, 2019 (Shift-I)]

- Neo-pentylamine
- n-butylamine
- Triethylamine
- t-butylamine

Sol. (b) Gabriel phthalimide reaction is used to prepare primary aliphatic amine. n-butyl amine is primary amine among the following option.

Separation of Amines

Separation of Amines

Hinsberg Test

Fractional distillation

Diethyl Oxalate Test

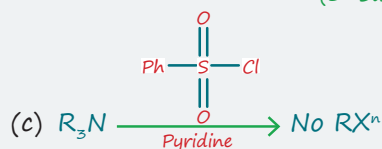
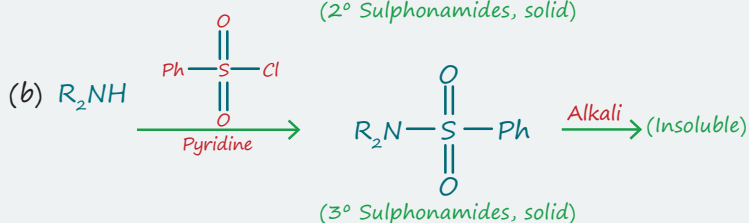
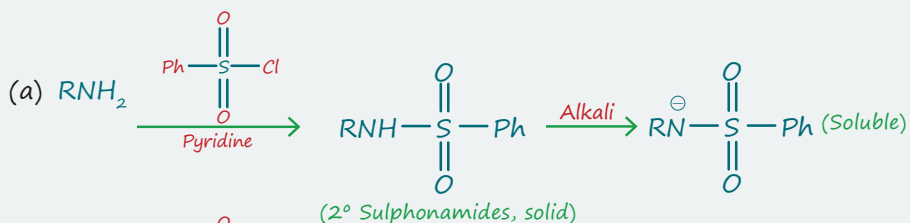
Hinsberg Test

- The Hinsberg test helps to distinguish primary, secondary, and tertiary amines.
- Amines react with benzenesulphonyl chloride (Hinsberg reagent) in the presence of a base.
- Primary amines form soluble sulphonamides, secondary amines form insoluble sulphonamides, and tertiary amines do not react.

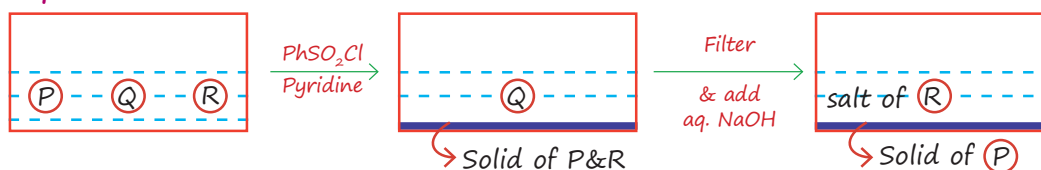
➤ We can also use tosyl chloride



जुगाड



➤ Used to separate 1°, 2° & 3° amine



Find P, Q and R ⇒

Practice Questions

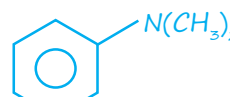
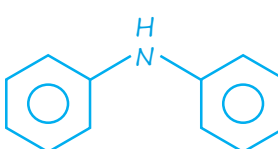
4. Which of the following compounds on reaction with Hinsberg reagent produces base insoluble sulphonamide

- (a) PhNH_2 (b) CH_3NH_2 (c) PhCONMe_2 (d) $(\text{CH}_3)_2\text{NH}$ (e) CH_3CONH_2

- (f)  (g) PhNHMe (h) PhCH_2NH_2 (i) EtCONHMe

Previous Year Questions

32. Which among the following react with Hinsberg's reagent?
[23 Jan, 2025 (Shift-I)]

- A.  B. 
C. $\text{CH}_3 - \text{NH}_2$ D. $\text{N}(\text{CH}_3)_3$
E. 

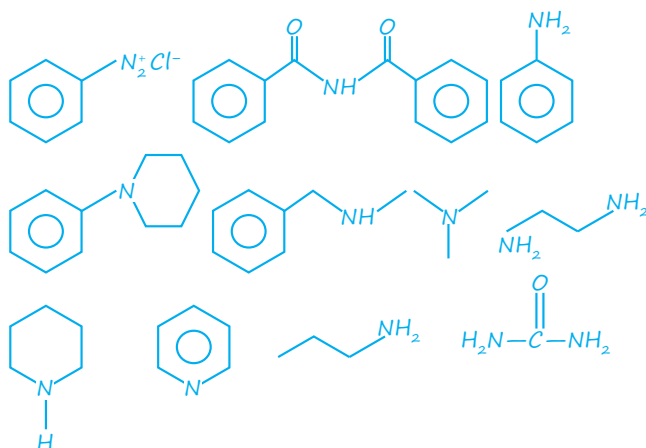
Choose the correct answer from the option given below:

- (a) A, B and E only (b) C and D only
(c) A, C and E only (d) B and D only

Sol. (c) Hinsberg reagent reacts with primary and secondary amines. Compounds given in B and D are tertiary amines, hence, do not react with Hinsberg reagent.

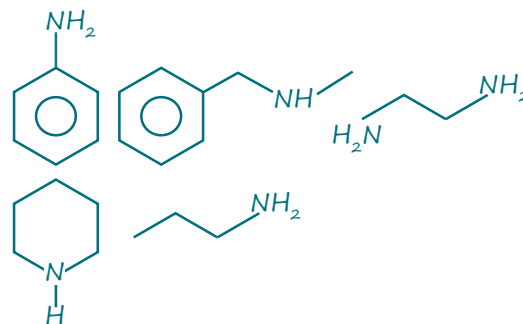
37. Number of compounds which give reaction with Hinsberg's reagent is ____.

[01 Feb, 2024 (Shift-II)]



Amines

Sol. [5]

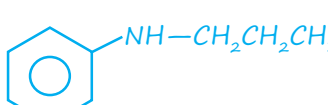


59. Which statement is NOT correct for p-toluenesulphonyl chloride?
[26 Jun, 2022 (Shift-II)]

- (a) It is known as Hinsberg's reagent
(b) It is used to distinguish primary and secondary amines.
(c) On treatment with secondary amine, it leads to a product, that is soluble in alkali.
(d) It doesn't react with tertiary amines.

Sol. (c) Product formed from secondary amine is insoluble in alkali.

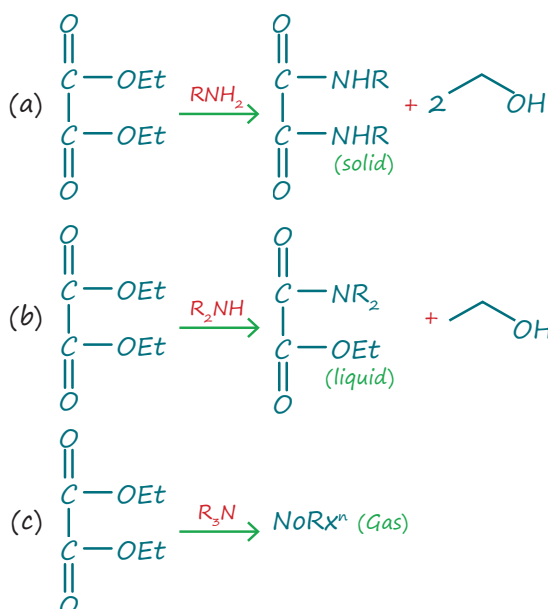
71. An amine on reaction with benzenesulphonyl chloride produces a compound insoluble in alkaline solution. This amine can be prepared by ammonolysis of ethyl chloride. The correct structure of amine is: [26 Feb, 2021 (Shift-I)]

- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}-\text{CH}_3$
(b) 
(c) $\text{CH}_3\text{CH}_2\text{NH}_2$
(d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NHCH}_2-\text{CH}_3$

Sol. (d) Since, given amine on reaction with PhSO_2Cl produces a compound insoluble in Alkaline solution, it is a 2° Amine.

Diethyl Oxalate Test

- The Diethyl oxalate test is used to detect & separate amines.
- Primary amines react with diethyl oxalate to form a white precipitate of oxamide, confirming their presence.
- Secondary amines do not form this precipitate because they cannot react the same way.
- Tertiary amines do not react with diethyl oxalate and give no precipitate.



Diethyl's Rule: No N-H? No Precipitate. No Entry.

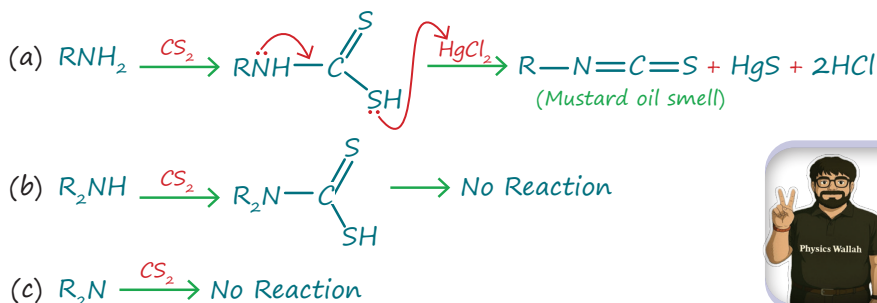
Try Yourself

A chemistry student performs the Diethyl Oxalate Test and the Hinsberg Test on three unknown organic compounds: X, Y and Z.

- X does not react with diethyl oxalate or Hinsberg reagent.
- Y gives no precipitate with diethyl oxalate but forms a solid product with Hinsberg reagent that is insoluble in alkali.
- Z gives a white precipitate when treated with diethyl oxalate.

Mustard Oil Test

- The test detects the presence of primary amines.
- Primary amines react with mustard oil (allyl isothiocyanate) to form a yellow solid.
- Secondary and tertiary amines do not give any yellow precipitate in this test.



"Primary amine's yellow party is lit — secondary and tertiary? They didn't get the invite."

Recap: Distinguish Test Comparison

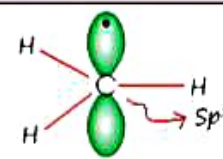
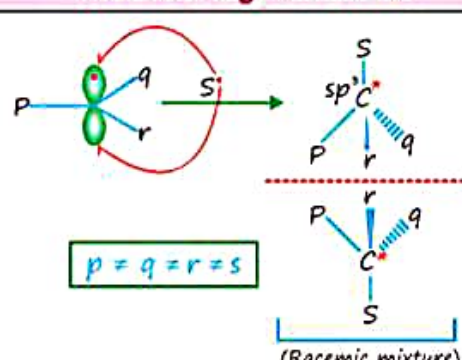
Test	1° amine	2° amine	3° amine
Hinsberg	Alkali Soluble	Alkali Insoluble	No reaction
Diethyl Oxalate	Solid	Liquid	gas
Mustard Oil	Yellow ppt	No reaction	No reaction

4

Hydrocarbons

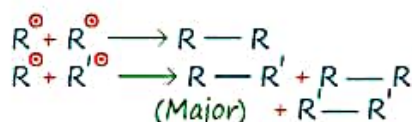
Free Radical

A free radical is a molecule or atom with an unpaired electron, making it highly reactive.

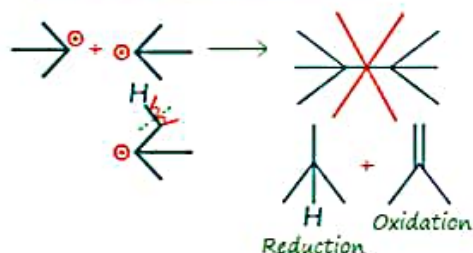
Properties	
➤ Octet incomplete ➤ Highly unstable ➤ Paramagnetic ➤ 1 U.P. (U.P. : unpaired electron) ➤ $\sqrt{1(1+2)} \text{ B.M.} \Rightarrow \sqrt{3} \text{ B.M.}$	➤ 0 I.P. (I.p. : lone pair) ➤ 3 B.P. (B.P. : Bond Pair) ➤ sp^2, Planar ➤ 0 A.E. for $Rx^{\bullet}$ (A.E. (Activation energy)) 
Factors	Attack by Radical
➤ Sunlight ($h\nu$, uv rays) ➤ High temperature (Δ) ➤ Metal ➤ Electrolysis ➤ Gaseous phase ➤ Another radical	 (Racemic mixture)

Reaction By Free Radical

Combination



Disproportionation



Rearrangement

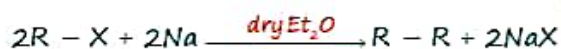


Just as free radicals initiate powerful chains, your small steps today start unstoppable progress.

Wurtz Reaction

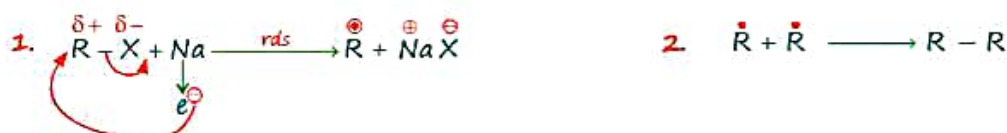
The Wurtz reaction is an organic chemical reaction used to couple two alkyl halides to form a higher alkane. It involves the reaction of alkyl halides with sodium metal in dry ether.

Reaction



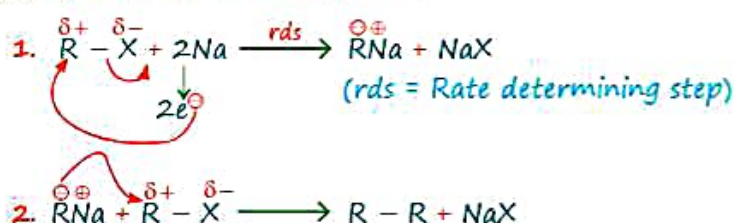
Mechanism 1

Involves homolytic cleavage of R-X by sodium metal, forming alkyl radicals (R•), which then couple to form R-R.



Mechanism 2

Sodium donates electrons to R-X forming R-Na⁺, which acts as a nucleophile and displaces halide from another R-X to form R-R.

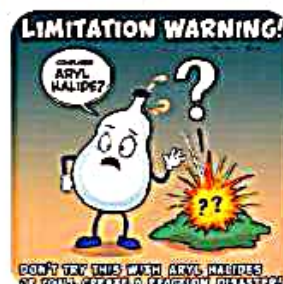
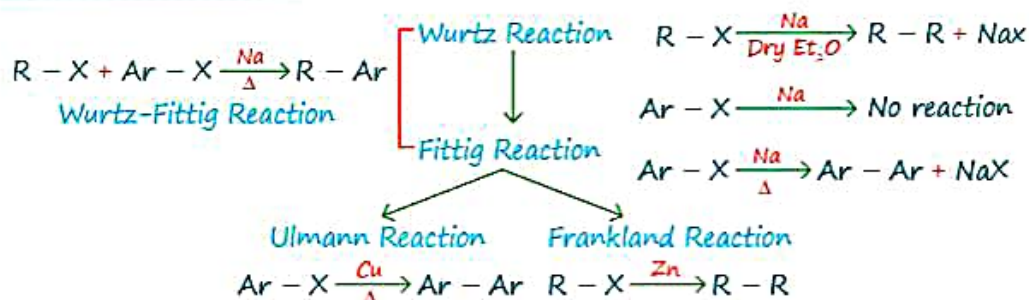


Wurtz: Two alkyl halides walk into a room, sodium says 'Let's team up!' and bam – a new alkane is born!

Baba Tea-Stall (BTS)

- (i) Both Carbanion and free intermediate obtained
- (ii) As the stability of (•) ↑ ⇒ Rate of reaction ↑
- (iii) Rate of reaction for different R-X is :
R-I > R-Br > R-Cl > R-F
- (iv) Symmetrical alkane obtained as major product
 - $\text{CH}_3\text{CH}_2\text{X} \xrightarrow{\quad} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$ (even) → (even)
 - $\text{CH}_3\text{CH}_2\text{CH}_2\text{X} \xrightarrow{\quad} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ (odd) → (even)
- (v) In case of 3°-X, major product is disproportionation reaction.

Wurtz ka khandaan



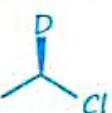
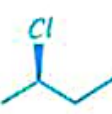
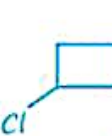
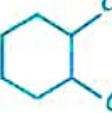
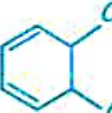
Practice Questions

Free Radical

- $\text{CH}_3^\bullet \rightarrow$
- $\text{CH}_3^\bullet + \text{Cl}^\bullet \rightarrow$
- $\text{Cl}^\bullet \rightarrow$
-  + $\text{Br}^\bullet \rightarrow$
-  $\rightarrow$
-  + $\text{CH}_3^\bullet \rightarrow$
-  + $\text{Cl}^\bullet \rightarrow$
-  + $\text{Br}^\bullet \rightarrow$
-  $\rightarrow$
-  $\rightarrow$
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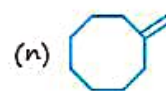
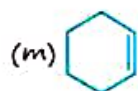
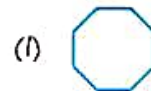
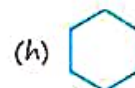
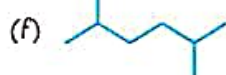
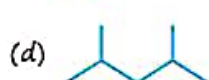
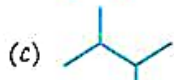
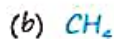
Wurtz Reaction

- $\text{CH}_3\text{Cl} \xrightarrow[\text{dry Et}_2\text{O}]{\text{Na}}$
- $\text{PhCH}_2\text{Br} \xrightarrow[\text{dry Et}_2\text{O}]{\text{Na}}$

-  $\xrightarrow[\text{Dry Et}_2\text{O}]{\text{Na}}$
-  $\xrightarrow[\text{Dry Et}_2\text{O}]{\text{Na}}$
-  $\xrightarrow[\text{Dry Et}_2\text{O}]{\text{Na}}$
-  $\xrightarrow[\text{Dry Et}_2\text{O}]{\text{Na}}$
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-  $\xrightarrow[\text{Dry Et}_2\text{O}]{\text{Na}}$
-  $\xrightarrow[\text{Dry Et}_2\text{O}]{\text{Na}}$
-  $\xrightarrow[\text{Dry Et}_2\text{O}]{\text{Na}}$



54. How many of the following compounds can be obtained in good yield by wurtz reaction using single reactant?



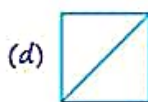
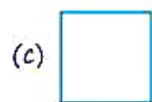
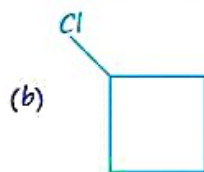
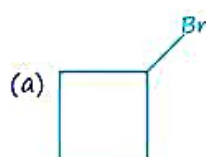
Previous Year Questions

1. Which of the following alkane cannot be made in good yield by Wurtz reaction? (NEET, 2020)

- (a) 2, 3-Dimethylbutane
- (b) n-Heptane
- (c) n-Butane
- (d) n-Hexane

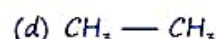
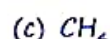
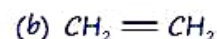
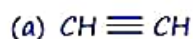
Sol. (b)

2. 1-bromo-3-chlorocyclobutane when treated with two equivalents of Na, in the presence of ether which of the following will be formed? (IIT JEE 2005)



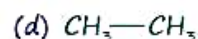
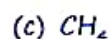
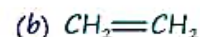
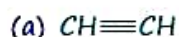
Sol. (d)

3. Hydrocarbon (A) reacts with bromine by substitution to form an alkyl bromide which by Wurtz reaction is converted to gaseous hydrocarbon containing less than four carbon atoms. (A) is (NEET, 2018)

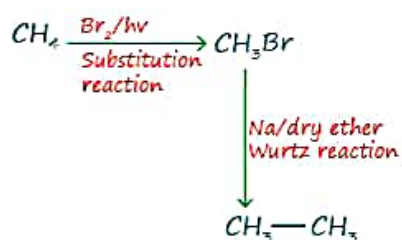


Sol. (c)

4. Hydrocarbon (A) reacts with bromine by substitution to form an alkyl bromide which by Wurtz reaction is converted to gaseous hydrocarbon containing less than four carbon atoms. (A) is (2018)



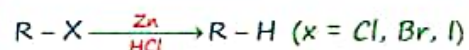
Sol. (c) Wurtz reaction gives symmetrical alkane with even no. of carbon atoms and since hydrocarbon containing less than four carbon atoms need to be obtained, therefore, the hydrocarbon A would be CH_4 . The reaction takes place as:



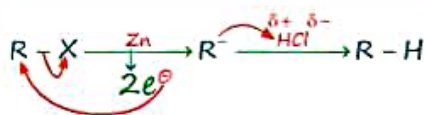
Reaction with Metal and Acid

When an alkyl halide reacts with zinc and hydrochloric acid (HCl), the halide is reduced, resulting in the formation of an alkane (R-H).

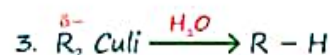
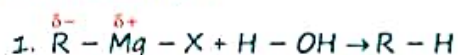
Reaction



Mechanism



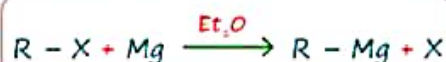
Other Example



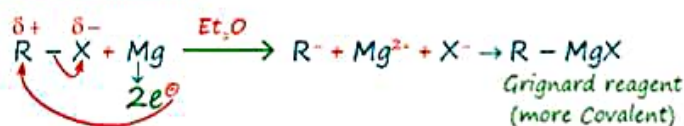
Formation of Grignard Reagent

When an alkyl halide (R-X) reacts with magnesium (Mg) in dry ether, it forms a Grignard reagent (R-Mg-X), which is an organometallic compound.

Reaction



Mechanism



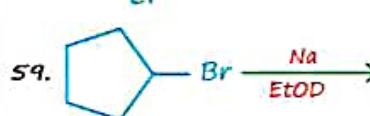
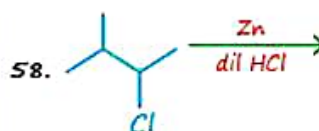
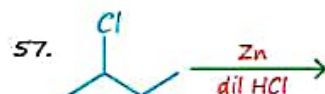
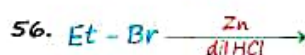
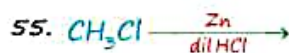
When life gives you alkyl halides, throw in some Mg and make a Grignard!

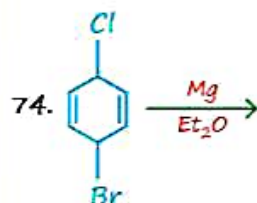
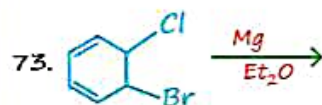
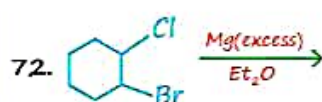
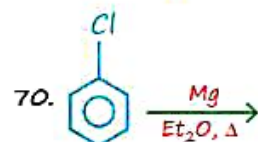
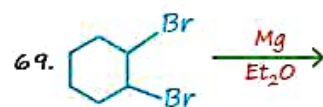
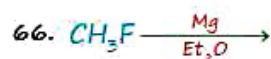
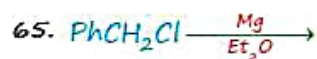
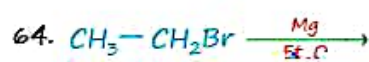
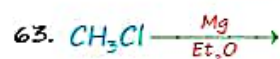
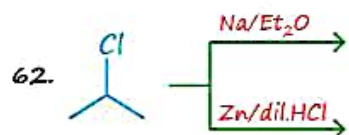
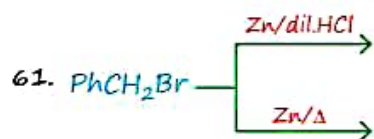
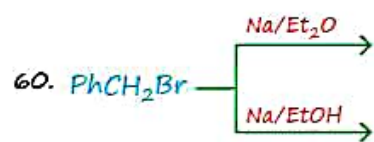
Baba Tea-Stall (BTS)

- (i) Carbanion and free radical Intermediate
- (ii) Organometallic compound obtained

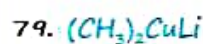
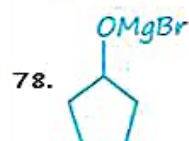
(iii) R-Mg-X known as Grignard reagent

Practice Questions





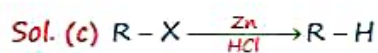
Q. Which of the following is (are) Grignard reagents?



Previous Year Questions

5. Grignard reagent is prepared by the reaction between: (1994)

- (a) Magnesium and alkane
- (b) Magnesium and aromatic hydrocarbon
- (c) Zinc and alkyl halide
- (d) Magnesium and alkyl halide

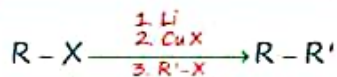


Grignard reagent is prepared by heating an alkyl halide with dry magnesium powder in dry ether.

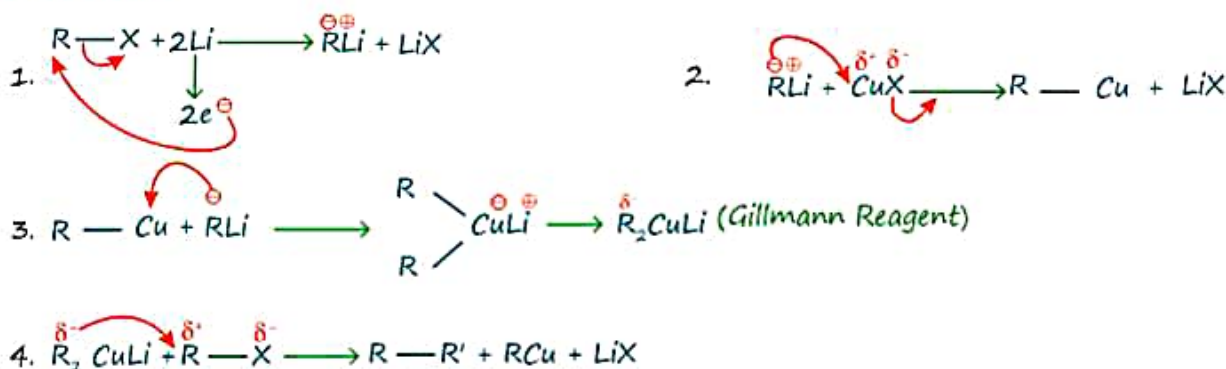
Corey House Synthesis

The Corey-House synthesis involves reacting an alkyl halide ($R-X$) with lithium (Li) to form an alkyl lithium ($R-Li$), which then reacts with copper halide (CuX) to form an organocuprate ($R-Cu-Li$). The organocuprate then reacts with another alkyl halide ($R'-X$) to form a new carbon-carbon bond ($R-R'$).

Reaction



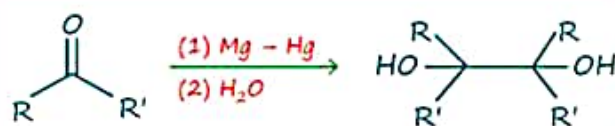
Mechanism



Pinacol Formation

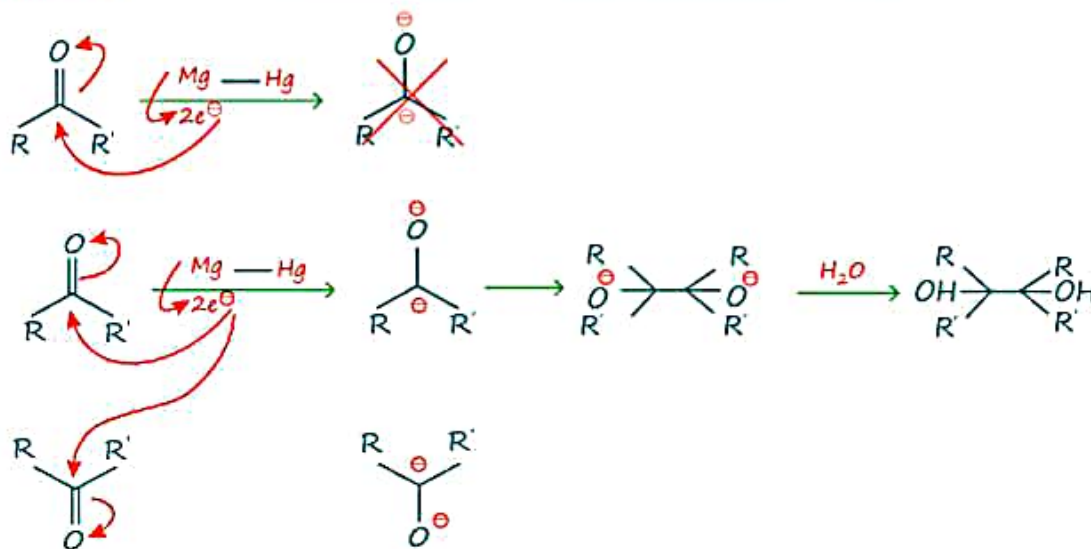
Pinacol formation is a reduction reaction where the carbonyl group ($-C=O$) is reduced to a hydroxyl group ($-OH$), leading to the formation of a vicinal diol (pinacol).

Reaction



- If $R = R' \Rightarrow 1$ pinacol
- If $R \neq R' \Rightarrow 3$ pinacol

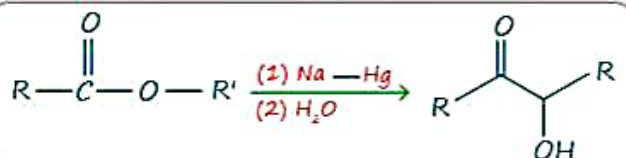
Mechanism



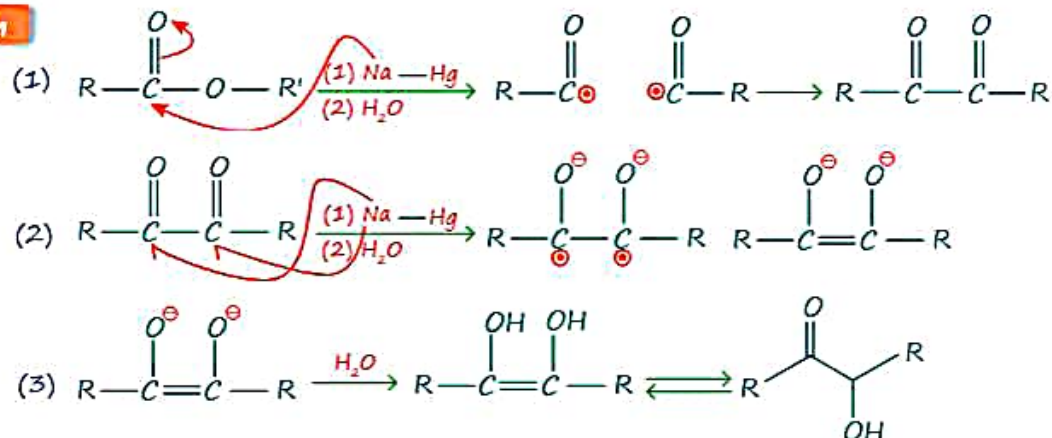
Acyloin Condensation

The Acyloin Condensation involves the reaction of two molecules of aldehydes or ketones in the presence of a Na-Hg/H₂O, resulting in the formation of an α-hydroxy ketone (acyloin).

Reaction

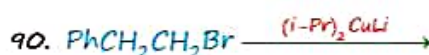
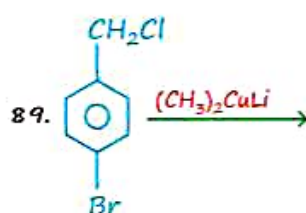
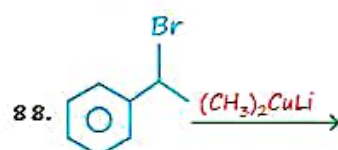
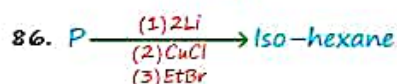
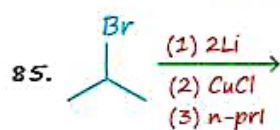
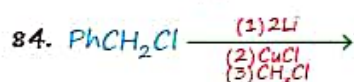
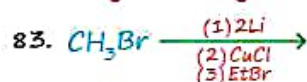


Mechanism

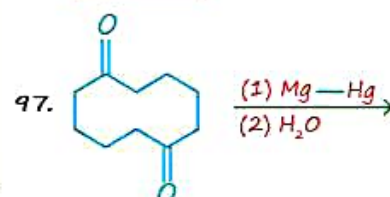
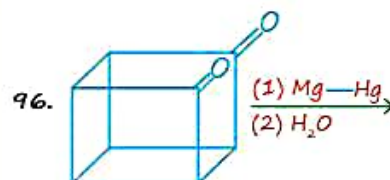
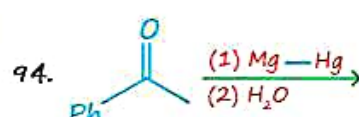
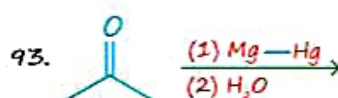
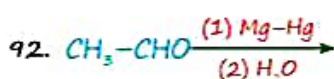
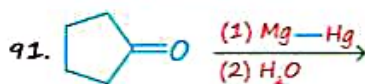


Practice Questions

Corey House Synthesis



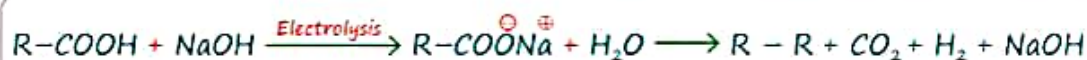
Pinacol Formation



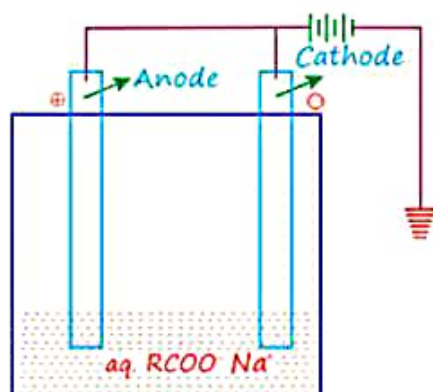
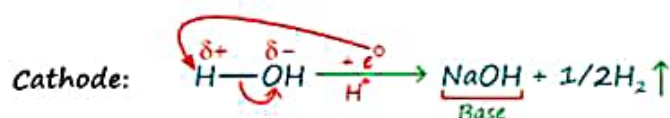
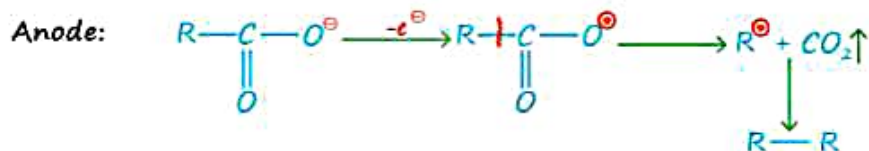
Kolbe's Electrolysis

Kolbe's electrolytic method: An aqueous solution of sodium or potassium salt of a carboxylic acid on electrolysis gives alkane containing even number of carbon atoms at the anode.

Reaction



Mechanism

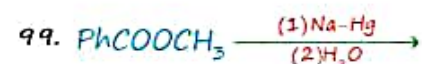
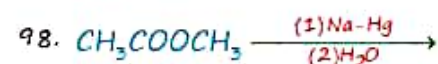


Baba Tea-Stall (BTS)

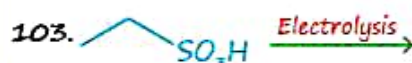
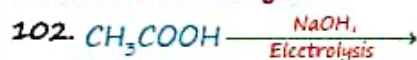
- (i) Free radical Obtained as intermediate
- (ii) CO_2 and $R-R$ at anode, H_2 at cathode obtained.
- (iii) pH of the medium increases (due to NaOH)
- (iv) CH_4 can never be obtained
- (v) In case of $3^{\circ}-COOH$, ester ($RCOOR$) will be obtained.

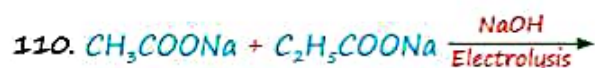
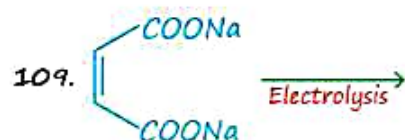
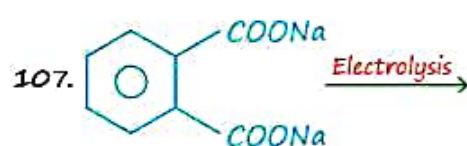
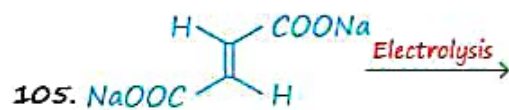
Practice Questions

> Acyloin Condensation

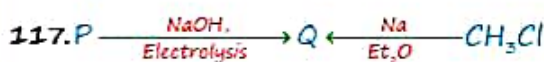
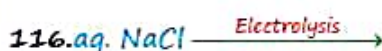
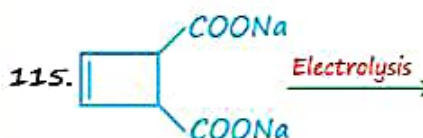
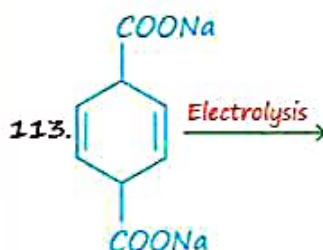
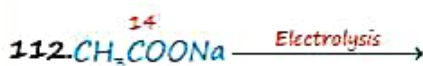


> Kolbe's Electrolysis





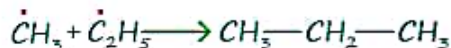
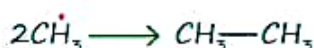
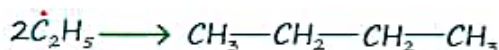
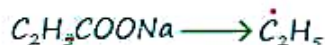
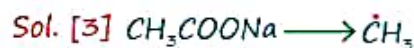
Hydrogen act both anode and cathode.



Previous Year Questions

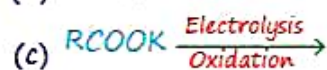
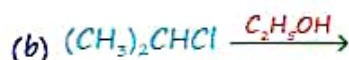
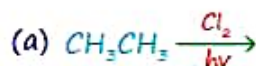
6. Number of alkanes obtained on electrolysis of a mixture of CH_3COONa and $\text{C}_2\text{H}_5\text{COONa}$ is ____

[31 Jan, 2024 (Shift-I)]

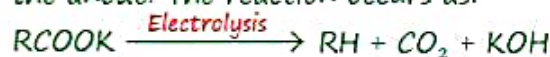


7. Which of the following reaction is expected to readily give a hydrocarbon product in good yields?

(1997)



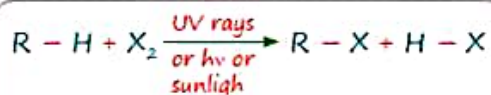
Sol. (c) When an aqueous solution of sodium or potassium salt of carboxylic acid is electrolysed, then an alkane containing even number of carbon atoms is formed at the anode. The reaction occurs as:



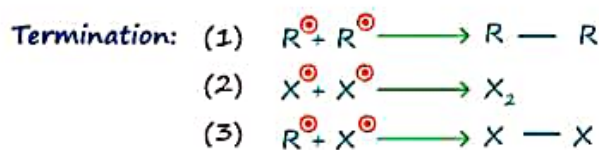
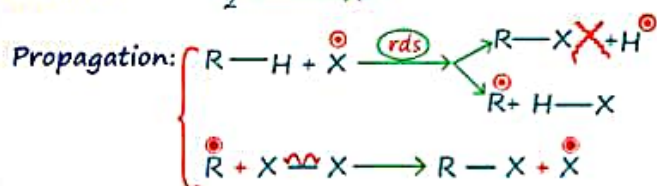
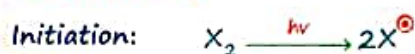
Photohalogenation

Halogenation takes place either at higher temperature (573–773 K) or in the presence of diffused sunlight or ultraviolet light. Lower alkanes do not undergo nitration and sulphonation reactions. These reactions in which hydrogen atoms of alkanes are substituted are known as substitution reactions

Reaction



Mechanism



Baba Tea-Stall (BTS)

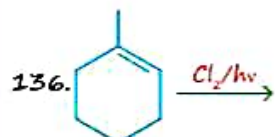
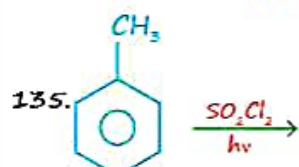
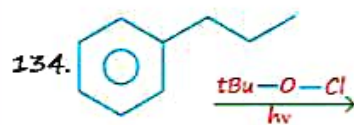
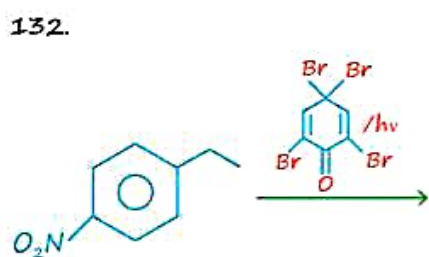
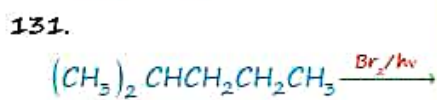
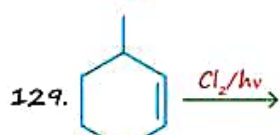
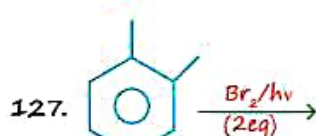
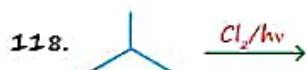
- (i) Example of Substitution Reaction
- (ii) Free radical intermediate
- (iii) UV rays/ $h\nu$ is used to initiate the reaction

- (iv) Rate of reaction for different X_2 is $F_2 > Cl_2 > Br_2 > I_2$
- (v) Kinetic isotopic effect ($R-H > R-D$)
- (vi) RDS is formation of $R^\bullet$

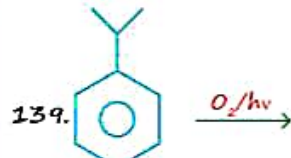
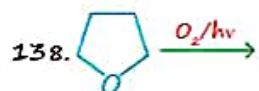
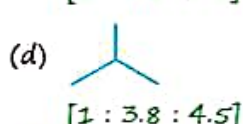
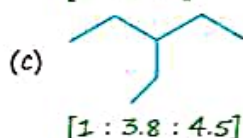
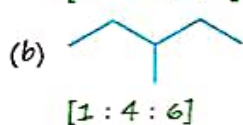
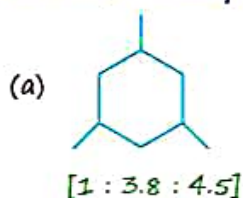
Types of Photohalogenation

Fluorination	Iodination
Highly Exothermic and explosive $CH_4 + 2F_2 \rightarrow C_{\text{black}} + 4HF$	Least reactive and reversible $CH_4 + I_2 \rightleftharpoons CH_3-I + H-I$ O.A. (HIO_3, HNO_3) $\downarrow$ I_2
Chlorination	Bromination
- More reactive - Less Selective $\downarrow$ $1:3.8:4.5 \rightarrow 3^\circ - H$ $\swarrow \searrow$ $1^\circ - H \quad 2^\circ - H$	- Less Reactive - More selective $\downarrow$ $1:80:1600$ $\swarrow$ $3^\circ - H$ or Reso/BB/Aro

Practice Questions

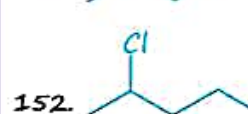
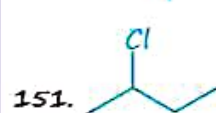
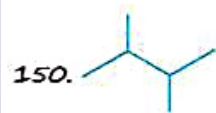
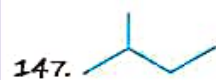


137. Find % yield of each monochlorinated product



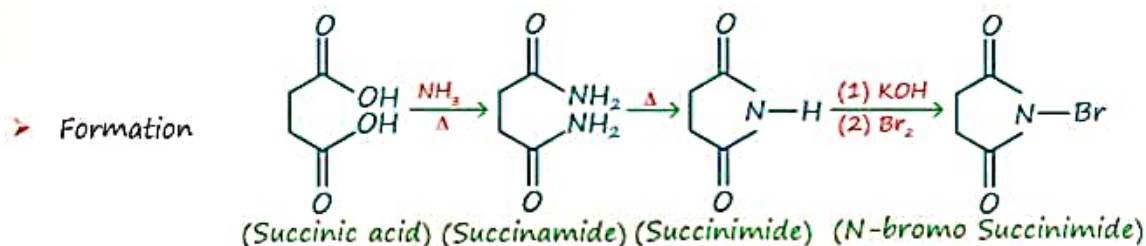
142. After monochlorination reaction on given compound, find

- Structures obtained
- Total products
- Optically active
- Meso
- Fractions

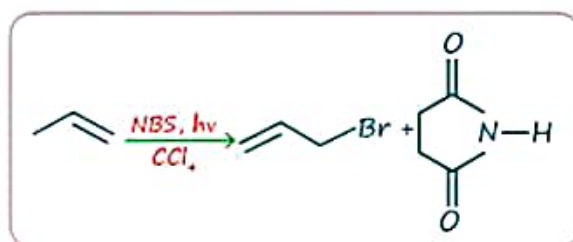


N-bromo succinimide (NBS)

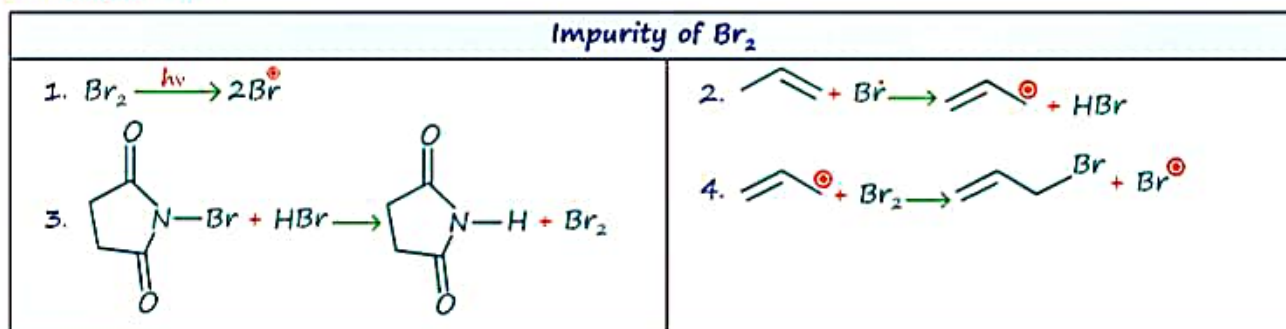
N-Bromosuccinimide (NBS) is a mild brominating agent used mainly for allylic and benzylic bromination via a radical mechanism, and for α -bromination of carbonyl compounds under acidic conditions.



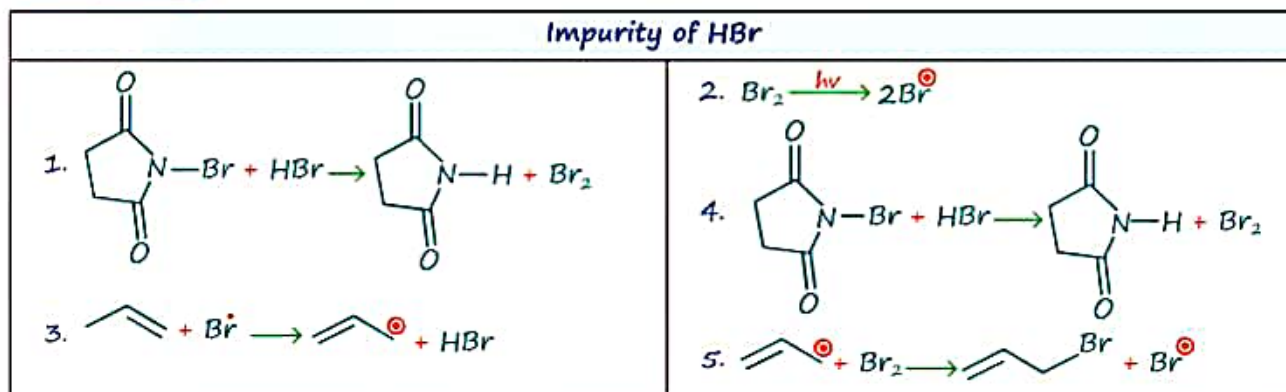
Reaction



Mechanism 1.



Mechanism 2.



Baba Tea-Stall (BTS)

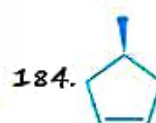
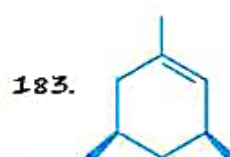
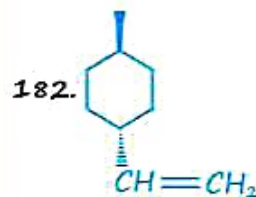
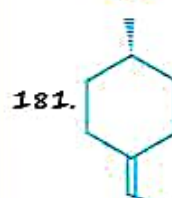
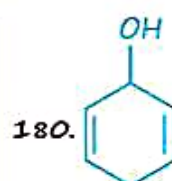
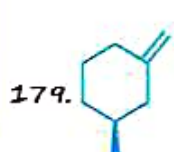
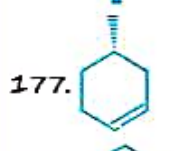
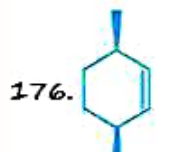
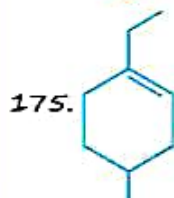
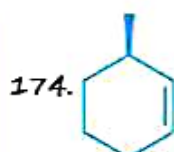
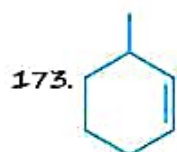
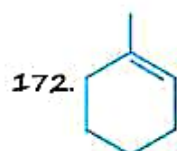
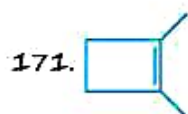
- (i) NBS is inert in nature
- (ii) trace amount of impurity Br_2 or HBr is added

- (iii) Reacts only with allylic ($\text{C}=\text{C}-\dot{\text{C}}$) and benzylic ($\text{Ph}-\dot{\text{C}}\text{H}_2$) positions.

Practice Questions

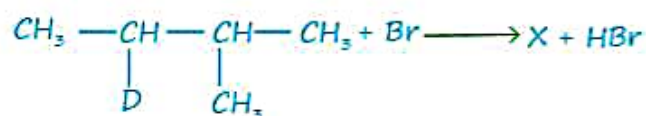
170. When following compound treated with NBS/h ν then find:

- Structure of monobrominated product
- Total monobrominated product
- optically active product
- Fractions obtained.

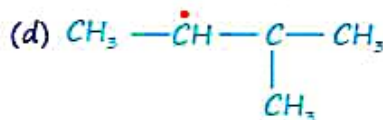
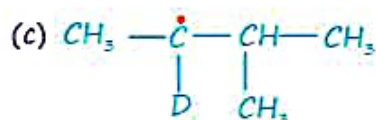
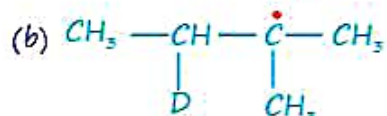
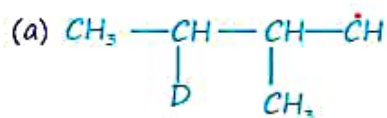


Previous Year Questions

10. Consider the following reaction



Identify the structure of the major product X
(IIT JEE 2002)



Sol. (b)

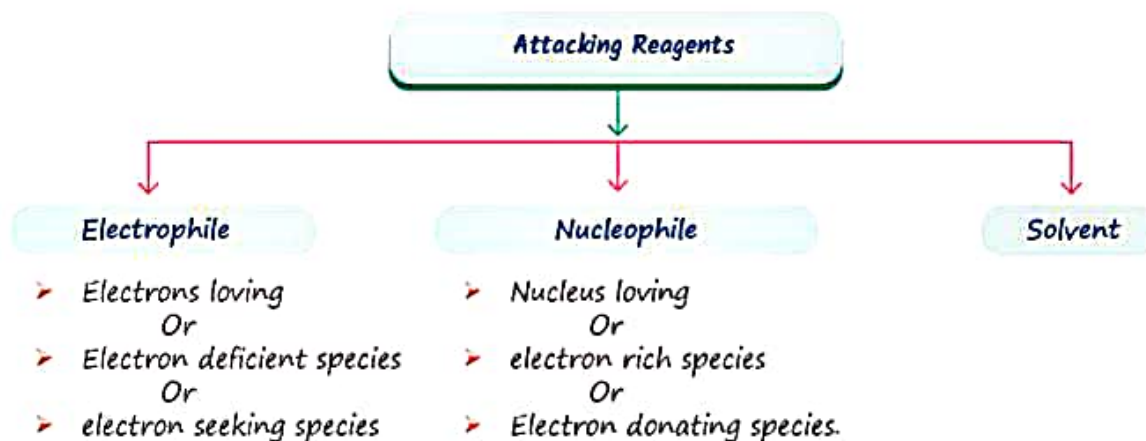
11. Which of the following is a free radical substitution reaction?
(2020-Covid)

- Acetylene with HBr
- Methane with Br₂/h ν
- Propene with HBr/(C₆H₅COO)₂
- Benzene with Br₂/AlCl₃

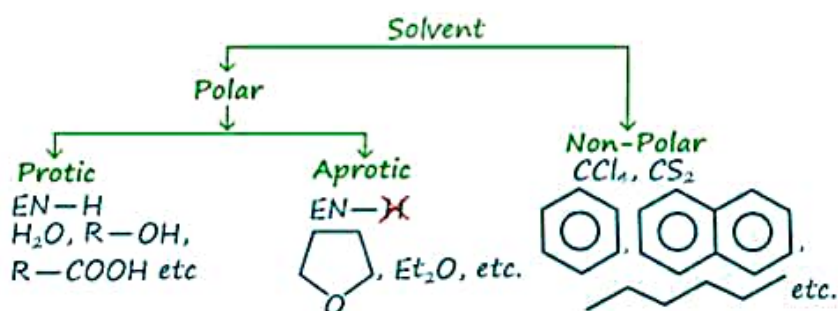
Sol. (b)

Attacking Reagents

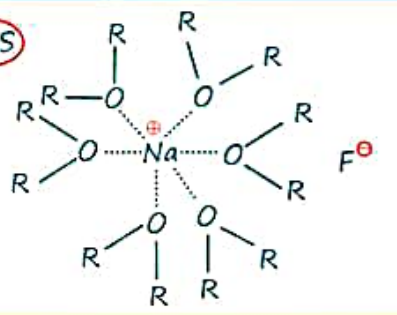
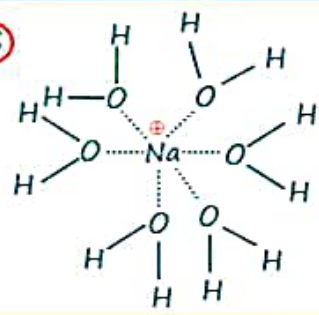
In chemistry, an attacking reagent is a molecule or ion that initiates a chemical reaction by interacting with a substrate molecule or an intermediate. It acts as an electron donor or electron acceptor, leading to the formation of a new chemical bond.



Nucleophilicity	Basicity
➤ Electron donating tendency to any E^+	➤ Electron donating tendency to H^+
$Nu^- + E^+ \rightleftharpoons E - Nu$	$B^- + H^+ \rightleftharpoons B - H$
➤ Kinetic property	➤ Thermodynamics properties



Nucleophilicity in different solvent

➤ NaF NaCl NaBr NaI	➤ NaF NaCl NaBr NaI	➤ NaF NaCl NaBr NaI
(PAS) 	(PPS) 	(NPS)
$F^- > Cl^- > E^- > I^-$	$NaI > NaBr > NaCl > NaF$	$NaI > NaBr > NaCl > NaF$

Electrophile	Nucleophile
1. Positive charged incomplete octet CH_3^+, F^+, Cl^+, Br^+, I^+, NO^+, NO_2^+, OH^+, H^+, D^+, T^+, $\text{R}-\overset{+}{\text{C}}=\text{O}$ etc. 2. Neutral & incomplete octet AlCl_3, AlBr_3, BF_3, BBR_3, Fe etc. 3. Neutral & Complete octet (π^*, σ^*) FeX_3, SbF_5, ZnCl_2, $\text{R}-\overset{\text{O}}{\underset{\text{H}}{\text{C}}}=\text{O}$, $\text{R}-\overset{\text{O}}{\underset{\text{X}}{\text{C}}}=\text{O}$, CO_2, SO_3, Br_2, I_2, $\text{I}-\text{Cl}$ etc.	1. Negatively Charged & complete octet CH_3^-, Cl^-, F^-, Br^-, I^-, OH^-, SH^-, CN^-, N_3^-, RCOO^-, H^-, D^-, NO_2^- ($\text{O}^--\text{N}=\text{O}$) 2. Neutral & Complete octet (lone pair) H_2O, NH_3, NR_3, $\text{R}-\ddot{\text{N}}\text{H}_2$, $\text{H}_2\text{N}-\ddot{\text{N}}\text{H}_2$ $\text{Ph}-\text{NH}-\text{NH}_2$, $\text{H}_2\text{N}-\text{NH}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$ 3. π - electrons Alkene, Alkyne,  Napthalene  etc.

Practice Questions

185. Which of the following statements are correct for nucleophile?

- (a) All negatively charged species are nucleophiles
 (b) Nucleophiles are Lewis bases
 (c) Alkenes, alkynes, benzene and pyridine are nucleophiles
 (d) All are correct

186. Consider the species.

- (I) OH^- (II) CH_3-O^-
 (III) CH_3^- (IV) NH_2^-

Arrange these nucleophilic species in their decreasing order of nucleophilicity:

- (a) III > IV > I > II (b) II > I > III > IV
 (c) I > II > III > IV (d) III > I > II > IV

187. The nucleophilicity of CH_3^- , NH_2^- , OH^- and F^- decrease in which order?

- (a) $\text{CH}_3^- > \text{NH}_2^- > \text{OH}^- > \text{F}^-$
 (b) $\text{OH}^- > \text{NH}_2^- > \text{CH}_3^- > \text{F}^-$
 (c) $\text{NH}_2^- > \text{OH}^- > \text{CH}_3^- > \text{F}^-$
 (d) $\text{CH}_3^- > \text{OH}^- > \text{F}^- > \text{NH}_2^-$

188. Arrange the following nucleophiles in the order of their nucleophilic strength:

- (a) $\text{OH}^- > \text{CH}_3\text{COO}^- > \text{OCH}_3^- > \text{C}_6\text{H}_5\text{O}^-$
 (b) $\text{CH}_3\text{COO}^- < \text{C}_6\text{H}_5\text{O}^- < \text{OCH}_3^- < \text{OH}^-$
 (c) $\text{C}_6\text{H}_5\text{O}^- > \text{CH}_3\text{COO}^- < \text{CH}_3\text{O}^- < \text{OH}^-$
 (d) $\text{CH}_3\text{COO}^- < \text{C}_6\text{H}_5\text{O}^- < \text{OH}^- < \text{CH}_3\text{O}^-$

189. Correct order of leaving group tendency is:

- (a) $\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$ (b) $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$
 (c) $\text{Cl}^- > \text{F}^- > \text{Br}^- > \text{I}^-$ (d) $\text{I}^- > \text{Cl}^- > \text{Br}^- > \text{F}^-$

190. Match the following

Column - I	Column - II
(a) I^-	(P) A good leaving group
(b) F^-	(Q) Very weak base
(c) $\text{CH}_3-\text{C}_6\text{H}_4-\text{SO}_3^-$	(R) A good nucleophile in polar protic solvent
(d) OH^-	(S) A poor leaving group

191. The most reactive nucleophile among the following is

- (a) CH_3O^- (b) $\text{C}_6\text{H}_5\text{O}^-$
 (c) $(\text{CH}_3)_2\text{CHO}^-$ (d) $(\text{CH}_3)_2\text{CO}^-$

192. Which of the following is not an electrophile?

- (a) CH_2CH_2 (b) BF_3 (c) $^+\text{NO}_2$ (d) Fe^{+3}

193. Which is the MOST basic nucleophile in the following series?

- (a) F^- (b) $\text{CH}_3\text{CH}_2\text{OH}$
 (c) H_2O (d) $\text{CH}_3\text{CH}_2\text{O}^-$

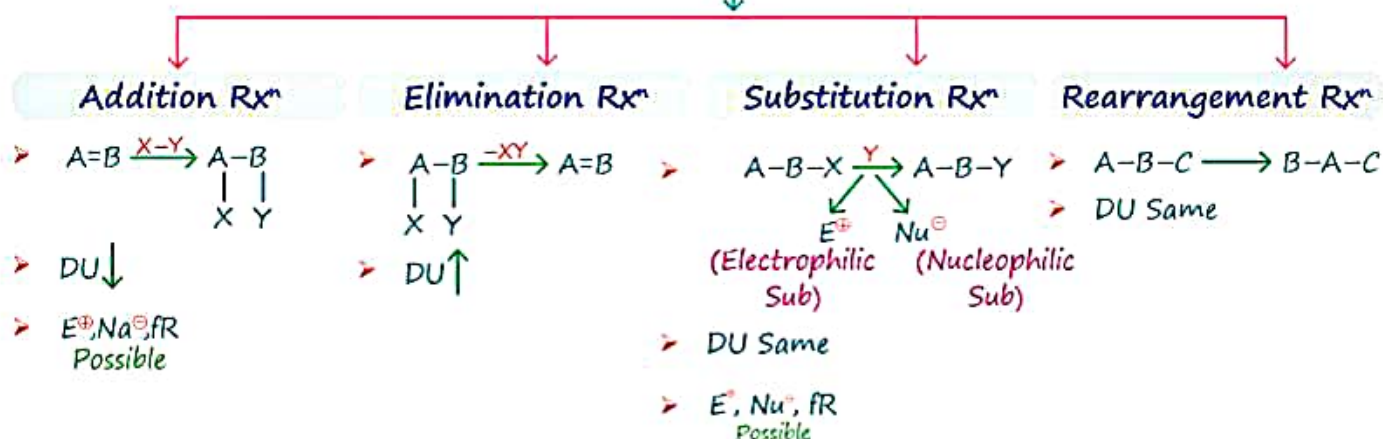
194. How are basicity and leaving group ability related?

- (a) They are not related to each other
 (b) Good leaving groups are strong bases
 (c) Good leaving groups are weak bases
 (d) Leaving group $\propto$ Basic strength

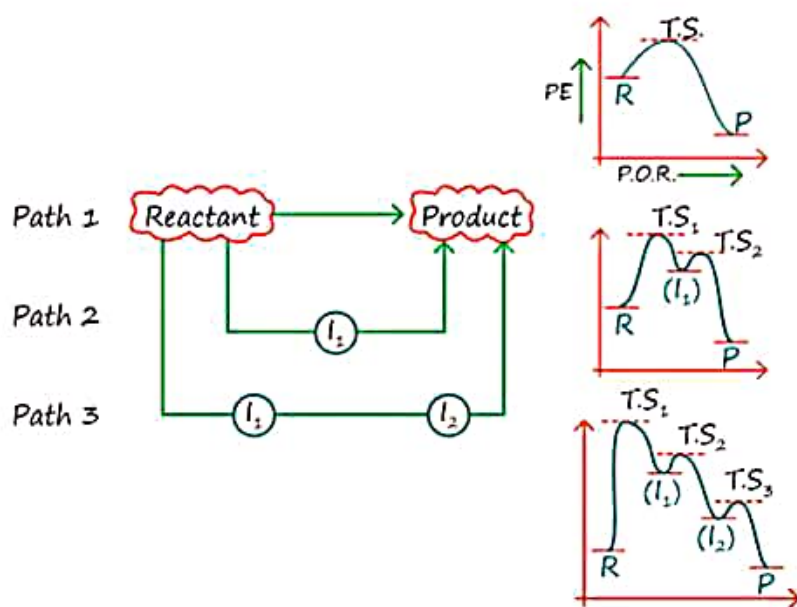
195. Correct order of leaving group character is:

- (a) $\text{CCl}_3\text{COO}^- > \text{CH}_3\text{COO}^- > \text{CH}_3\text{O}^- > \text{PhO}^-$
 (b) $\text{OTs}^- > \text{CH}_3\text{SO}_3^- > \text{PhO}^- > \text{OH}^-$
 (c) $\text{CCl}_3\text{COO}^- > \text{PhO}^- > \text{CH}_2=\text{CH}-\text{O}^- > \text{CH}_3\text{O}^-$
 (d) $\text{HCOO}^- > \text{PhO}^- > \text{CH}_3\text{COO}^- > \text{CH}_3\text{O}^-$

Types of Reaction

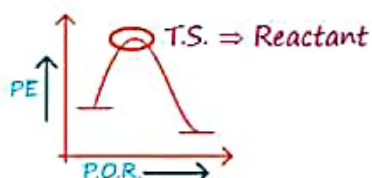


Basics of Rxⁿ



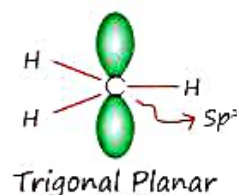
Hammond's Postulate

- Reaction is exothermic if T.S. will look like Reactant
- Reaction is endothermic if T.S. will look like Product



Basics of Carbocation

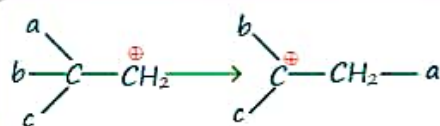
A species having a carbon atom possessing sextet of electrons and a positive charge is called a carbocation (earlier called carbonium ion).



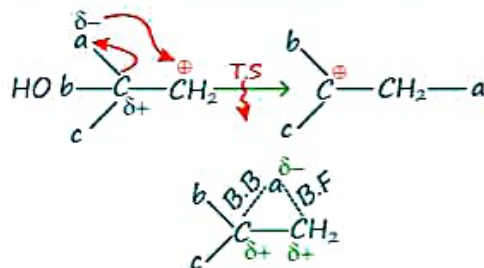
Properties	Reaction of Carbocation
➤ CH_3^+ ➤ $\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$, sp^2 ➤ Magnetic moment = $\sqrt{n(n+2)}$ B.M. = 0 M.M. ➤ $SM = 2s + 1 = 1$ SM, singlet, ➤ Vacant p_π orbital ➤ Octet incomplete ➤ EDG (+R, +H, +I) ➤ 3 Bond Pair ➤ 0 Unpaired electron	(Racemic mixture) If $a \neq b \neq c \neq d$

Rearrangement of Carbocation

Reaction



Mechanism

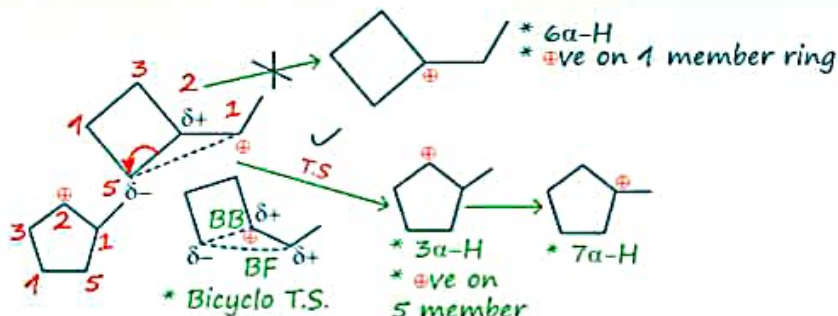


Baba Tea-Stall (BTS)

- | | |
|---|---|
| <ul style="list-style-type: none"> (i) 3 Member cyclic T.S involved (ii) $3C - 2e^-$ bond (iii) 1, 2- shift involved (iv) Carbocation Rearrangement (v) Stability increases | <ul style="list-style-type: none"> (vi) In case of rearrangement, carbocation gets stability in each step. (vii) During rearrangement, Configuration doesn't change (viii) In case of B. B. following migratory aptitude is applied. $H > Ar > R$ |
|---|---|

Ring Expansion

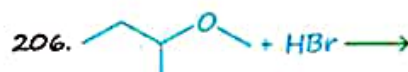
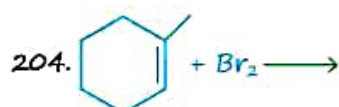
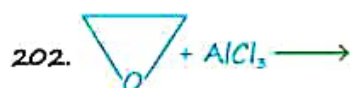
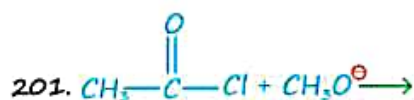
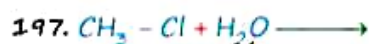
Rearrangement of carbocation is a process in which a carbocation shifts its structure (via hydride shift or alkyl shift) to form a more stable carbocation, usually moving from a less stable (1° or 2°) to a more stable (2° or 3° or resonance-stabilized) form.



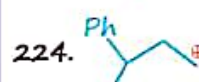
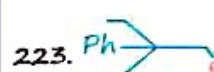
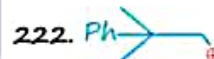
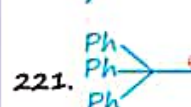
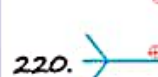
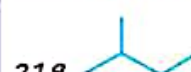
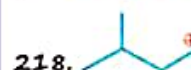
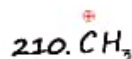
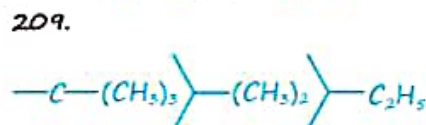
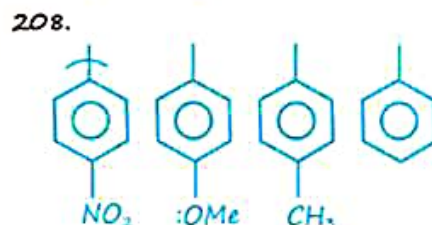
Practice Questions

➤ Attacking Reagents

196. Find the E^+ and Nu^- in the following reactions.



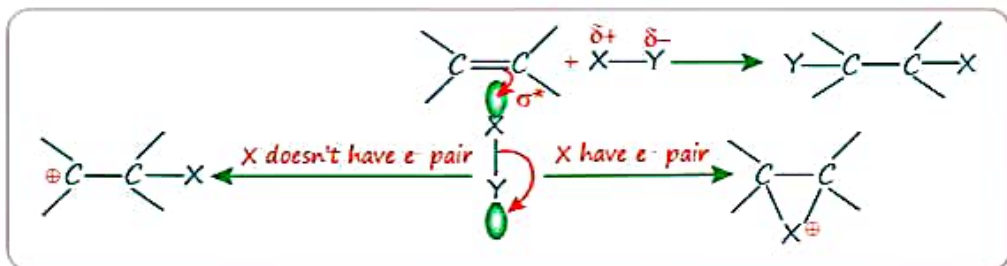
➤ Carbocation



Electrophilic Addition Reaction

Electrophilic addition reaction is a reaction where an electrophile attacks a double or triple bond in an unsaturated compound, leading to the addition of atoms or groups across the π bond.

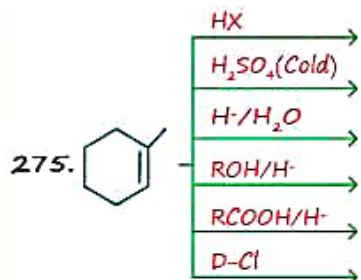
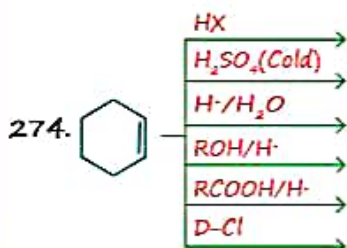
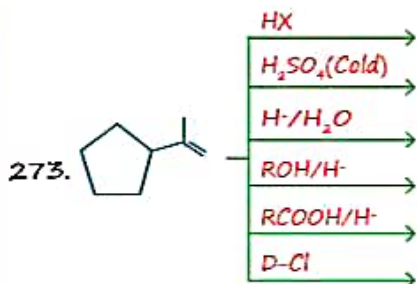
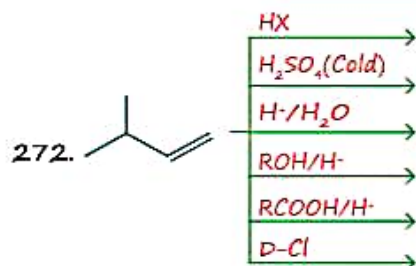
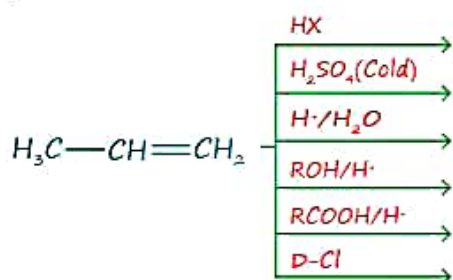
Reaction



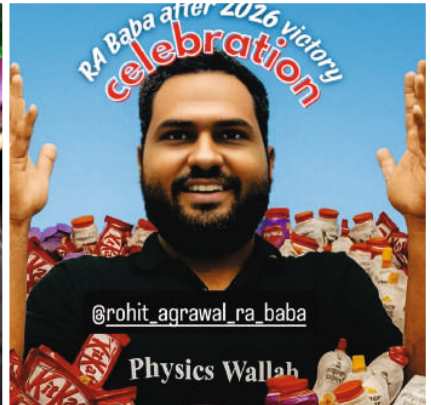
Reagent	E^+	Nu^-	Path
$H-Cl/H-Br/H-I$	H^+	X^- (Hydrohalogenation)	1(cc)
$D-Cl$	D^+	Cl^-	1(cc)
H^+/H_2O or H_3O^+ dilute H_2SO_4	H^+	H_2O (Hydration)	1(cc)
ROH/H^+	H^+	$R-OH_2^+$	1(cc)
$RCOOH/H^+$	H^+	$R-C(=O)OH$	1(cc)

Practice Questions

271.



Glimpse of BTS



About The Author



RA Baba

Ex-Director and Head of Organic Chemistry Department, Nucleus, Kota

Popularly known as **RA Baba**, he is a name synonymous with excellence in Organic Chemistry. With over **14 years of teaching experience**, he has mentored countless aspirants, including top JEE Advanced rankers such as **AIR 3, 8, 10, 12, 16, 18, 23, 24, and many others.**

A **two-time IIT-JEE qualifier** himself, he holds a B.Tech degree from **NIT Surathkal.**

He has **authored more than 5 books** for competitive exams, known for their conceptual clarity and student-focused design.

His mentorship extends to **Olympiad achievers** like Mrinal Sharan (currently at Oxford University) and Sushant Padha (IIT Bombay) and many more.

Famous for his **12-hour live Organic Chemistry sessions** on the **JEE Wallah channel**, he is known for teaching difficult topics in a fun and simple way that keeps students interested and motivated.

He believes in the philosophy: **"A student learns better while enjoying."** His intuitive approach and visual tools for reaction mechanisms make complex concepts accessible and enjoyable for learners.



SKC Sir

With **17+ years of teaching experience** in Organic Chemistry, he is a well-known name in **JEE and NEET preparation.** He has guided top NEET rankers including **AIR 1, 2, 3, 5, 6, 9, 11, 15, 20, 21, 25** and many more.

Former **#1 Legend Category Educator** at Unacademy (5 years)

Held senior leadership roles such as: **Vice President** of Sarvottam Career Institute Pvt. Ltd., Kota, **Director & HOD** of Chemistrio Institute Pvt. Ltd., Kota, **Former Senior Chemistry Faculty** at Allen Career Institute, Kota.

Author of **4 best-selling books**, known for their simple and effective explanations.

A registered author with top academic publishers: **Pearson India** – for NCERT Mantra and **Universities Press** – for Fundamental Problems in Organic Chemistry.



Majid Husain (Air 3 JEE Adv 2025)

My experience, learning organic chemistry under Rohit Agrawal sir was amazing. The depth of concepts he teaches and how it applies perfectly to JEE problems helps a lot. I can attest to the effectiveness and completeness of his notes. His notes made the most important part of the preparation for the subject which I followed through out helping me perform well and boost my performance. Apart from this his selected problems were perfect to prepare from scratch and reach JEE advanced level. The problem sets cover all relevant cases and strengthen the concepts. These notes along with the problems are a complete and sufficient course for JEE organic chemistry.

I recommend serious JEE aspirants to use these to learn organic chemistry the way it is intended for JEE advanced.



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